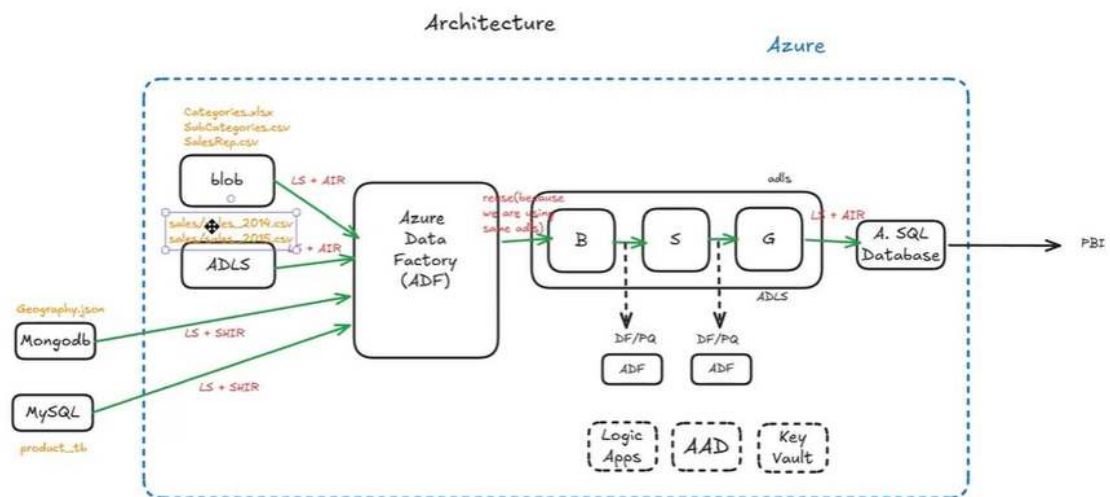


DENIS PROJECT:

Project Objective:

Design and implement a centralized **Azure Data Warehousing System** using the **Medallion Architecture (Bronze, Silver, and Gold layers)** to integrate sales-related data from multiple sources, including **Flat Files, Excel, CSV, MySQL, and MongoDB**. The solution will ingest raw data into the **Bronze layer**, cleanse, standardize, and transform it in the **Silver layer**, and create business-ready, curated datasets in the **Gold layer** for analytics and reporting. This architecture will provide a scalable, reliable, and governed single source of truth, enabling accurate sales insights, interactive dashboards, and data-driven decision-making across the organization.

Architecture:



DATA SOURCES:

- **ADLS** - Sales (Files in folder by year: csv)
- **Blob Container:** Categories.Xlsx, SubCategories.csv, SalesRep.csv
- **Mongodb:** Geography.json
- **MYSQL:** Product_Tbl

DATA MODELING APPROACH:

Fact Table:

- SalesFact

Dimension Tables:

- DimProduct
- DimCategory
- DimSubCategory

- DimGeography
- DimSalesRep

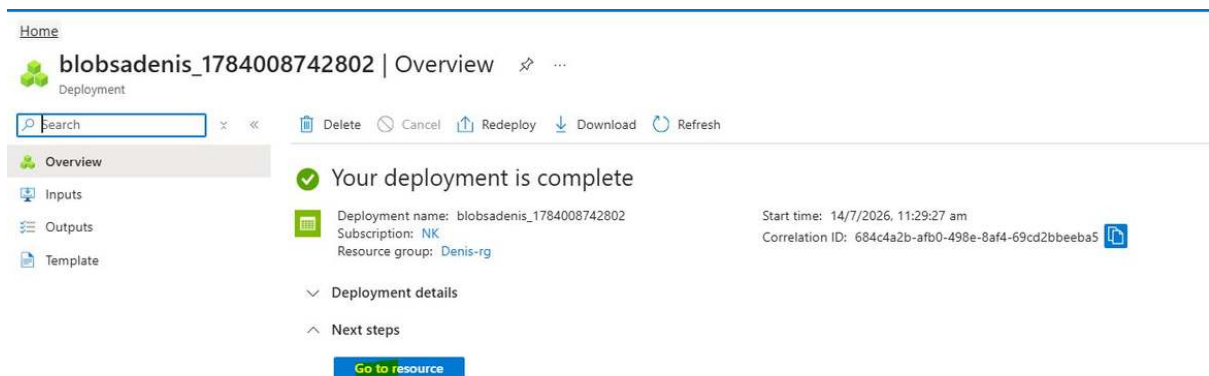
Resources Required:

- Azure Subscription
- Resource Group
- Blob Storage
- Data Lake Storage Gen2
- MySQL
- Azure Cosmos DB
- Azure Data factory
- Logic Apps(For scheduling pipelines)
- Azure SQLDatabase
- Azure Keyvault(For storing passwords)

Creating Services:

1.Storage account(blob):

Create resource---storage account---fill in all the basic details---create



Go to storage browser---create a container---denisdimraw---upload files category,subcategory,salesrep

Home > blobsadenis_1784008742802 | Overview > blobsadenis

blobsadenis | Storage browser

Storage account

Search

blobsadenis

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Settings

Configuration

Data Lake Gen2

Add or remove favorites by pressing Ctrl+Shift+F

blobsadenis

Favorites

Recently viewed

Blob containers

Slogs

Classic file shares

Queues

Tables

+ Add container

Upload

Refresh

Delete

Change access level

Restore containers

Search containers by prefix

Showing all 1 items

Name	Last modified	Anonymous access level
slogs	14/07/2026, 11:30:14	Private

New container

Name *

denisdimraw

Anonymous access level

Private (no anonymous access)

The access level is set to private because anonymous access is disabled on this storage account.

Advanced

Create

Give feedback

Home > blobsadenis

blobsadenis | Storage browser

Storage account

Search

blobsadenis

Overview

Activity log

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Add or remove favorites by pressing Ctrl+Shift+F

blobsadenis

Favorites

Recently viewed

Blob containers

Slogs

denisdimraw

Classic file shares

Queues

Tables

+ Add Directory

Upload

Change access level

Refresh

Delete

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Showing all 0 items

No items found

Upload blob

3 file(s) selected: Categories.csv, SalesRep.csv, SubCategories.csv

Drag and drop files here or Browse for files

Overwrite if files already exist

Advanced

Upload

Give feedback

Home > blobsadenis

blobsadenis | Storage browser

Storage account

Search

blobsadenis

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

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Add or remove favorites by pressing Ctrl+Shift+F

blobsadenis

Favorites

Recently viewed

Blob containers

Slogs

denisdimraw

Classic file shares

Queues

Tables

+ Add Directory

Upload

Change access level

Refresh

Delete

Copy

Paste

Rename

Acquire lease

Break lease

Edit columns

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 3 items

Name	Last modified	Access tier	Blob type	Size	Lease state
Categories.csv	14/07/2026, 12:38:38	Cold (Inferred)	Block blob	44 B	Available
SalesRep.csv	14/07/2026, 12:38:38	Cold (Inferred)	Block blob	160 B	Available
SubCategories.csv	14/07/2026, 12:39:23	Cold (Inferred)	Block blob	18.09 KiB	Available

2.Storage account(ADLS):

Create resource---storage account—fill in all the basic details --enable hierarchical name space---create

Home

denisadlssa_1784009146610 | Overview

Deployment

Search

Delete Cancel Redeploy Download Refresh

Overview

Inputs

Outputs

Template

Your deployment is complete

Deployment name: denisadlssa_1784009146610
Subscription: NK
Resource group: Denis-rg

Start time: 14/7/2026, 11:36:00 am
Correlation ID: 5298ba10-5f92-42d7-89cc-7362a0bc18a3

Deployment details

Next steps

Go to resource

Home > denisadlssa

denisadlssa | Storage browser

Storage account

Search

Overview

Activity log

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Diagnose and solve problems

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Add or remove favorites by pressing Ctrl+Shift+F

denisadlssa

Favorites

Recently viewed

Blob containers

Slogs

View all

Classic file shares

Queues

Tables

+ Add container Upload Refresh Delete Change access level Restore containers

Blob containers

Search containers by prefix

Showing all 1 items

Name	Last modified	Anonymous access level
Slogs	14/07/2026, 11:36:24	Private

New container

Name *

denisfactraw

Anonymous access level

Private (no anonymous access)

The access level is set to private because anonymous access is disabled on this storage account.

Advanced

Create Give feedback

Create a folder inside a denisfactraw(container)---sales

Home > denisadlssa

denisadlssa | Storage browser

Storage account

Search

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Add or remove favorites by pressing Ctrl+Shift+F

denisadlssa

Favorites

Recently viewed

Blob containers

Slogs

denisfactraw

View all

Classic file shares

Queues

Tables

+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Blob containers > denisfactraw

Authentication method: Access key (Switch to Microsoft Entra user account)

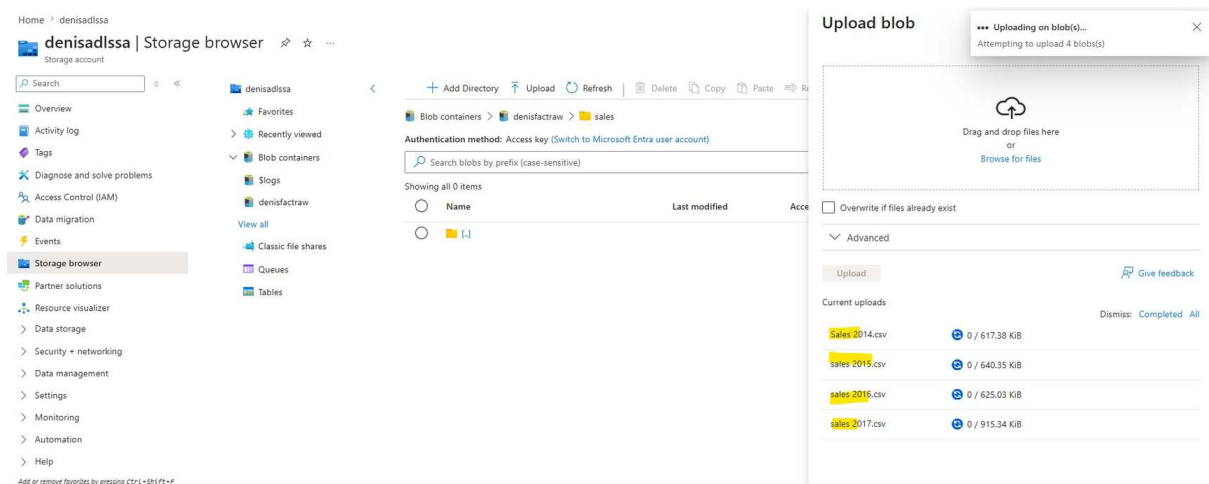
Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 1 items

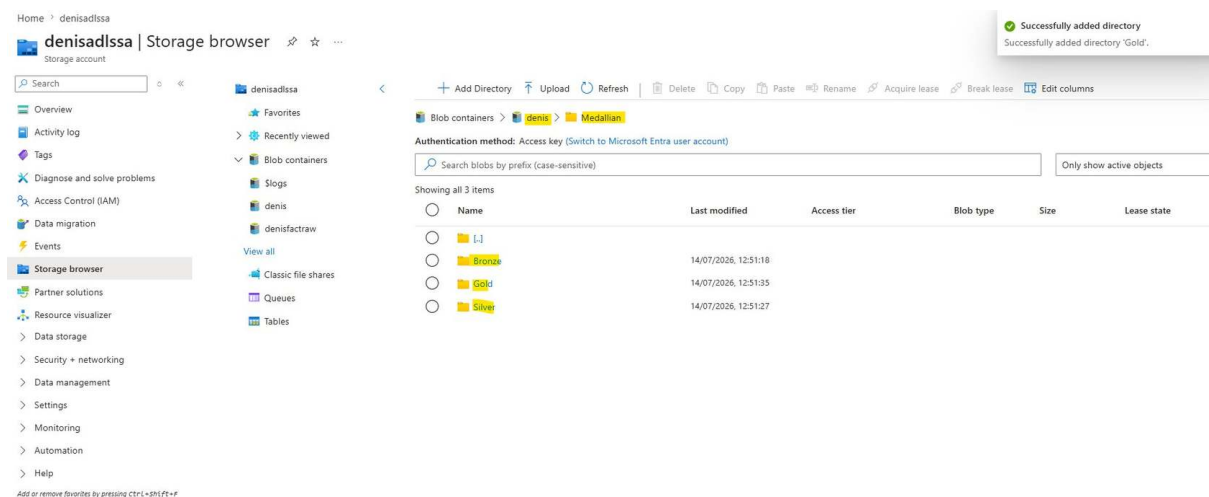
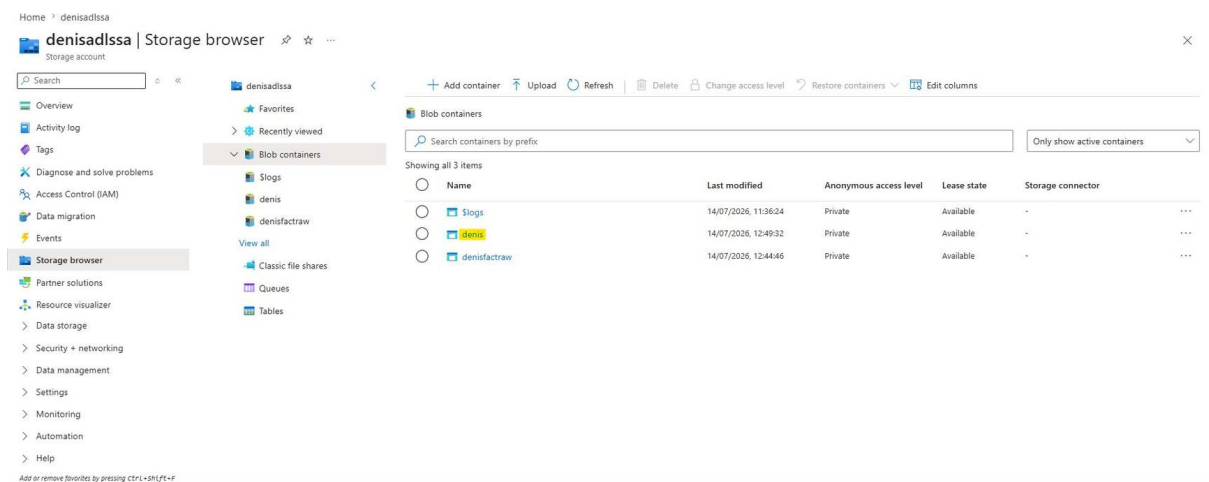
Name	Last modified	Access tier	Blob type	Size	Lease state
sales	14/07/2026, 12:46:39				...

Upload all the sales files in the folder



Hence we are using adls for medallion architecture

Let's create a container called denis--- medallion---bronze,silver,gold



3.Create ADF:

Home > Microsoft.DataFactory-20260714113847 | Overview > Denis-rg

nkdenisprojectadf
Data factory (V2)

Search [] Delete

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Resource visualizer
- Settings
 - Networking
 - Managed identities
 - Properties
 - Locks
- Getting started
- Monitoring
- Automation
- Help

Essentials

Resource group (move) : [Denis-rg](#) Type : Data factory (V2)
Status : Succeeded Getting started : [Quick start](#)
Location : West US 2
Subscription (move) : [NK](#)
Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f

Azure Data Factory Studio

[Launch studio](#)

Quick Starts Tutorials Template Gallery Training Modules

4.Create Azure SQL DB:

Home

Microsoft.SQLDatabase.newDatabaseNewServer_eb00c596744c49bc80c02 | Overview ...

Deployment

Search [] Delete Cancel Redeploy Download Refresh

Overview

- Inputs
- Outputs
- Template

Your deployment is complete

Deployment name : Microsoft.SQLDatabase.newDatabaseNewServer_eb00c596744c49bc80c... Start time : 14/7/2026, 11:52:25 am
Subscription : [NK](#) Correlation ID : 15caa5bd-da0f-464b-a993-903
Resource group : [Denis-rg](#)

> Deployment details

< Next steps

[Go to resource](#)

5.Create Azure key vault:

To secure the passwords/keys

Create a key vault ...

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

NK

Resource group *

Denis-rg

[Create new](#)

Instance details

Key vault name * ⓘ

nkdeniskeyvault

Region *

West US 2

Pricing tier * ⓘ

Standard

Recovery options

Soft delete protection will automatically be enabled on this key vault. This feature allows you to recover or permanently delete a key vault and secrets for the duration of the retention period. This protection applies to the key vault and the secrets stored within the key vault.

To enforce a mandatory retention period and prevent the permanent deletion of key vaults or secrets prior to the retention period elapsing, you can turn on purge protection. When purge protection is enabled, secrets cannot be purged by users or

[Previous](#)

[Next](#)

[Review + create](#)

Create a key vault

- Basics
- Access configuration
- Networking
- Tags
- Review + create

Configure data plane access for this key vault

To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Aut

Permission model

Grant data plane access by using a [Azure RBAC](#) or [Key Vault access policy](#)

- ☐ Azure role-based access control (recommended) ⓘ
- ☒ Vault access policy ⓘ

Resource access

- ☐ Azure Virtual Machines for deployment ⓘ
- ☐ Azure Resource Manager for template deployment ⓘ
- ☐ Azure Disk Encryption for volume encryption ⓘ

Access policies

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

- Previous
- Next
- Review + create

Create a key vault

You can connect to this key vault either publicly, via public IP addresses or service endpoints, or privately, using a private endpoint.

Enable public access ☒

Public Access

Allow access from:

- ☒ All networks
- ☐ Selected networks
- i** Traffic from all public networks can access this resource. This is not recommended for private applications or environments. [Learn more about key vault network security configuration](#)

Private endpoint

Create a private endpoint to allow a private connection to this resource. Additional private endpoint connections can be created within the key vault or private link center. [Learn more](#)

+ Create a private endpoint

Name	Subscription	Resource group	Region	Subnet	Private DNS Zone
Click on add button to add private endpoint					

- Previous
- Next
- Review + create

Successfully created

Home

nkdeniskeyvault Key vault

How do I troubleshoot issues with this Key Vault? Show me total service API hits metrics of this Key Vault. Show me overall vault saturation percentage metrics of this Key Vault.

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Access policies
- Resource visualizer
- Events
- Objects
- Settings
- Monitoring
- Automation
- Help

Essentials

Resource group (move) : [Denis-rg](#)

Location : West US 2

Subscription (move) : [NK](#)

Subscription ID : 5bd8f1cd-03ad-48fa-8410-6e58274fc11f

Tags (edit) : [Add tags](#)

Vault URI : <https://nkdeniskeyvault.vault.azure.net/>

Sku (Pricing tier) : Standard

Directory ID : d8c0f1cb-2b36-4bb9-bae3-19c3c1d0d2

Directory Name : Default Directory

Soft-delete : [Enabled](#)

Purge protection : [Disabled](#)

Get started Properties Monitoring Tools + SDKs Tutorials

Manage keys and secrets used by apps and services

Our recommendation is to use a vault per application per environment (Development, Pre-Production and Production). This helps you to not share secrets across environments and also reduces the threat in case of a breach.

6.Create MYSQL:

Create database denis_db

Import table---product

SCHEMAS

Filter objects

- amazon_db
- assignment2_db
- assignment3
- box_office_db
- denis_db**
 - Tables
 - Views
 - Stored Procedures
 - Functions
- flipkart_db
- guess_db
- join1_db
- joins_db
- ksr_db
- Tables

Administration Schemas

Information

Schema: **denis_db**

Limit to 1000 rows

```

98
99
100
101
102
103 • create database Denis_DB;
104 • use denis_Db;
105 • select * from product;

```

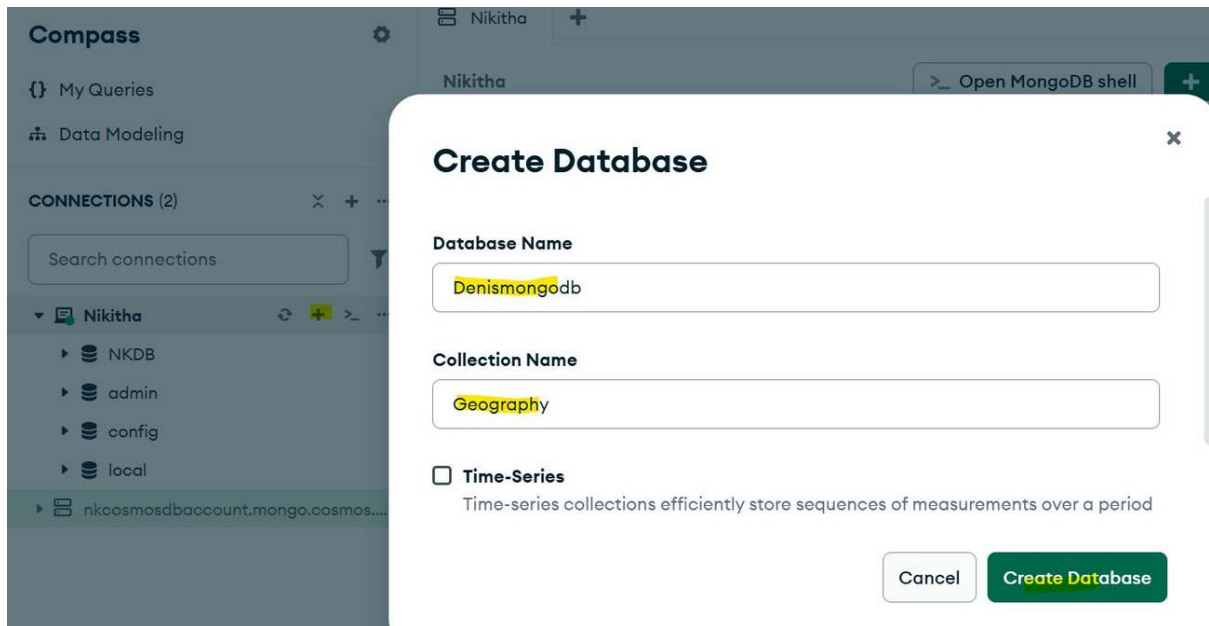
Result Grid

ProductID	Sub Category Key	Color	ProductName	RetailPrice	StandardCost
1	3	Red	Alder	23.95	7.55
2	2	Blue	Linder	23.95	7.55
3	2	Green	Magnum	23.95	7.55
4	1	Red	Quad	43.95	13.75
5	1	Blue	Black Monk	43.95	13.75
6	4	Green	Quad	43.95	13.75

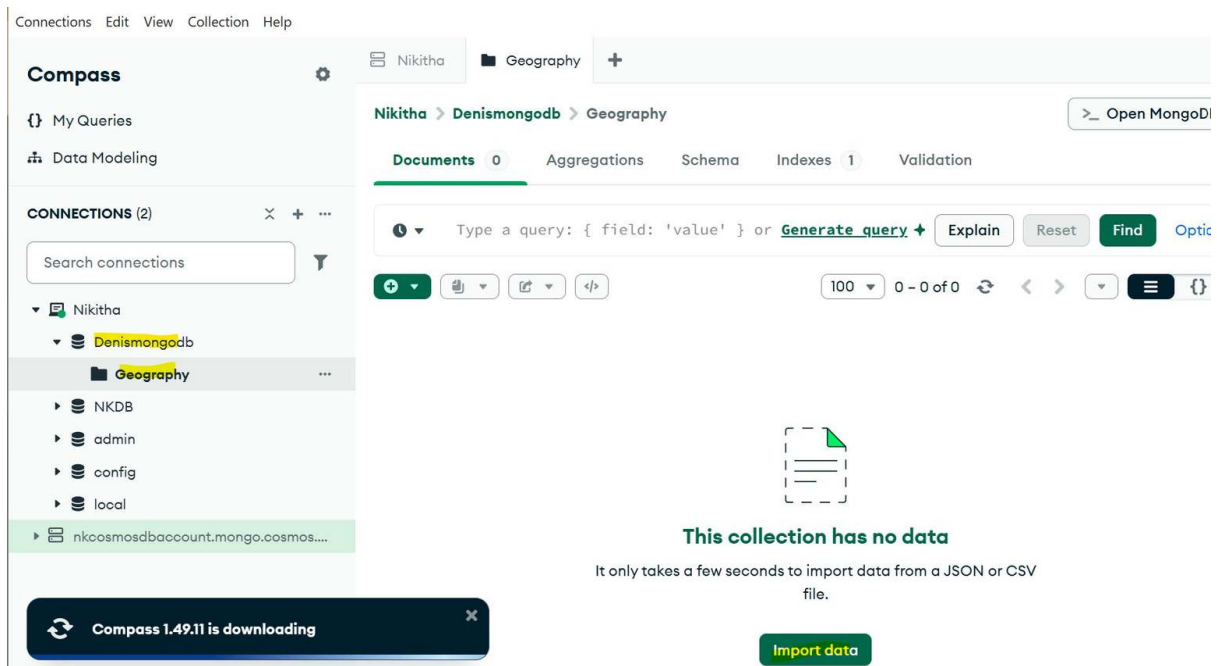
product 1 x

7.Create Mongodb:

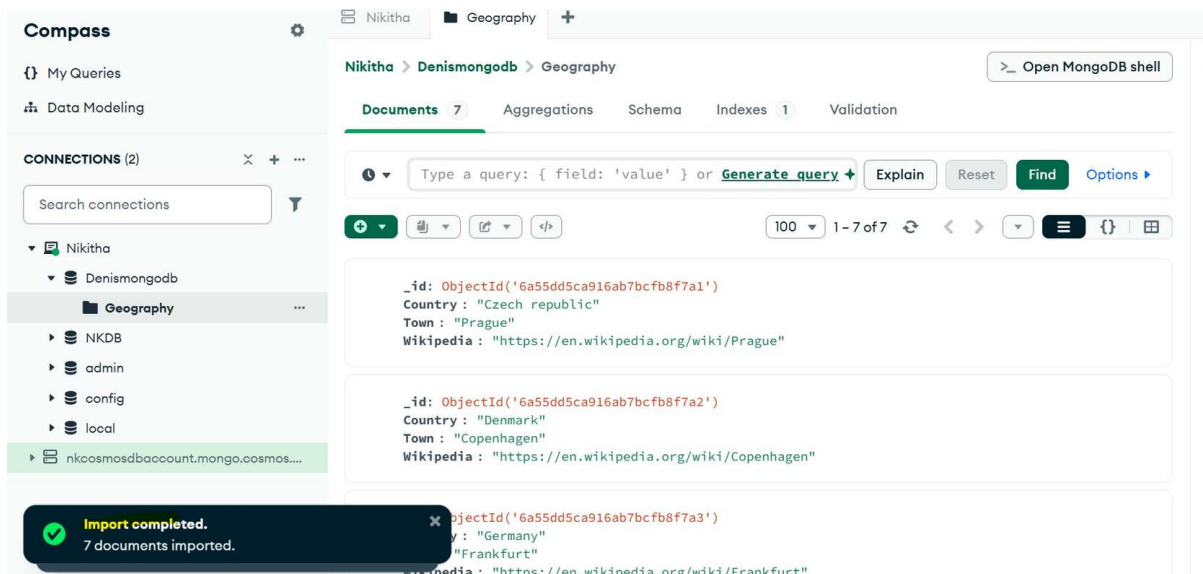
Open mongodb---create db---collection name---import the data



Import data



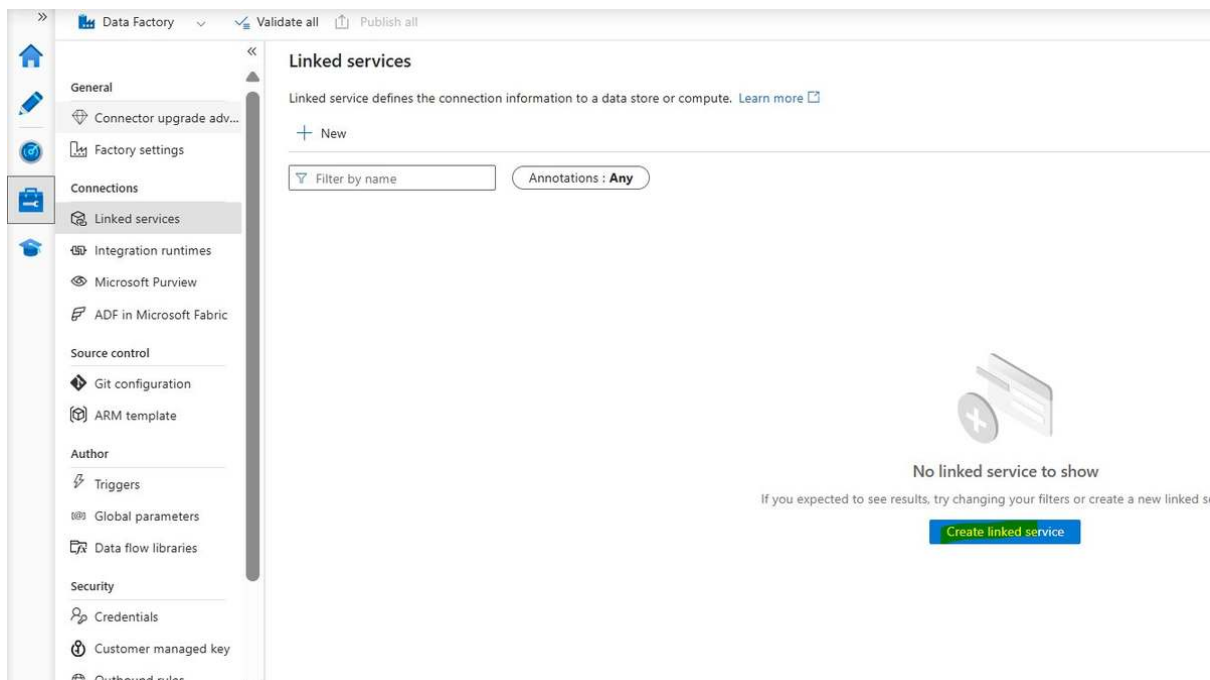
Successfully imported data



Implementation:

Steps Inside ADF Studio:

1)creating linked services inside ADF:



New linked service

Data store

Compute

 Jira	 MariaDB	 Microsoft 365 (Office 365)
 Microsoft Access	 Microsoft Fabric Lakehouse	 Microsoft Fabric Warehouse
 MongoDB	 MongoDB Atlas	 MySQL
 Snowflake	 Amazon Redshift	 Databricks

Continue

Cancel

New linked service

 MySQL [Learn more](#) 

Name *

MySql2adf_Is

Description

Connect via integration runtime * 

 SHintegrationRuntime1  

 The credentials are stored in the machines of self-hosted integration runtime if you don't choose to store them in Azure Key Vault.

Server name *

localhost

Port

3306

Database name *

denis_db

User name *

root

Password

[Azure Key Vault](#)

Password *

 Connection successful

Create

Back

 Test connection

Cancel

But I don't wanted to give hardcoded values and the details to visible everyone----now the only option is to use the azure key vault (to secure the password)

New linked service

MySQL [Learn more](#)

Name *
MySql2adf_Is

Description

Connect via integration runtime * ⓘ
SHIntegrationRuntime1

Server name *
localhost

Port
3306

Database name *
denis_db

User name *
root

Password **Azure Key Vault**

AKV linked service * ⓘ
Select...

Create Back Test connection Cancel

Now---go to azure key vault service----go to secrets---right side click-->+ generate/import

Home > nkdeniskeyvault

nkdeniskeyvault | Secrets

Key vault

Search + Generate/import Refresh Restore Backup Manage deleted secrets View sample code

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Access policies
- Resource visualizer
- Events
- Objects
- Keys
- Secrets**
- Certificates
- Settings
- Monitoring
- Automation
- Help

Name	Type	Status
There are no secrets available.		

Give the name and the password of mysql

Home > nkdeniskeyvault | Secrets



Create a secret

Upload options	Manual
Name *	mysqlpass
Secret value *	*****
Content type (optional)	
Set activation date	☐
Set expiration date	☐
Enabled	☒ Yes ☐ No
Tags	0 tags

Create

Cancel



Successfully created

Home > nkdeniskeyvault



nkdeniskeyvault | Secrets

Key vault

Search

+ Generate/Import Refresh Restore Backup Manage deleted secrets View sample code

Overview

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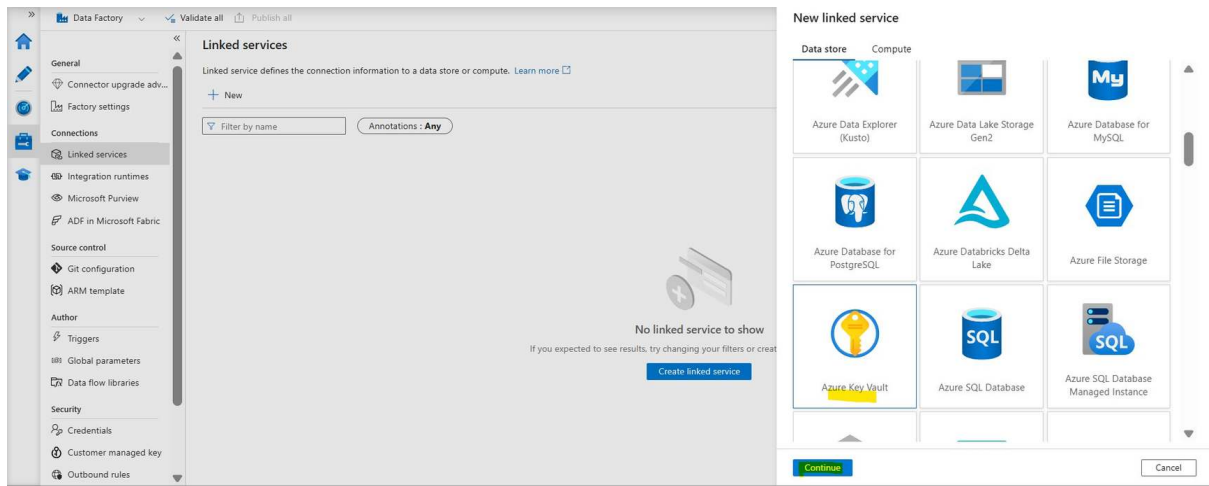
Automation

Help

The secret 'mysqlpass' has been successfully created.

Name	Type	Status	Expiration date
mysqlpass		✓ Enabled	

*****need to create linkedservice from azure keyvault to adf



New linked service

Azure Key Vault

Name *

AzureKeyVault2adf_ls

Description

Azure key vault selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Azure key vault name *

nkdeniskeyvault

[Edit key vault](#)

Authentication method

System-assigned managed identity

Managed identity name: **nkdenisprojectadf**
Managed identity object ID: **3bfd7943-7312-4af8-a787-cf08706823f2**
Grant Data Factory service managed identity access to your Azure Key Vault. [Learn more](#) ⓘ

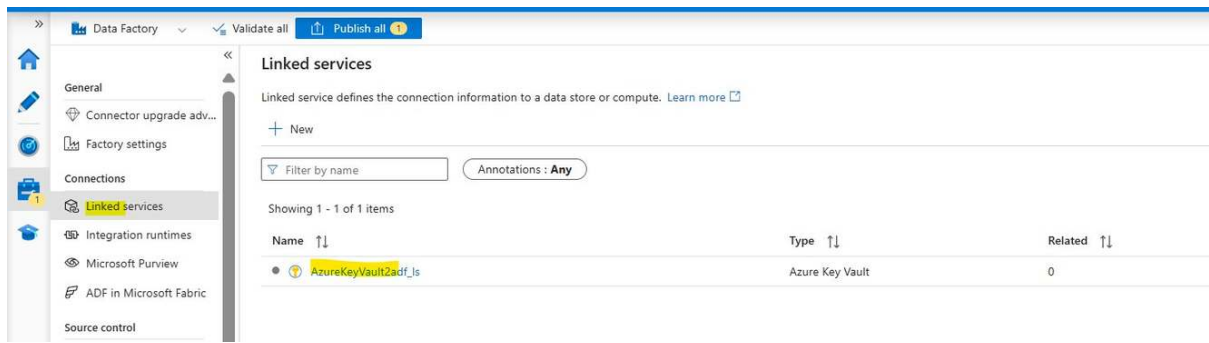
Test connection

☒ To linked service ☐ To secret

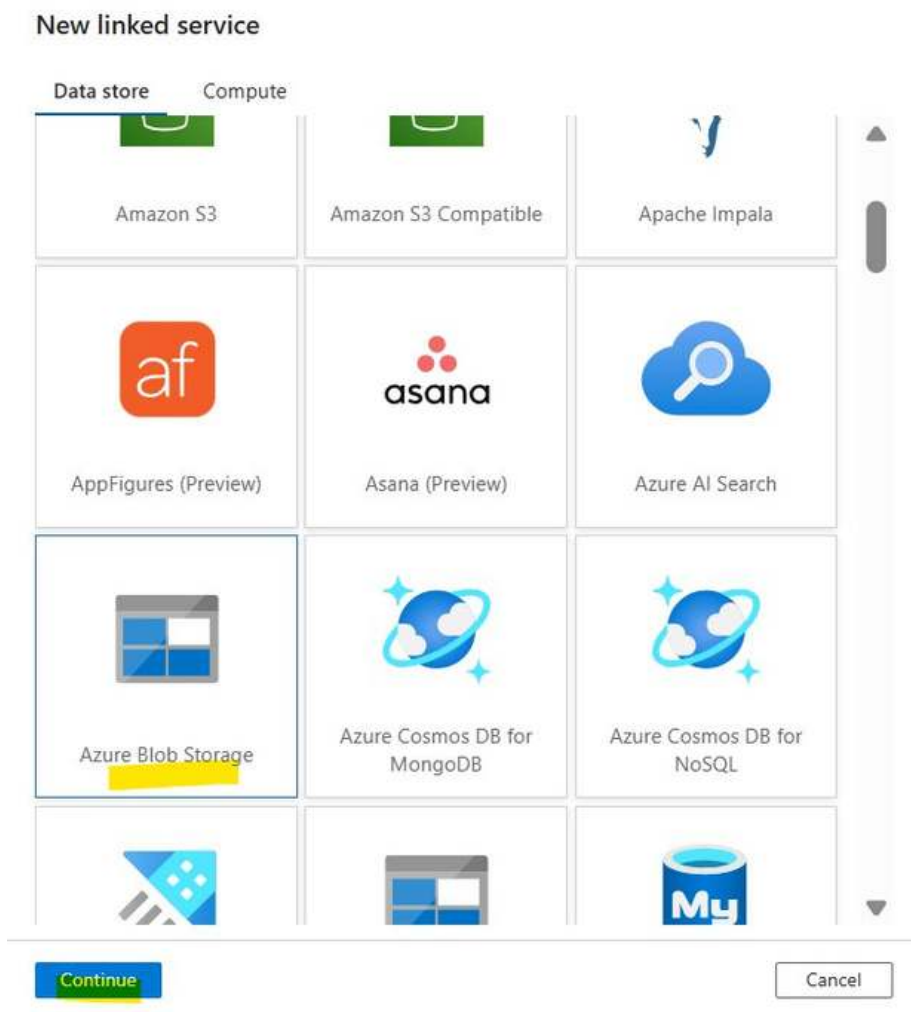
Connection successful

[Create](#) [Back](#) [Test connection](#) [Cancel](#)

Successfully now we have 1 linked service from azure keyvault-----adf



2)Creating linked service between azure blob sa---adf



New linked service

Azure Blob Storage [Learn more](#)

Name *

AzureBlob2adf_ls

Description

Connect via integration runtime * ⓘ

✓

 AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string

Azure Key Vault

Account selection method ⓘ

☒ From Azure subscription

☐ Enter manually

Azure subscription ⓘ

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

blobsadenis

Additional connection properties

✓ Connection successful

Create

Back

Test connection

Cancel

Inorder to secure the password we will use the azure key vault---secret---generate/import

Home > nkdeniskeyvault | Secrets

Create a secret ...

Upload options

Manual

Name * ⓘ

Blobaccesskey

Secret value * ⓘ

Enter the secret.

Content type (optional)

Set activation date ⓘ

☐

Set expiration date ⓘ

☐

Enabled

Yes

No

Tags

0 tags

Create

Cancel

Inorder to provide the secret value---go to blob sa---security+networking---access---connection string---copy it and give in the azure key vault secret value

Home > blobsadenis

blobsadenis | Access keys ☆ ...

Storage account

Search

Activity log

Tags

Diagnose and solve problems

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Data migration

Events

Storage browser

Storage Mover

Partner solutions

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Data storage

Security + networking

Networking

Front Door and CDN

Access keys

Shared access signature

Rotation

Set rotation reminder Refresh Give feedback

Access keys authenticate your applications' requests to this storage account. Keep your keys in a secure location like Azure Key Vault, and replace them often with new keys. The two keys allow you to replace one while still using the other.

Remember to update the keys with any Azure resources and apps that use this storage account.
[Learn more about managing storage account access keys](#)

Storage account name
blobsadenis

key1 Rotate key
Last rotated: 14/07/2026 (0 days ago)

Key
..... Show

Connection string
..... Show

key2 Rotate key
Last rotated: 14/07/2026 (0 days ago)

Key
..... Show

Connection string
DefaultEndpointsProtocol=https;AccountName=blobsadenis;AccountKey=p2kQo... Hide

Paste in the secret value---azure key vault

Home > nkdeniskeyvault | Secrets

Create a secret ...

Upload options
Manual

Name * ⓘ
Blobaccesskey

Secret value * ⓘ
..... ✓

Content type (optional)
.....

Set activation date ⓘ
☐

Set expiration date ⓘ
☐

Enabled
☒ Yes ☐ No

Tags
0 tags

Create Cancel

Successfully created

Home > nkdeniskeyvault

nkdeniskeyvault | Secrets ☆ ...

Key vault

Search

Generate/Import Refresh Restore Backup Manage deleted secrets View sample code

The secret 'Blobaccesskey' has been successfully created.

Name	Type	Status	Expiration date
Blobaccesskey		✓ Enabled	
mysqlpass		✓ Enabled	

Overview

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Similarly for adls---security+networking---copy connection string---go to azure key vault---secret---generate/import---paste the connection string in the secret value

Home > nkdeniskeyvault | Secrets

Create a secret ...

Upload options Manual

Name * adlspasskey

Secret value * Enter the secret.

Content type (optional)

Set activation date

Set expiration date

Enabled Yes No

Tags 0 tags

Create Cancel

Home > denisadlssa



denisadlssa | Access keys

Storage account

Search

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- Diagnose and solve problems
- Access Control (IAM)
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- Events
- Storage browser
- Partner solutions
- Resource visualizer
- Data storage
- Security + networking
 - Networking
 - Access keys**
 - Shared access signature
 - Encryption

Add or remove favorites by pressing Ctrl+Shift+F

Set rotation reminder Refresh Give feedback

Access keys authenticate your applications' requests to this storage account. Keep your keys in a secure location like Azure Key Vault, and replace them often with new keys. The two keys allow you to replace one while still using the other.

Remember to update the keys with any Azure resources and apps that use this storage account.
[Learn more about managing storage account access keys](#)

Storage account name

denisadlssa

key1 Rotate key

Last rotated: 14/07/2026 (0 days ago)

Key

.....

Show

Connection string

.....

Show

key2 Rotate key

Last rotated: 14/07/2026 (0 days ago)

Key

.....

Show

Connection string

DefaultEndpointsProtocol=https;AccountName=denisadlssa;AccountKey=Tz+PF...

Hide

Home > nkdeniskeyvault | Secrets



Create a secret

Upload options

Manual

Name *

adlspasskey

Secret value *

.....

Content type (optional)

Set activation date

☐

Set expiration date

☐

Enabled

Yes No

Tags

0 tags

Create

Cancel

Successfully created adls passkey

Home > nkdeniskeyvault

nkdeniskeyvault | Secrets ☆ ...

Key vault

[+ Generate/Import](#) [Refresh](#) [Restore Backup](#) [Manage deleted secrets](#)

[Overview](#)

[Activity log](#)

[Access control \(IAM\)](#)

[Tags](#)

[Diagnose and solve problems](#)

[Access policies](#)

[Resource visualizer](#)

[Events](#)

[Objects](#)

[Keys](#)

[Secrets](#)

[Certificates](#)

[Settings](#)

[Monitoring](#)

[Automation](#)

[Help](#)

[i](#) The secret 'adlspasskey' has been successfully created.

Name	Type
adlspasskey	
Blobaccesskey	
mysqlpass	

Now I must able to see in the adf linked service ---secret name that created in the azure key vault

New linked service

 Azure Blob Storage [Learn more](#)

Name *

AzureBlob2adf_Is

Description

Connect via integration runtime * 

 AutoResolveIntegrationRuntime

Authentication type

Account key

Connection string

Azure Key Vault

AKV linked service * 

AzureKeyVault2adf_Is

Secret name * 

Loading failed

 Failed [More](#)

☐ Edit

Secret version 

Latest version

☐ Edit

Create

Back

 Test connection

Cancel

It is not showing----so we need to provide access----go to azure key vault---add role—key vault administrator—select members---adf

Home > nkdeniskeyvault

 nkdeniskeyvault | Access control (IAM)  

Search  Add  Download role assignments  Edit columns  Refresh  Delete  Feedback

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 Diagnose and solve problems

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> Settings

> Monitoring

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> Help

Check access Role assignments Roles Deny assignments

My access

View my level of access to this resource.

[View my access](#)

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

[Check access](#)

Grant access to this resource

Grant access to resources by assigning a role.

[Learn more](#)

[Add role assignment](#)

View access to this resource

View the role assignments that grant access to this and other resources.

[Learn more](#)

[View](#)

View deny assignments

View the role assignments that have been denied access to specific actions at this scope.

[Learn more](#)

[View](#)

Add role assignment

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

[Copilot can help pick a role](#)

Job function roles

Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Name	Description	Type	Category	Details
App Configuration Contributor	Grants permission for all management operations, except purge, for App Configuration resources. This role does not grant access to data pla...	BuiltInRole	None	View
App Configuration Reader	Grants permission for read operations for App Configuration resources. This role does not grant access to data plane resources such as key...	BuiltInRole	None	View
Azure Managed Redis Reader	Read Azure Managed Redis resources and their configuration. Cannot modify resources, retrieve access keys, or read data stored in the cache.	BuiltInRole	None	View
Cognitive Services Content Understanding Contributor	Has access to read, create, update, analyze, and test operations for Azure AI Content Understanding. Can manage analyzers, classifiers, label...	BuiltInRole	None	View
Cognitive Services Contributor	Lets you create, read, update, delete and manage keys of Cognitive Services.	BuiltInRole	AI + Machine Learning	View
Cognitive Services User	Lets you read and list keys of Cognitive Services.	BuiltInRole	AI + Machine Learning	View
Cosmos DB Operator	Lets you manage Azure Cosmos DB accounts, but not access data in them. Prevents access to account keys and connection strings.	BuiltInRole	Databases	View
Key Vault Administrator	Perform all data plane operations on a key vault and all objects in it, including certificates, keys, and secrets. Cannot manage key vault resour...	BuiltInRole	Security	View
Key Vault Certificate User	Read certificate contents. Only works for key vaults that use the 'Azure role-based access control' permission model.	BuiltInRole	None	View

[Review + assign](#)[Previous](#)[Next](#)[Feedback](#)

Add role assignment

Role: **Members** Conditions Review + assign

Selected role: Key Vault Administrator

Assign access to: ☒ User, group, or service principal
☐ Managed identity

Members: [+ Select members](#)

Name	Object ID	Type
No members selected		

Description

Optional

[Review + assign](#)[Previous](#)[Next](#)

Select members

 nkdenisprojectadf /subscriptions/5bd8f1cd-03ad-48fa-8410-6e58274fc11f/resourcegroups/D...

Selected members:

 nkdenisprojectadf /subscriptions/5bd8f1cd-03ad-48fa-8410-6e58274fc11f/resourceeg...

[Select](#)[Close](#)

Review+assign

Now go to access policies---select

Create an access policy ...

nkdeniskeyvault

Key permissions

Key Management Operations

- ☒ Select all
- ☒ Get
- ☒ List
- ☒ Update
- ☒ Create
- ☒ Import
- ☒ Delete
- ☒ Recover
- ☒ Backup
- ☒ Restore

Cryptographic Operations

- ☐ Select all
- ☐ Decrypt
- ☐ Encrypt

Secret permissions

Secret Management Operations

- ☒ Select all
- ☒ Get
- ☒ List
- ☒ Set
- ☒ Delete
- ☒ Recover
- ☒ Backup
- ☒ Restore

Privileged Secret Operations

- ☐ Select all
- ☐ Purge

Certificate permissions

Certificate Management Operations

- ☒ Select all
- ☒ Get
- ☒ List
- ☒ Update
- ☒ Create
- ☒ Import
- ☒ Delete
- ☒ Recover
- ☒ Backup
- ☒ Restore
- ☒ Manage Contacts
- ☒ Manage Certificate Authorities
- ☒ Get Certificate Authorities
- ☒ List Certificate Authorities
- ☒ Set Certificate Authorities

Previous

Next

Select the adf

Create an access policy ...

nkdeniskeyvault

- 1 Permissions 2 Principal 3 Application (optional) 4 Review + create

Only 1 principal can be assigned per access policy.
Use the new embedded experience to select a principal. The previous popup experience can be accessed here. [Select a principal](#)



nkdenisprojectadf

c4123c15-2e6b-4939-bdd2-9b483babe69f

Selected item



nkdenisprojectadf

c4123c15-2e6b-4939-bdd2-9b483babe69f

Previous

Next

Create an access policy ...

nkdeniskeyvault

✓ Permissions
✓ Principal
✓ Application (optional)
4 Review + create

Key Permissions

Key Management Operations	All selected
Cryptographic Operations	None selected
Privileged Key Operations	None selected
Rotation Policy Operations	All selected

Secret Permissions

Secret Management Operations	All selected
Privileged Secret Operations	None selected

Certificate Permissions

Certificate Management Operations	All selected
Privileged Certificate Operations	None selected

Principal

Principal name	nkdenisprojectadf
Object ID	3bfd7943-7312-4af8-a787-cf08706823f2

Previous

Create

The screenshot shows the Azure Key Vault 'Access policies' page. The left sidebar contains navigation links: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Access policies (selected), Resource visualizer, Events, Objects, Keys, Secrets, Certificates, Settings, Monitoring, Automation, and Help. The main content area has a search bar and filters for 'Permissions: All' and 'Type: All'. It displays a table with 2 records:

Name	Email	Key Permissions	Secret Permissions	Certificate Permissions
APPLICATION				
nkdenisprojectadf		Get, List, Update, Create, Import, Delete, Rec...	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Rec...
USER				
Nikitha Kandula	nikithareddykandula_outlook.com#EXT#@ni...	Get, List, Update, Create, Import, Delete, Rec...	Get, List, Set, Delete, Recover, Backup, Restore	Get, List, Update, Create, Import, Delete, Rec...

Successfully by using the azure key vault created alinked service for blob sa to adf

New linked service

 Azure Blob Storage [Learn more](#)

Name *

Description

Connect via integration runtime * ⓘ
☒ AutoResolveIntegrationRuntime

Authentication type

AKV linked service * ⓘ

Secret name * ⓘ

☐ Edit

Secret version ⓘ

☐ Edit

☒ Connection successful

Data Factory | Validate all | Publish all

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations: Any

Showing 1 - 2 of 2 items

Name	Type	Related
AzureBlob2adf_Is	Azure Blob Storage	0
AzureKeyVault2adf_Is	Azure Key Vault	1

3.create linked service mysql2adf

New linked service

Data store

Compute

Jira	MariaDB	Microsoft 365 (Office 365)
 Microsoft Access	 Microsoft Fabric Lakehouse	 Microsoft Fabric Warehouse
 MongoDB	 MongoDB Atlas	 MySQL
		

Continue

Cancel

New linked service

 MySQL [Learn more](#)

Connect via integration runtime * ⓘ

✓ SHintegrationRuntime1

Server name *

localhost

Port

3306

Database name *

denis_db

User name *

root

Password Azure Key Vault

AKV linked service * ⓘ

AzureKeyVault2adf_Is

Secret name *

mysqlpass

☐ Edit

✓ Connection successful

Create Back Test connection Cancel

Data Factory Validate all Publish all

General

Connector upgrade adv...

Factory settings

Connections

Linked services

Integration runtimes

Microsoft Purview

ADF in Microsoft Fabric

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Security

Credentials

Customer managed key

Outbound rules

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

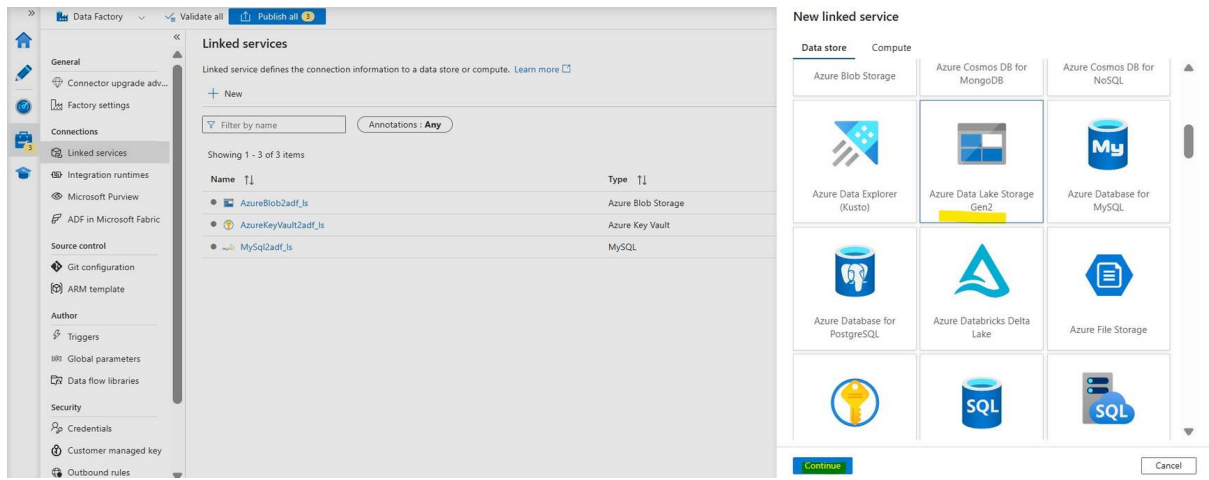
+ New

Filter by name Annotations: Any

Showing 1 - 3 of 3 items

Name ↑↓	Type ↑↓	Related ↑↓
AzureBlob2adf_Is	Azure Blob Storage	0
AzureKeyVault2adf_Is	Azure Key Vault	2
MySQL2adf_Is	MySQL	0

4:create linkedservice adls2adf:



New linked service

Azure Data Lake Storage Gen2 [Learn more](#)

Name *

ADLS2adf_ls

Description

Connect via integration runtime * ¹

AutoResolveIntegrationRuntime

Authentication type

Account key

Account selection method ¹

☒ From Azure subscription ☐ Enter manually

Azure subscription ¹

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f)

Storage account name *

denisadlssa

Test connection ¹

☒ To linked service ☐ To file path

Annotations

Connection successful

Create

Back

Test connection

Cancel

» Data Factory Validate all Publish all Migrate to...

General
Connector upgrade adv...
Factory settings

Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric

Source control
Git configuration
ARM template

Author

Linked services

Linked service defines the connection information to a data store or compute. Learn more

+ New

Filter by name Annotations: Any

Showing 1 - 4 of 4 items

Name	Type	Related	Annotations
ADLS2adf_ls	Azure Data Lake Storage Gen2	0	
AzureBlob2adf_ls	Azure Blob Storage	0	
AzureKeyVault2adf_ls	Azure Key Vault	2	
MySQL2adf_ls	MySQL	0	

Successfully created
Successfully created ADLS2adf_ls (L)

5.create azure sql db 2 adf linked service

General
Connector upgrade adv...
Factory settings

Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric

Source control
Git configuration
ARM template

Author

Triggers
Global parameters
Data flow libraries

Security
Credentials
Customer managed key
Outbound rules

Linked services

Linked service defines the connection information to a data store or compute. Learn more

+ New

Filter by name Annotations: Any

Showing 1 - 4 of 4 items

Name	Type
ADLS2adf_ls	Azure Data Lake Storage Gen2
AzureBlob2adf_ls	Azure Blob Storage
AzureKeyVault2adf_ls	Azure Key Vault
MySQL2adf_ls	MySQL

Data store Compute

Azure Database for PostgreSQL Azure Databricks Delta Lake Azure File Storage

Azure Key Vault Azure SQL Database Azure SQL Database Managed Instance

Azure Synapse Analytics Azure Table Storage Cassandra

DB2

Continue Cancel

New linked service

 Azure SQL Database [Learn more](#)

Version

☒ 2.0 (Recommended) ☐ 1.0

[Import from connection string](#)

Account selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f) ▼

Server name *

denisazuresqlserver ▼ 

Database name *

denisazuresqlldb ▼ 

Authentication type *

SQL authentication ▼

User name *

nkadmin

Password

Azure Key Vault

Password *

Always encrypted ⓘ ☐

Create

Back

 Test connection

Cancel

Go to azure key vault service---secret---generate/import---give password



Create a secret ...

Upload options

Manual

Name * ⓘ

Azuresqldbpass

Secret value * ⓘ

Content type (optional)

Set activation date ⓘ

☐

Set expiration date ⓘ

☐

Enabled

Yes

No

Tags

0 tags

Create

Cancel



nkdeniskeyvault | Secrets ☆ ...

Key vault

Search

+ Generate/Import

↻ Refresh

↶ Restore Backup

🔗 Manage deleted secrets

</> View sample code

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Keys

Secrets

Certificates

> Settings

> Monitoring

> Automation

The secret 'Azuresqldbpass' has been successfully created.

Name	Type	Status
Azuresqldbpass		✓ Enable
adlspaskey		✓ Enable
Blobaccesskey		✓ Enable
mysqlpass		✓ Enable

Go to adf

New linked service

 Azure SQL Database [Learn more](#)

☒ From Azure subscription ☐ Enter manually

Azure subscription

NK (5bd8f1cd-03ad-48fa-8410-6e58274fc11f) ▼

Server name *

denisazuresqlserver ▼ 

Database name *

denisazuresqlldb ▼ 

Authentication type *

SQL authentication ▼

User name *

nkadmin

Password **Azure Key Vault**

AKV linked service * ⓘ

AzureKeyVault2adf_js ▼ 

Secret name *

Azuresqldbpass ▼ 

☐ Edit

Secret version ⓘ

 Connection successful

Create

Back

 Test connection

Cancel

Successfully created linkedservice

>> Data Factory Validate all Publish all 4

Home Edit Monitor Connections 4 Integration runtimes Microsoft Purview ADF in Microsoft Fabric Source control Git configuration ARM template Author Triggers Global parameters Data flow libraries Security Credentials

General
Connector upgrade adv...
Factory settings

Connections
Linked services
Integration runtimes
Microsoft Purview
ADF in Microsoft Fabric

Source control
Git configuration
ARM template

Author
Triggers
Global parameters
Data flow libraries

Security
Credentials

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

Filter by name Annotations : Any

Showing 1 - 5 of 5 items

Name ↑↓	Type ↑↓
ADLS2adf_ls	Azure Data L
AzureBlob2adf_ls	Azure Blob S
AzureKeyVault2adf_ls	Azure Key Va
AzureSqlDatabase2adf_ls	Azure SQL D:
MySQL2adf_ls	MySQL

6.Create linkedservice mongodb2adf

New linked service

Data store Compute

mon

< All Azure Database File Generic protocol



Azure Cosmos DB for MongoDB



Dataverse (Common Data Service for Apps)



MongoDB



MongoDB Atlas

Continue

Cancel

New linked service

 MongoDB [Learn more](#)

Name *

MongoDb2adf_Is

Description

Connect via integration runtime * ⓘ

✓ SHintegrationRuntime1

Connection string

Azure Key Vault

AKV linked service * ⓘ

Select...

Secret name * ⓘ

No secret

☐ Edit

Secret version ⓘ

Latest version

☐ Edit

Database name *

Annotations

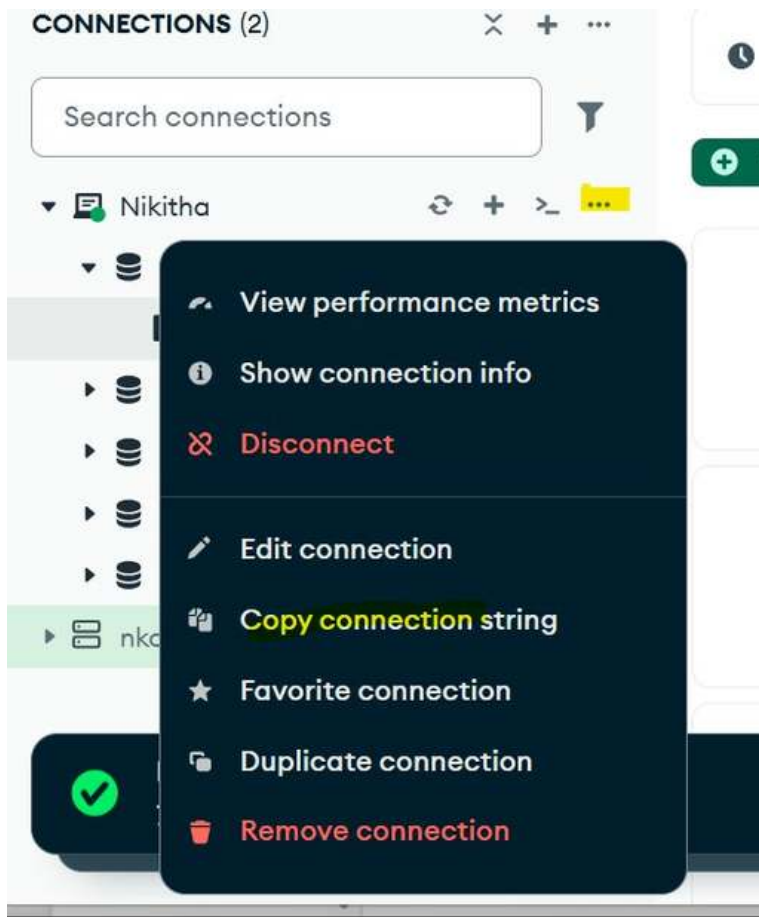
Create

Back

 Test connection

Cancel

Let's copy the connection string from mongodb



Go to azure key vault---secret---generate/import



Create a secret ...

Upload options

Manual

Name * ⓘ

Mongodbconnectionstring

Secret value * ⓘ

.....

Content type (optional)

Set activation date ⓘ

☐

Set expiration date ⓘ

☐

Enabled

Yes

No

Tags

0 tags

Create

Cancel



nkdeniskeyvault | Secrets ☆ ...

Key vault

Search

+ Generate/Import

Refresh

Restore Backup

Manage deleted secrets

View sample code

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Resource visualizer

Events

Objects

Keys

Secrets

Certificates

> Settings

> Monitoring

> Automation

> Help

The secret 'Mongodbconnectionstring' has been successfully created.

Name	Type	Status	Ex
Mongodbconnectionstring		✓ Enabled	
Azuresqlpass		✓ Enabled	
adlspasskey		✓ Enabled	
Blobaccesskey		✓ Enabled	
mysqlpass		✓ Enabled	

Go to adf

New linked service

 MongoDB [Learn more](#)

Description

Connect via integration runtime * ^①

 SHintegrationRuntime1

Connection string

Azure Key Vault

AKV linked service * ^①

AzureKeyVault2adf_ls

Secret name * ^①

Mongodbconnectionstring

☐ Edit

Secret version ^①

6a4e822e598f43e5b952ad707ce7d048 (Current version)

☐ Edit

Database name *

Denismongodb

Annotations

1

 Connection successful

Create

Back

 Test connection

Cancel

Data Factory | Validate all | Publish all

General

Connector upgrade adv...

Factory settings

Connections

Linked services

Integration runtimes

Microsoft Purview

ADF in Microsoft Fabric

Source control

Git configuration

ARM template

Author

Triggers

Global parameters

Data flow libraries

Security

Credentials

Customer managed key

Linked services

Linked service defines the connection information to a data store or compute. [Learn more](#)

+ New

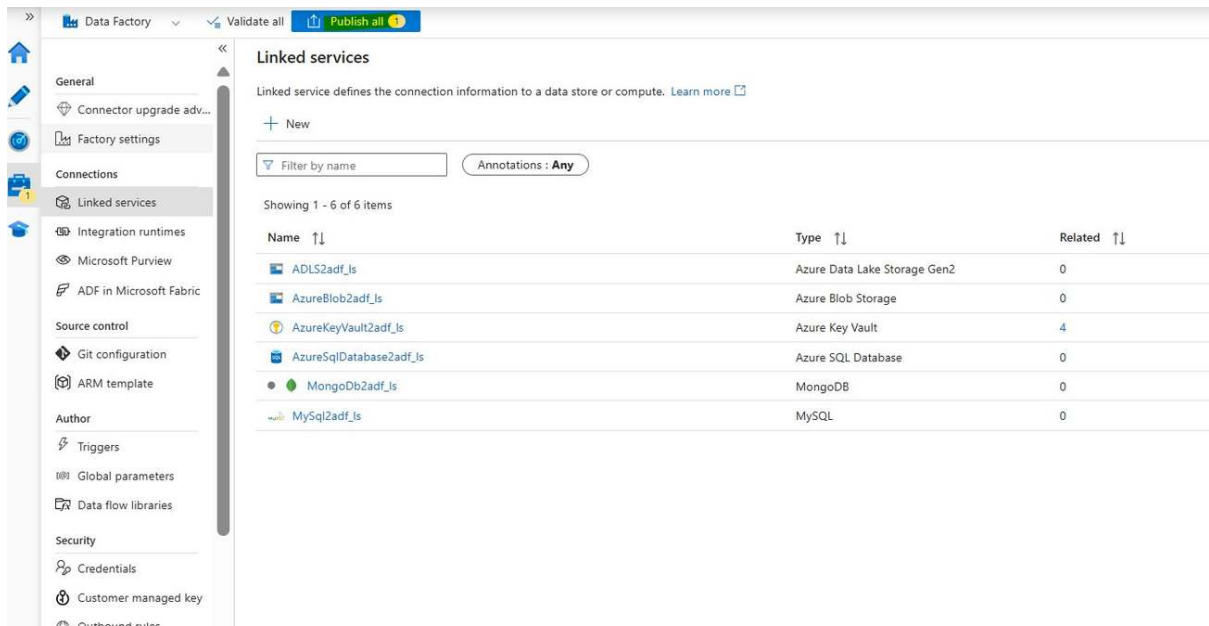
Filter by name

Annotations: Any

Showing 1 - 6 of 6 items

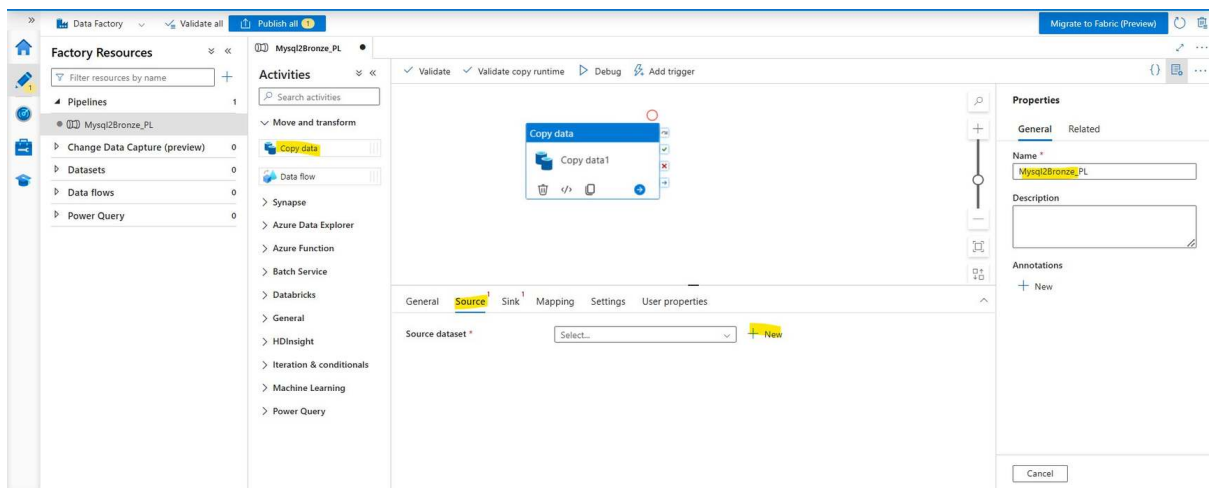
Name ↑↓	Type ↑↓	Related ↑↓
ADLS2adf_ls	Azure Data Lake Storage Gen2	0
AzureBlob2adf_ls	Azure Blob Storage	0
AzureKeyVault2adf_ls	Azure Key Vault	4
AzureSqlDatabase2adf_ls	Azure SQL Database	0
MongoDb2adf_ls	MongoDB	0
MySQL2adf_ls	MySQL	0

Publish all the changes in adf



Let's copy all the raw data to bronze layer using copy activity:

1)create pipeline for mysql2bronze:



New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

< All Azure Database File Generic protocol >

Microsoft Fabric Warehouse


MySQL


OData

MongoDB


Netezza


Oracle

MongoDB Atlas


ODBC


Oracle Cloud Storage (S3 API)

Continue

Cancel

Set properties

Name

Product_Raw

Linked service *

MySQL2adf_ls

Connect via integration runtime * ⓘ

✓ SHintegrationRuntime1

Table name

product (product)

☐ Enter manually

> Advanced

OK

Back

Cancel

Set properties

Name: Product_Raw

Linked service *: MySQL2adf_ls

Connect via integration runtime * ⓘ: ✓ SHintegrationRuntime1

Table name: product (product)

☐ Enter manually

> Advanced

Factory Resources

- Pipelines (1)
 - MySQL2Bronze_PL
- Change Data Capture (preview) (0)
- Datasets (1)
- Data flows (0)
- Power Query (0)

Activities

- Move and transform
 - Copy data (1)
 - Copy data1
 - Data flow (0)
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset *: Select...

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All Azure Database File Generic protocol

- Azure AI Search
- Azure Blob Storage
- Azure Cosmos DB for MongoDB
- Azure Cosmos DB for NoSQL
- Azure Data Explorer (Kusto)
- Azure Data Lake Storage Gen2

Continue Cancel

Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Iceberg	 JSON	 ORC
 Parquet		

Continue

Back

Cancel

Set properties

Name

Product_Bronze

Linked service *

ADLS2adf_ls

File path

denis

/ Medallian/Bronze

/ File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

The screenshot displays the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane shows a hierarchy with 'Pipelines' expanded, containing 'Mysql2Bronze_PL'. The 'Activities' pane on the right shows a 'Copy data' activity. The 'Sink' tab is selected, showing the 'Sink dataset' as 'Product_Bronze'. Other settings include 'Copy behavior' set to 'Select...', 'Max concurrent connections' set to 1, 'Block size (MB)' set to 1, 'Metadata' set to '+ New', 'Quote all text' checked, 'File extension' set to '.csv', and 'Max rows per file' set to 1. The 'Properties' pane on the far right shows the 'General' tab with 'Name' set to 'Mysql2Bronze_PL' and 'Description' empty. The 'Annotations' section shows '+ New'. The top bar includes 'Data Factory', 'Validate all', 'Publish all', and 'Migrate to Fabric (Preview)'.

Publish all changes

Publish all

You are about to publish all pending changes to the live environment. [Learn more](#)

Pending changes (3)

NAME	CHANGE	EXISTING
Pipelines		
Mysql2Bronze_PL	(New)	-
Datasets		
Product_Raw	(New)	-
Product_Bronze	(New)	-

Copy data

Copy data1

General Source Sink Mapping Settings User properties

Sink dataset * Product_Bronze

Copy behavior Select...

Max concurrent connections

Block size (MB)

Metadata + New

Quote all text ☒

File extension .CSV

Max rows per file

Publish **Cancel**

2)create pipeline for mongodb2bronze:

Factory Resources

Filter resources by name

Pipelines 1

Mysql2Bronze_PL

Change Data Capture (preview)

Datasets

Data flows 0

Power Query 0

Activities

Search activities

New pipeline

Pipeline from template

New folder

Synapse

Publish all

Migrate to Fabric (Preview)

Factory Resources

Filter resources by name

Pipelines 2

Mysql2Bronze_PL

Mongodb2Bronze_PL

Change Data Capture (preview)

Datasets

Data flows 0

Power Query 0

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Copy data

Copy data1

General Source Sink Mapping Settings User properties

Source dataset * Select... New

Properties

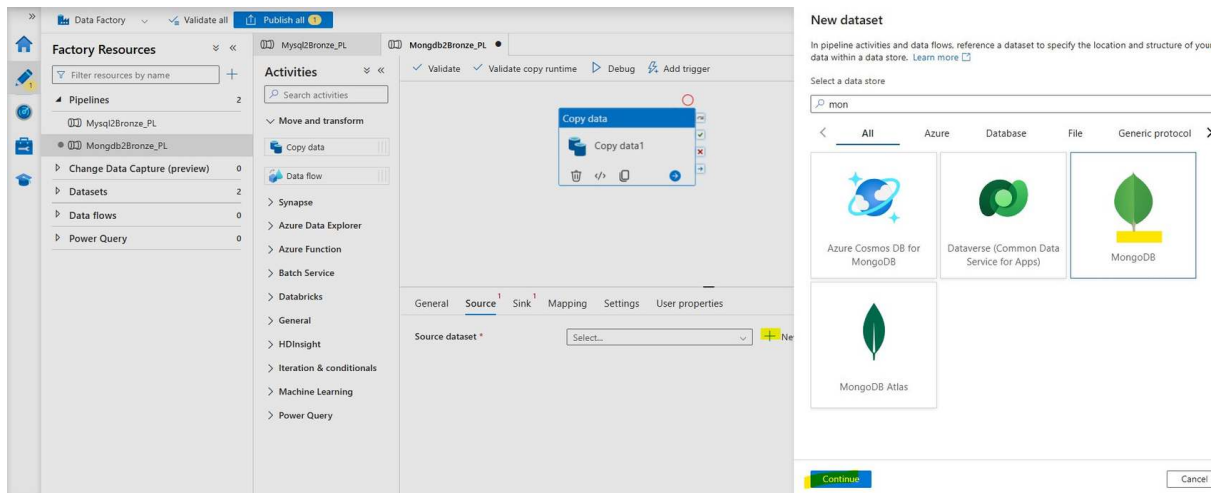
General Related

Name * Mongodb2Bronze_PL

Description

Annotations + New

Cancel



Set properties

Name

Geography_Raw

Linked service *

MongoDb2adf_Is

Connect via integration runtime * ⓘ

SHintegrationRuntime1

Collection name

Geography

☐ Enter manually

> Advanced

OK

Back

Cancel

» Data Factory Validate all Publish all Migrate to Fabric (Preview)

Factory Resources

- Pipelines (2)
 - Mysql2Bronze_PL
 - Mongdb2Bronze_PL
- Change Data Capture (preview) (0)
- Datasets (3)
 - Data flows (0)
 - Power Query (0)

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
- General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

Copy data

Copy data1

General Source Sink Mapping Settings User properties

Source dataset * GeographyRaw

Filter

Cursor methods

Properties

General Related

Name * Mongdb2Bronze_PL

Description

Annotations

+ New

Cancel

» Data Factory Validate all Publish all

Factory Resources

- Pipelines (2)
 - Mysql2Bronze_PL
 - Mongdb2Bronze_PL
- Change Data Capture (preview) (0)
- Datasets (3)
 - Data flows (0)
 - Power Query (0)

Activities

- Move and transform
 - Copy data
 - Data flow
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
- General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

Copy data

Copy data1

General Source Sink Mapping Settings User properties

Sink dataset * Select...

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All Azure Database File Generic protocol

- Azure AI Search
- Azure Blob Storage
- Azure Cosmos DB for MongoDB
- Azure Cosmos DB for NoSQL
- Azure Data Explorer (Kusto)
- Azure Data Lake Storage Gen2
- MySQL
- Microsoft SQL Server
- Amazon S3

Continue Cancel

Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Iceberg	 JSON	 ORC
 Parquet		

Continue

Back

Cancel

Set properties

Name

Geography_Bronze

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Bronze

/ File name

Import schema

☒ From connection/store ☐ From sample file ☐ None

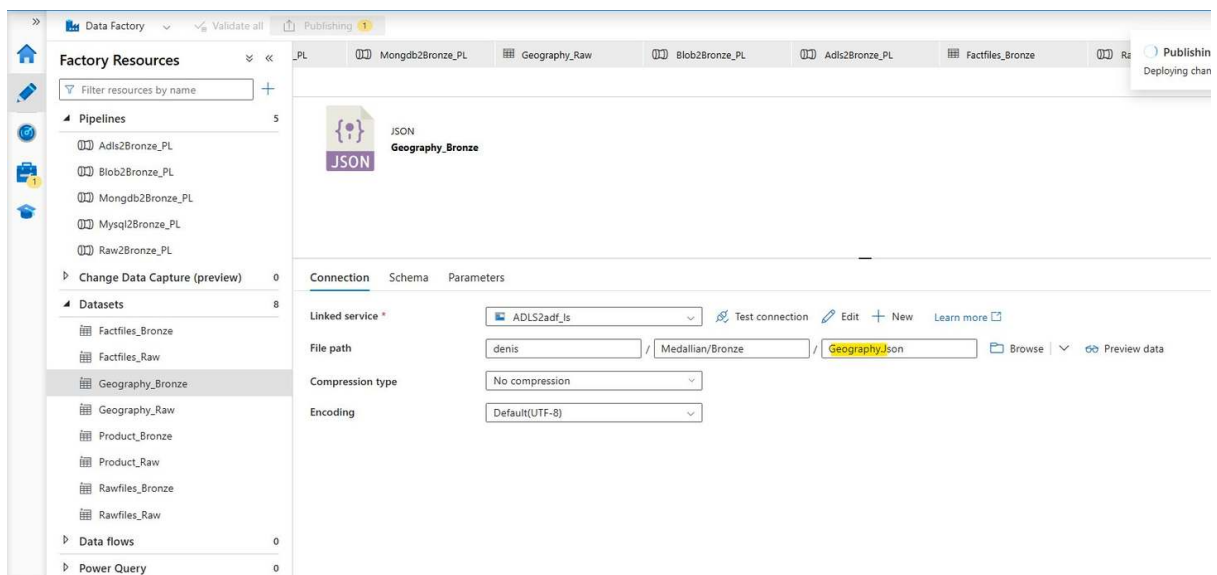
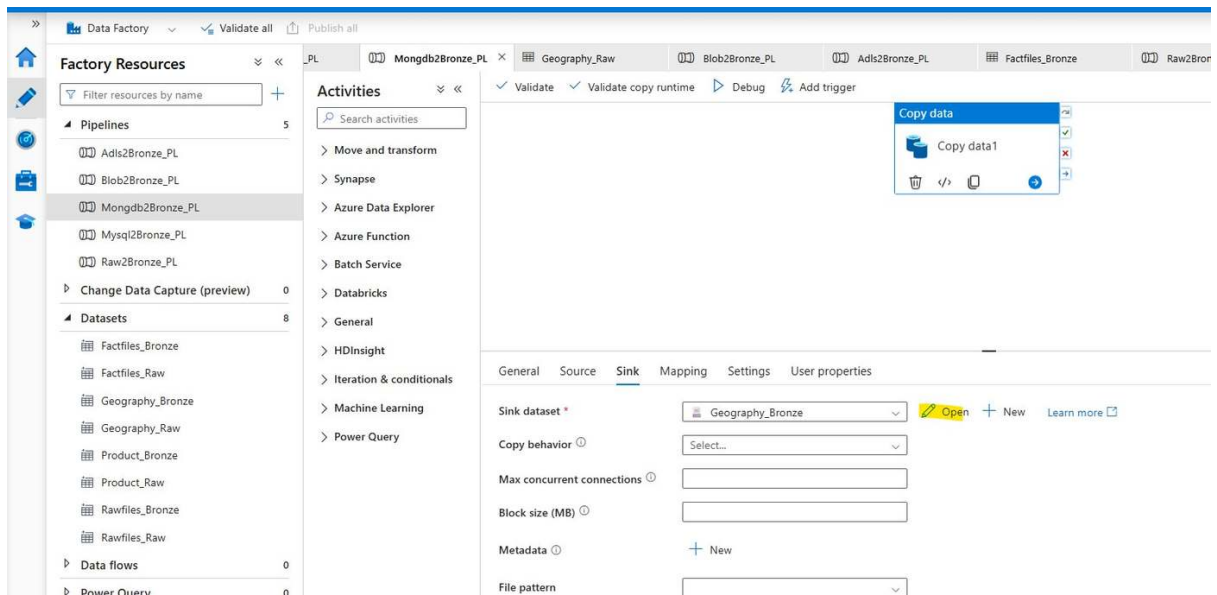
> Advanced

OK

Back

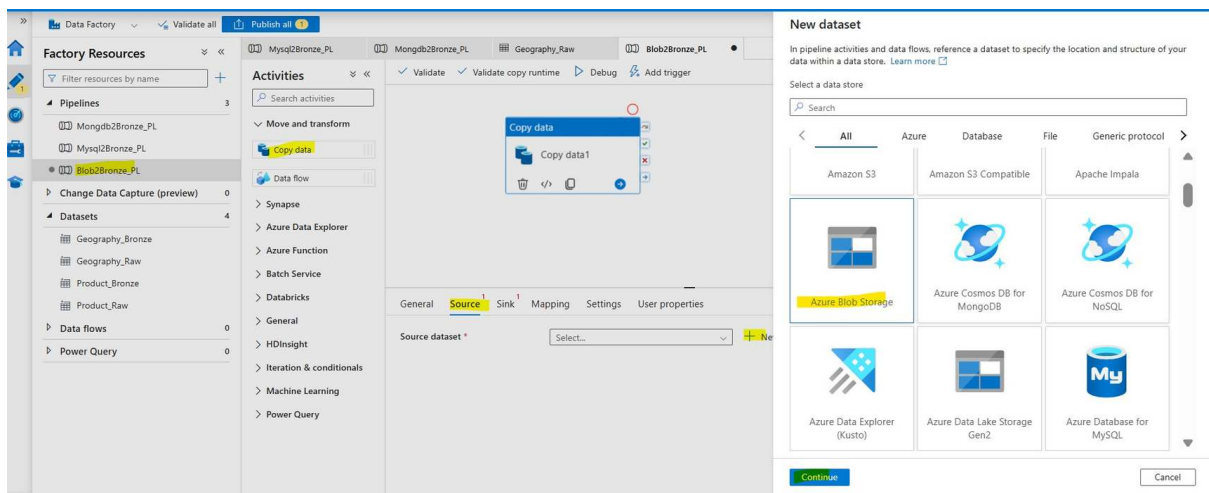
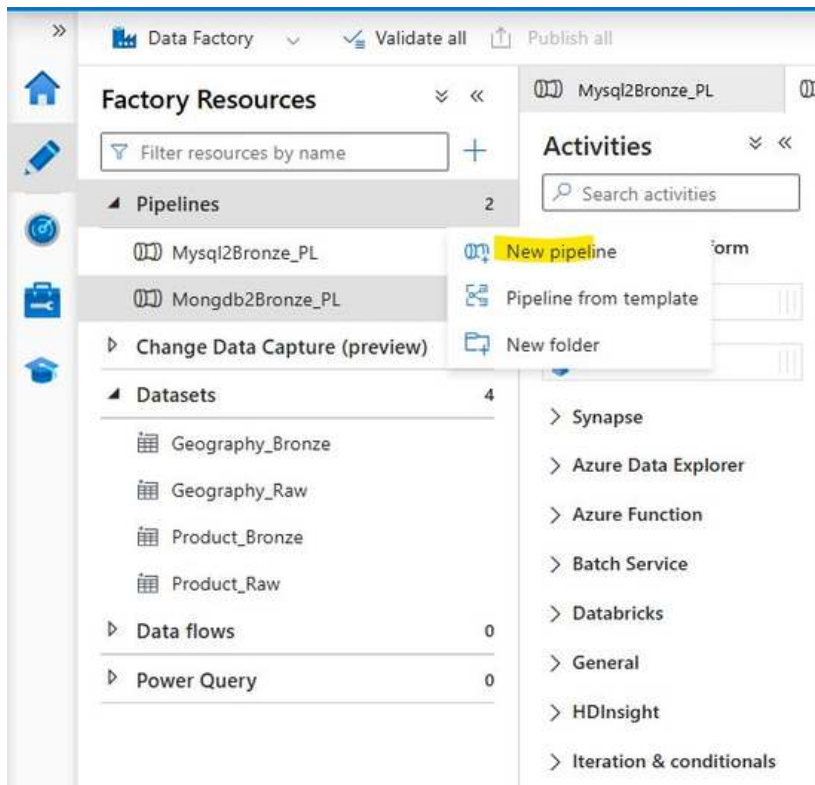
Cancel

The screenshot shows the 'Set properties' dialog for a 'Copy data' activity in Azure Data Factory. The dialog is titled 'Copy data' and has a 'Copy data1' sub-title. The 'Sink dataset' is set to 'Geography_Bronze'. The 'Copy behavior' is set to 'Select...'. The 'Max concurrent connections' is set to 1. The 'Block size (MB)' is set to 1. The 'Metadata' is set to '+ New'. The 'File pattern' is set to '/'. The 'Properties' panel on the right shows the 'General' tab with the 'Name' set to 'Mongodb2Bronze_PL' and a 'Description' field. The 'Factory Resources' panel on the left shows the 'Mongodb2Bronze_PL' dataset selected. The 'Activities' panel shows the 'Copy data' activity selected. The 'Data Factory' tab is active, and the 'Publish all' button is visible. The 'Migrate to Fabric (Preview)' button is also visible in the top right corner.



Publish all changes in adf

3)create pipeline for Blobsa2bronze:



Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 JSON	 ORC
 Parquet	 XML	

Continue

Back

Cancel

Here in blob storage we have multiple files so we will create a single raw data which will have all the files

Browse

Select a file or folder.

[Root folder](#) > denisdimraw

-  Categories.xlsx
-  SalesRep.csv
-  SubCategories.csv

Showing 1 - 3 of 3 items

Set properties

Name

Raw2Bronze

Linked service *

AzureBlob2adf_ls

File path

denisdimraw

denisdimraw

File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced



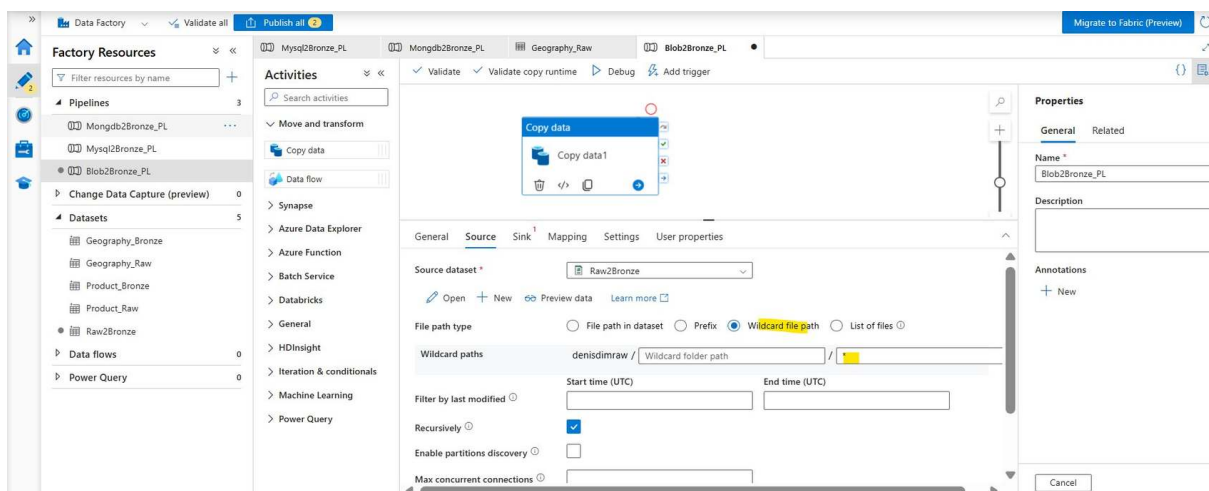
Back

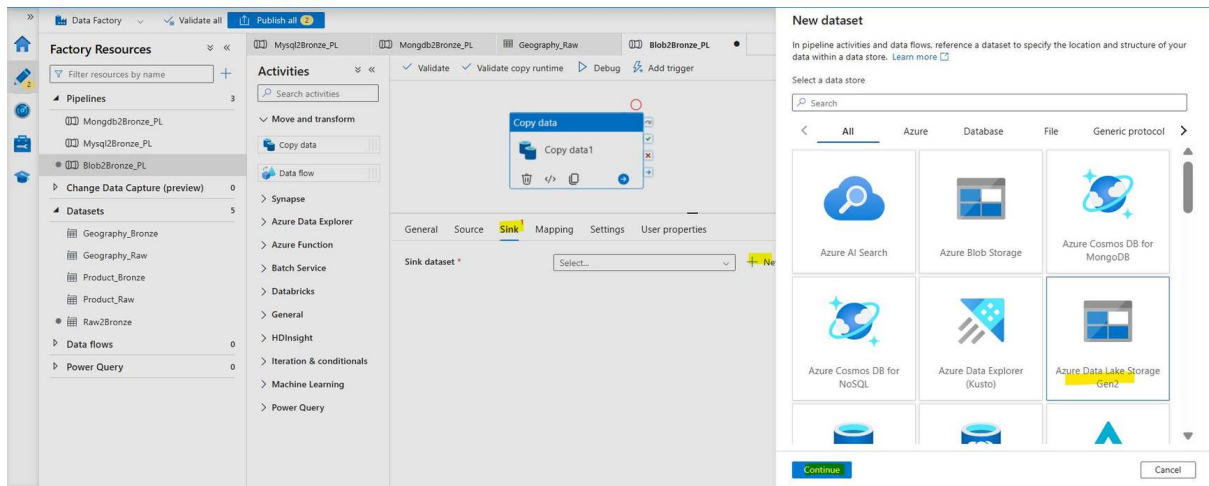
Cancel

Note:

Hence we have multiple files---in sink---we need to select wild card---* means will take all the files---if needed only csv files then *.csv

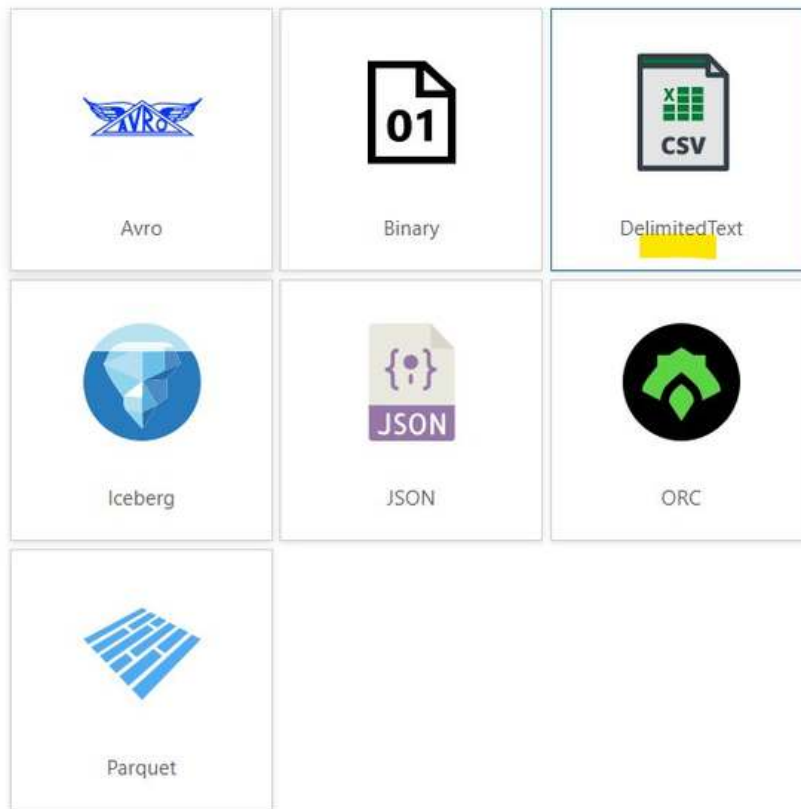
This will reduce creating multiple input and output data sets





Select format

Choose the format type of your data



Set properties

Name
Rawfiles_Bronze

Linked service *
ADLS2adf_Is

File path
denis / Medallian/Bronze / File name

First row as header ☒

Import schema
☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK Back Cancel

15:15

Publish all changes in adf

4)create pipeline for adlssa2bronze:

Data Factory | Validate all | Publish all

Factory Resources

Filter resources by name

Pipelines 3

- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Blob2Bronze_PL

Datasets 6

Geography Bronze

Activities

Search activities

- New pipeline
- Pipeline from template
- New folder

Synapse

Azure Data Explorer

» Data Factory Validate all Publish all

Factory Resources

Filter resources by name

Pipelines

- pipeline1
- Change Data Capture (preview)

Datasets

- Geography_Bronze
- Geography_Raw
- Product_Bronze
- Product_Raw
- Rawfiles_Bronze
- Rawfiles_Raw
- Data flows
- Power Query

Activities

Search activities

Move and transform

- Copy data
- Data flow

Synapse

- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General
- HDInsight
- Iteration & conditionals
- Machine Learning
- Power Query

Copy data

Copy data1

General Source Sink Mapping Settings User properties

Source dataset *

Select...

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All Azure Database File Generic protocol

- Azure Blob Storage
- Azure Cosmos DB for MongoDB
- Azure Cosmos DB for NoSQL
- Azure Data Explorer (Kusto)
- Azure Data Lake Storage Gen2
- Azure Database for MySQL

Continue Cancel

Select format

Choose the format type of your data

Avro

Binary

DelimitedText

Excel

JSON

ORC

Parquet

XML

Continue

Back

Cancel

Set properties

Name

Factfiles_Raw

Linked service *

ADLS2adf_ls

File path

denisfactraw

/ sales

File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced



Back

Cancel

The screenshot shows the Azure Data Factory (ADF) interface. On the left, the 'Factory Resources' pane lists various pipelines and datasets. The 'Activities' pane shows a 'Copy data' activity selected. The main workspace displays the 'Copy data' activity configuration. The 'Source dataset' is set to 'Factfiles_Raw'. The 'File path type' is set to 'Wildcard file path'. The 'Wildcard paths' are configured as 'denisfactraw / Wildcard folder path'. The 'Filter by last modified' checkbox is checked. The 'Enable partitions discovery' checkbox is unchecked. The 'Properties' pane on the right shows the 'General' tab with the 'Name' set to 'Adls2Bronze_PL'.

»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines

4

Blob2Bronze_PL

Mongodb2Bronze_PL

MySQL2Bronze_PL

Adis2Bronze_PL

Change Data Capture (preview)

0

Datasets

7

Factfiles_Raw

Geography_Bronze

Geography_Raw

Product_Bronze

Product_Raw

Rawfiles_Bronze

Rawfiles_Raw

Data flows

0

Power Query

0

Activities

Search activities

Move and transform

Copy data

Data flow

Synapse

Azure Data Explorer

Azure Function

Batch Service

Databricks

General

HDInsight

Iteration & conditionals

Machine Learning

Power Query

Copy data

Copy data1

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

Select...

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of y data within a data store. [Learn more](#)

Select a data store

Search

All

Azure

Database

File

Generic protocol

Azure AI Search

Azure Blob Storage

Azure Cosmos DB for MongoDB

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Continue

Cancel

Set properties

Name

Factfiles_Bronze

Linked service *

ADLS2adf_ls

File path

denis

/

Medallian/Bronze

/

File name

First row as header

☒

Import schema

☒ From connection/store

☐ From sample file

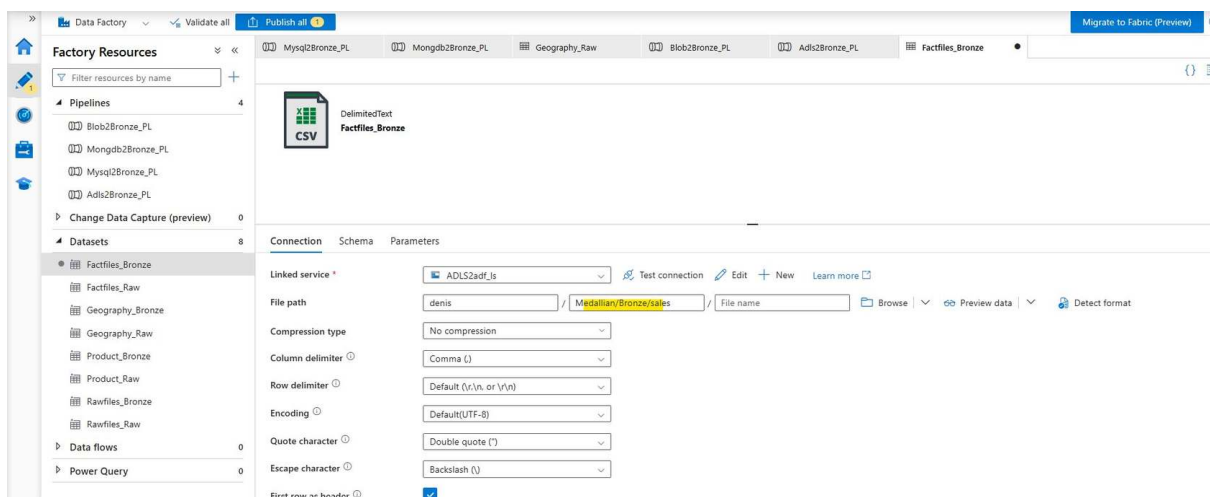
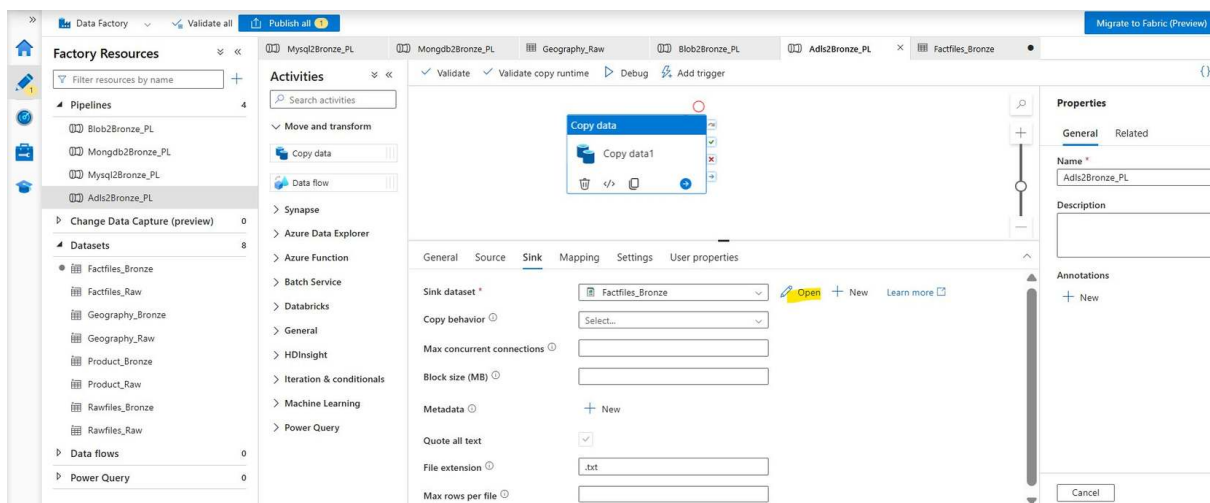
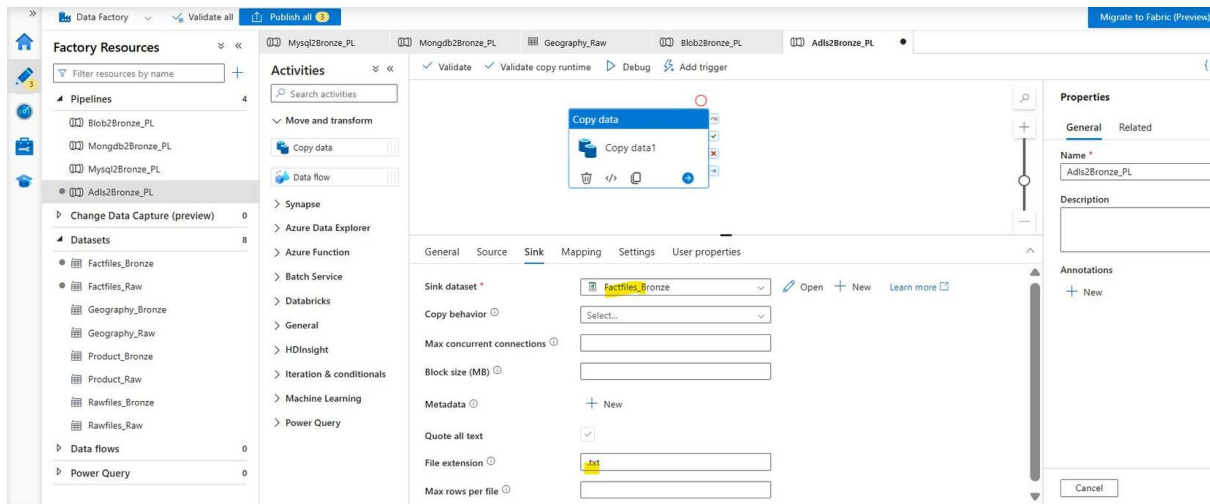
☐ None

> Advanced

OK

Back

Cancel



Publish all the changes in adf

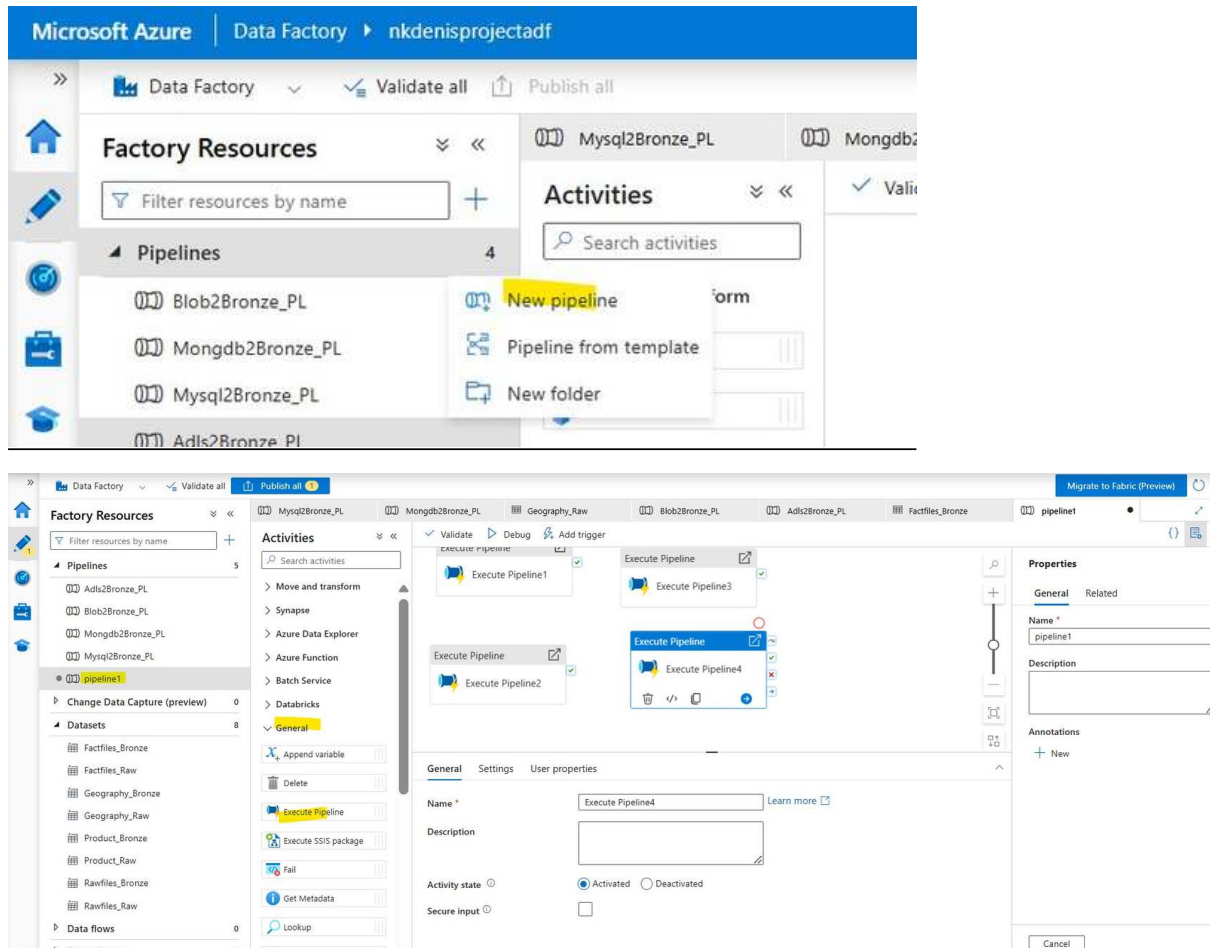
Note:

******Hence 4 pipelines are independent and they are not depending on each we can execute the 4 pipelines at once**

******In case if any dependent pipelines we need to connect end to end on completion**

MAIN PIPELINE:

Let's create a new pipeline:



Factory Resources

Filter resources by name

Pipelines

- Adls2Bronze_PL
- Blob2Bronze_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview)

Datasets

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Product_Bronze
- Product_Raw
- Rawfiles_Bronze
- Rawfiles_Raw

Data flows

Power Query

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General

Append variable

Delete

Execute Pipeline

Execute SSIS package

Fail

Get Metadata

Lookup

Stored procedure

Validate

Debug

Add trigger

Execute Pipeline1

Execute Pipeline3

Execute Pipeline2

Execute Pipeline4

General

Settings

User properties

Invoked pipeline *

Mysql2Bronze_PL

Open

New

Wait on completion

Factory Resources

Filter resources by name

Pipelines

- Adls2Bronze_PL
- Blob2Bronze_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview)

Datasets

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Product_Bronze
- Product_Raw
- Rawfiles_Bronze
- Rawfiles_Raw

Data flows

Power Query

Activities

Search activities

- Move and transform
- Synapse
- Azure Data Explorer
- Azure Function
- Batch Service
- Databricks
- General

Append variable

Delete

Execute Pipeline

Execute SSIS package

Fail

Get Metadata

Lookup

Stored procedure

Validate

Debug

Add trigger

Execute Pipeline1

Execute Pipeline3

Execute Pipeline2

Execute Pipeline4

General

Settings

User properties

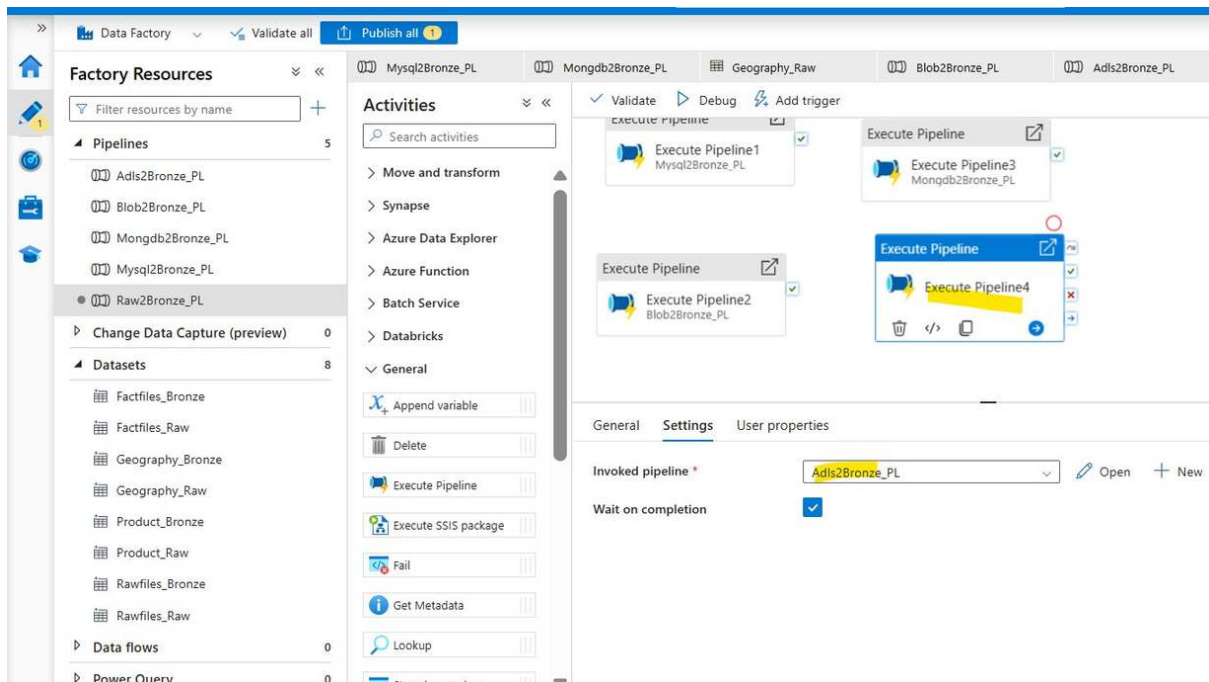
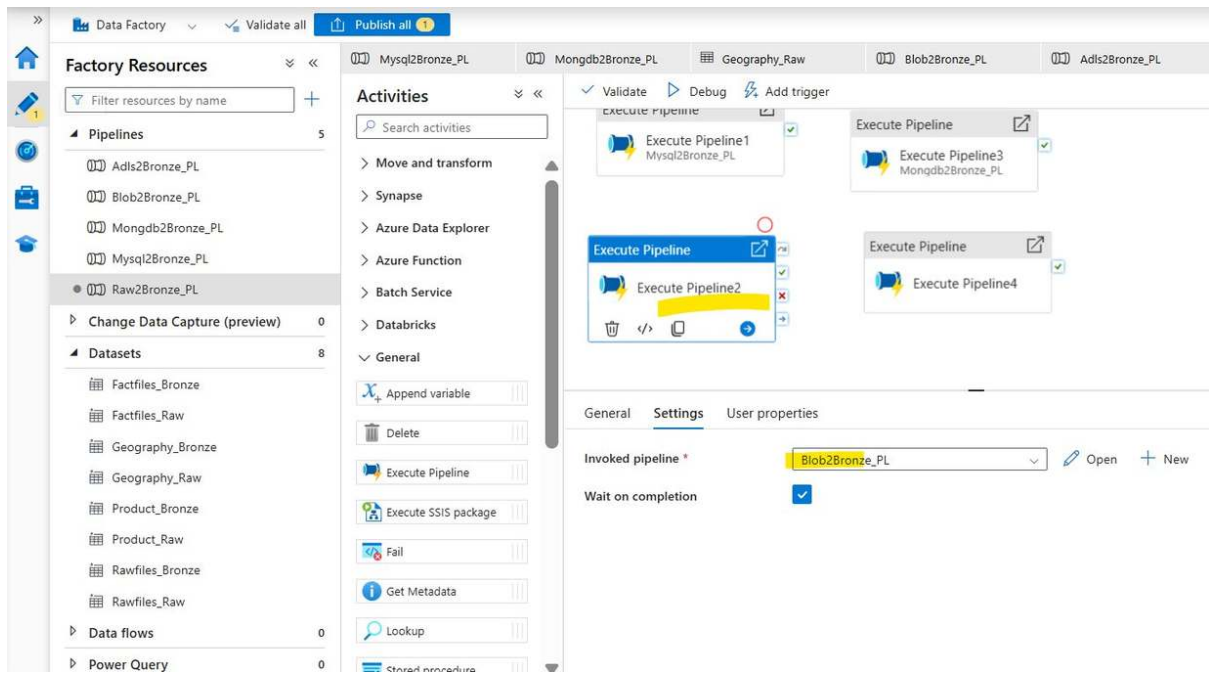
Invoked pipeline *

Mongodb2Bronze_PL

Open

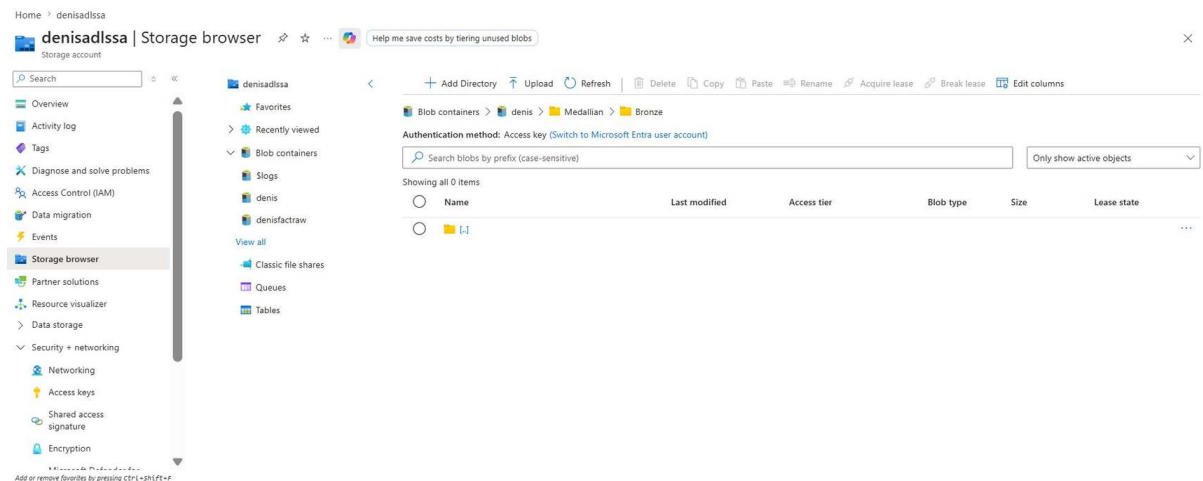
New

Wait on completion

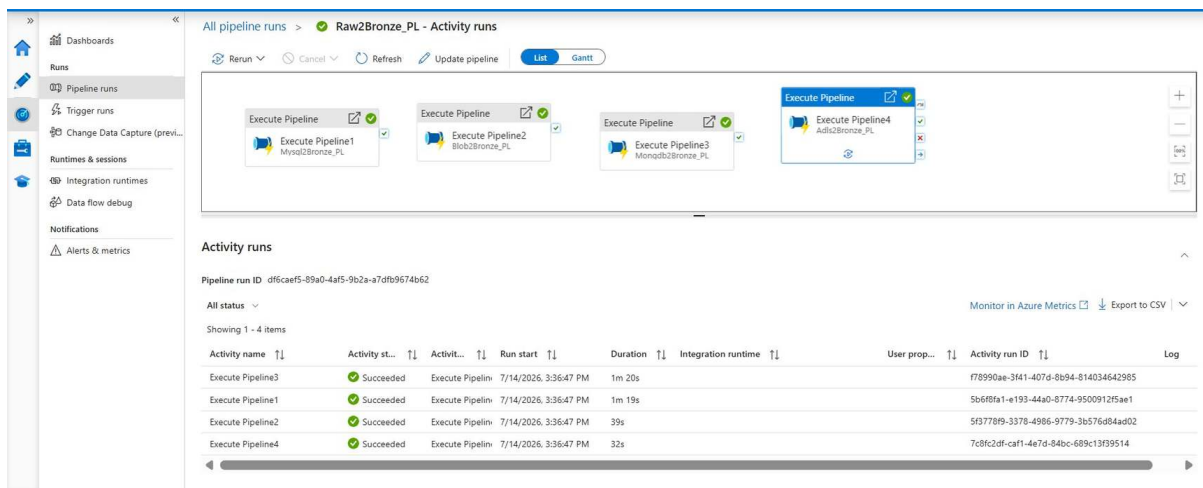


Publish all the changes

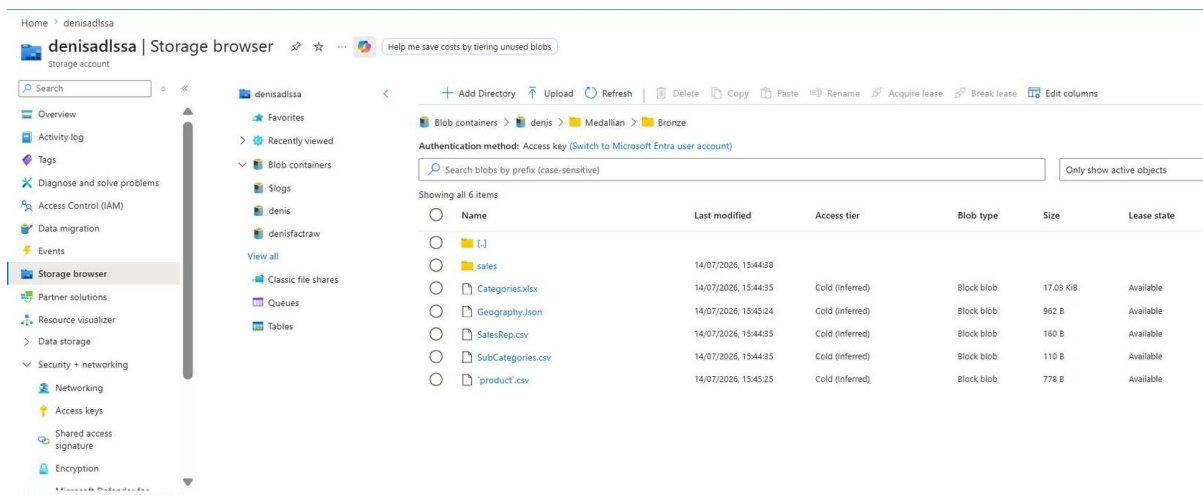
Before running the pipeline We don't have anything in medallion bronze

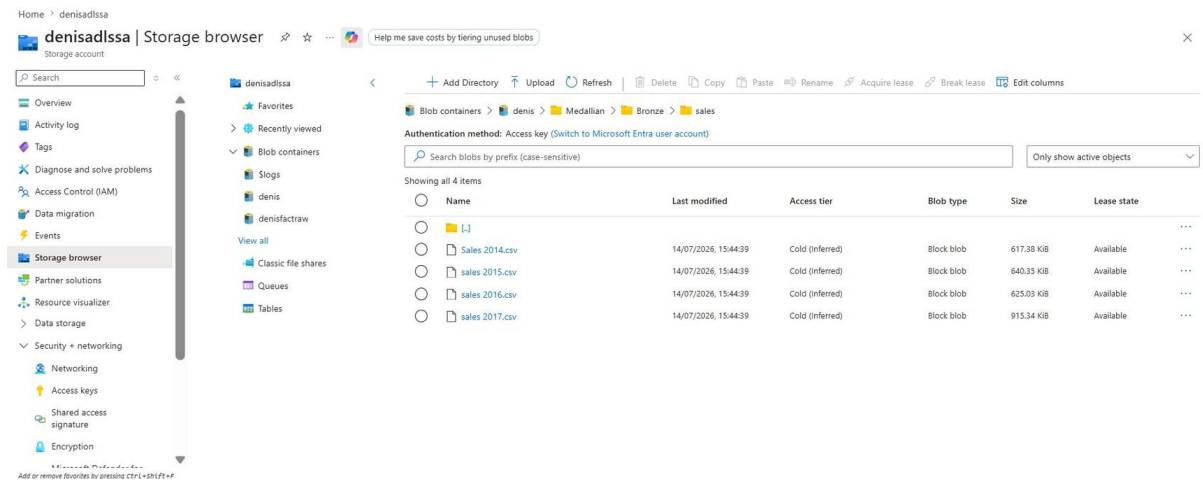


Add trigger---trigger now in adf



Successfully copied the raw data to the medallion---bronze layer





Note:

Now we have the all the raw data into bronze layer---let's transform the data using silver layer.

TRANSFORMATIONS USING DATAFLOW/PQ(IN SILVER LAYER):

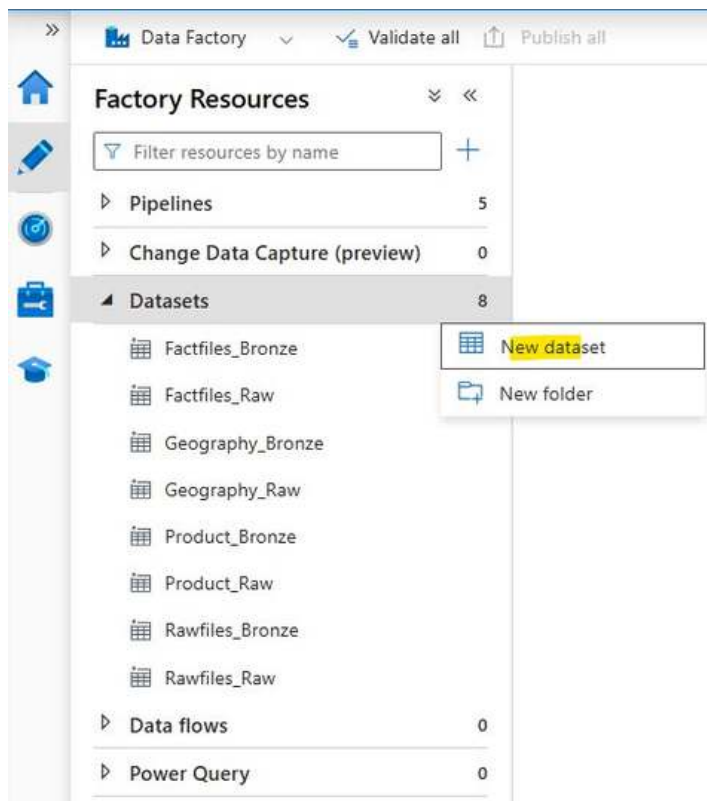
1)Salesrep dataflow:

Note:

We have copied all the 3 files together from blob 2 bronze that we have as rawfiles_bronze

We need to create a datasets for all the 3 subcategory,category,salesrep

Create dataset:



Hence we have the bronze data in adls we need to select adls

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

< All Azure Database File Generic protocol >

Amazon S3

Azure AI Search

Azure Cosmos DB for NoSQL

Amazon S3 Compatible

Azure Blob Storage

Azure Data Explorer (Kusto)

Apache Impala

Azure Cosmos DB for MongoDB

Azure Data Lake Storage Gen2

Continue

Cancel

Select format

Select format

Choose the format type of your data



Avro



Binary



DelimitedText



Excel



Iceberg



JSON



ORC



Parquet



XML

Continue

Back

Cancel

Browse

Select a file or folder.

[Root folder](#) > [denis](#) > [Medallian](#) > **Bronze**

-  [sales](#)
-  Categories.xlsx
-  Geography.Json
-  **SalesRep.csv**
-  SubCategories.csv
-  `product`.csv

Showing 1 - 6 of 6 items



Cancel

Set properties

Name

SalesRep_Bronze

Linked service *

ADLS2adf_ls

File path

denis

/ Medallian/Bronze

/ SalesRep.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

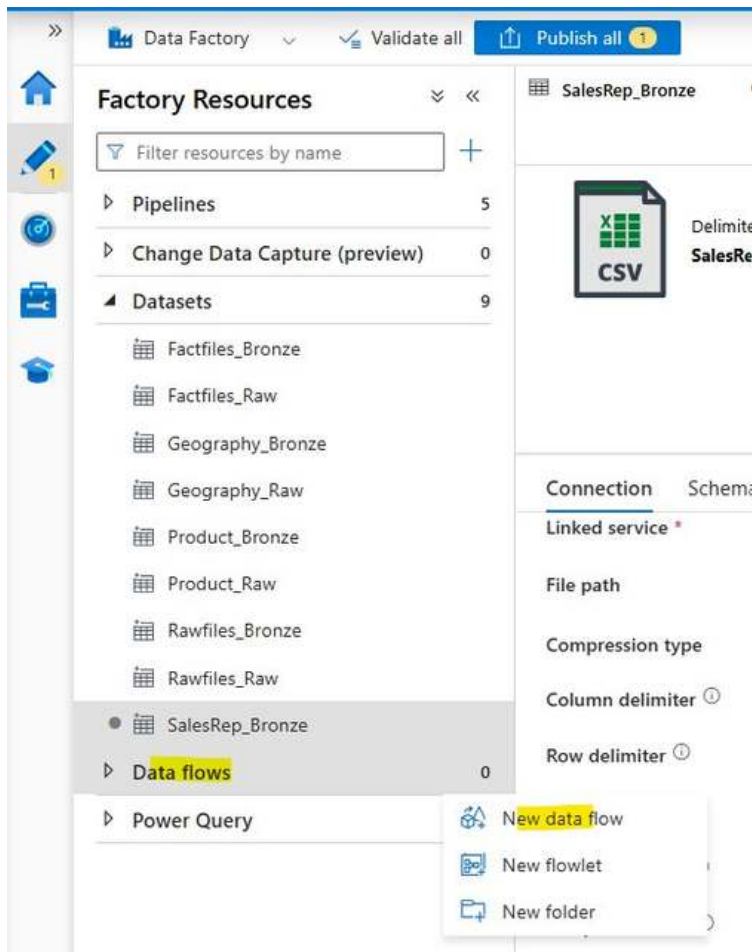
OK

Back

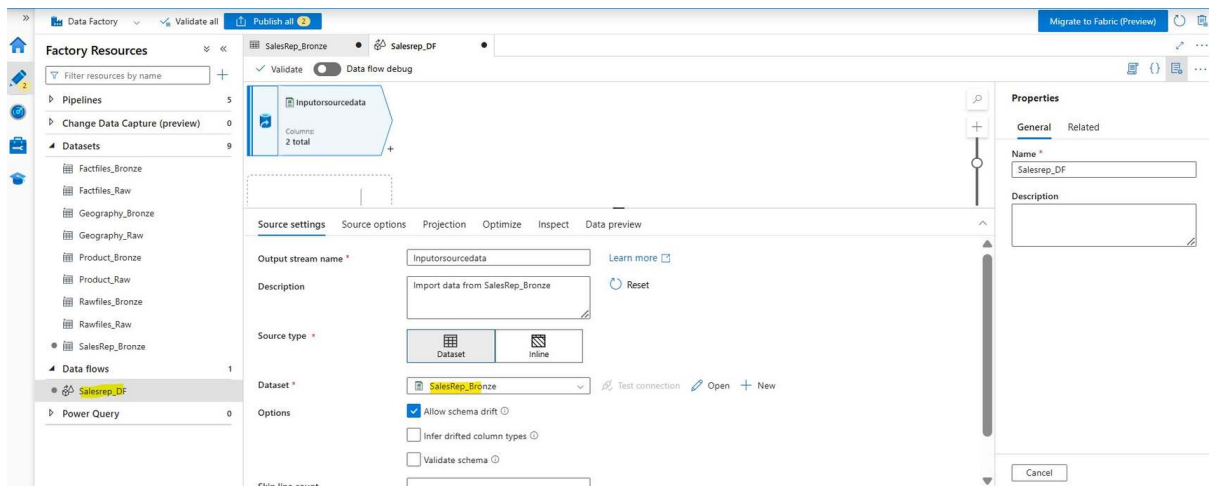
Cancel

The screenshot shows the 'Set properties' dialog for a new dataset named 'SalesRep_Bronze' in Azure Data Factory. The dialog is titled 'Set properties' and has a 'Publish all' button in the top right. The 'Name' field is 'SalesRep_Bronze'. The 'Linked service' is 'ADLS2adf_ls'. The 'File path' is 'denis / Medallian/Bronze / SalesRep.csv'. The 'First row as header' checkbox is checked. The 'Import schema' section shows 'From connection/store' selected. The 'Factory Resources' pane on the left shows the 'SalesRep_Bronze' dataset selected under 'Datasets'. The 'Properties' pane on the right shows the 'General' tab with the 'Name' field set to 'SalesRep_Bronze' and a 'Description' field. The 'Annotations' section shows a '+ New' button. The 'Connection' tab is selected in the 'Properties' pane, showing the 'Linked service' as 'ADLS2adf_ls' and the 'File path' as 'denis / Medallian/Bronze / SalesRep.csv'. The 'Parameters' tab is also visible, showing various settings like 'Compression type', 'Column delimiter', 'Row delimiter', 'Encoding', 'Quote character', 'Escape character', and 'First row as header'.

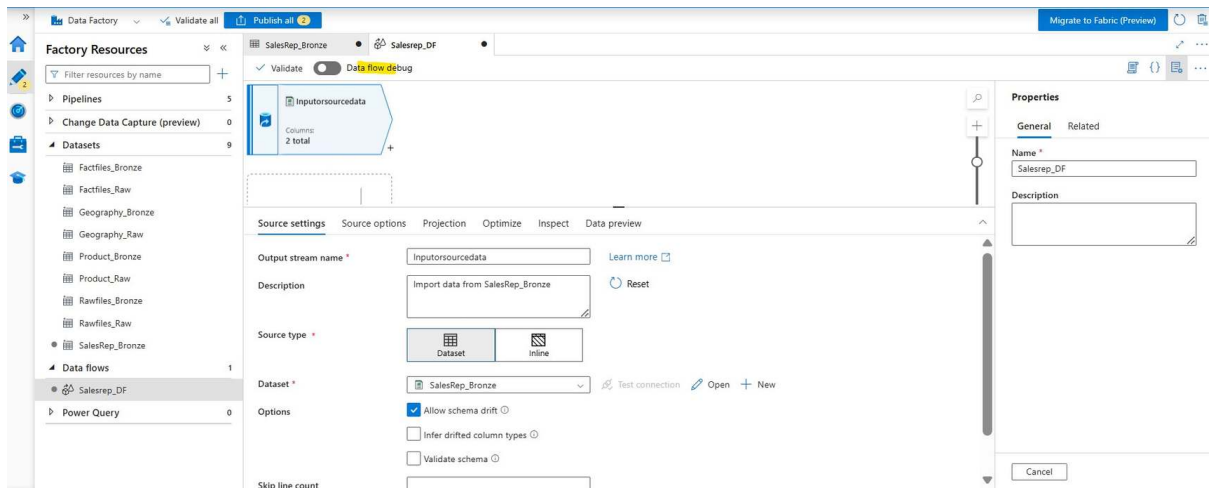
Create dataflow:



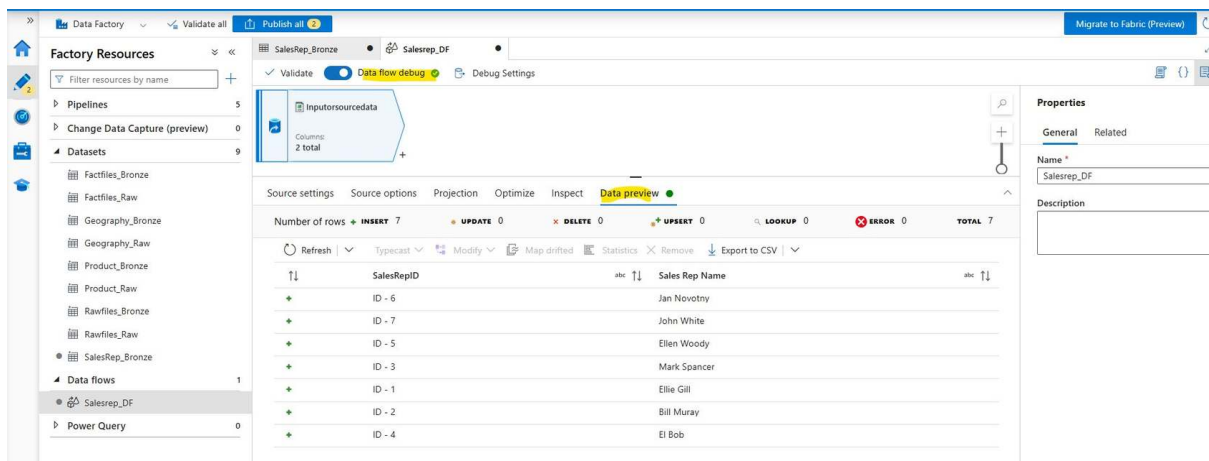
Add the source---inputdata



Turn on the debug mode---1 hour --to preview the data



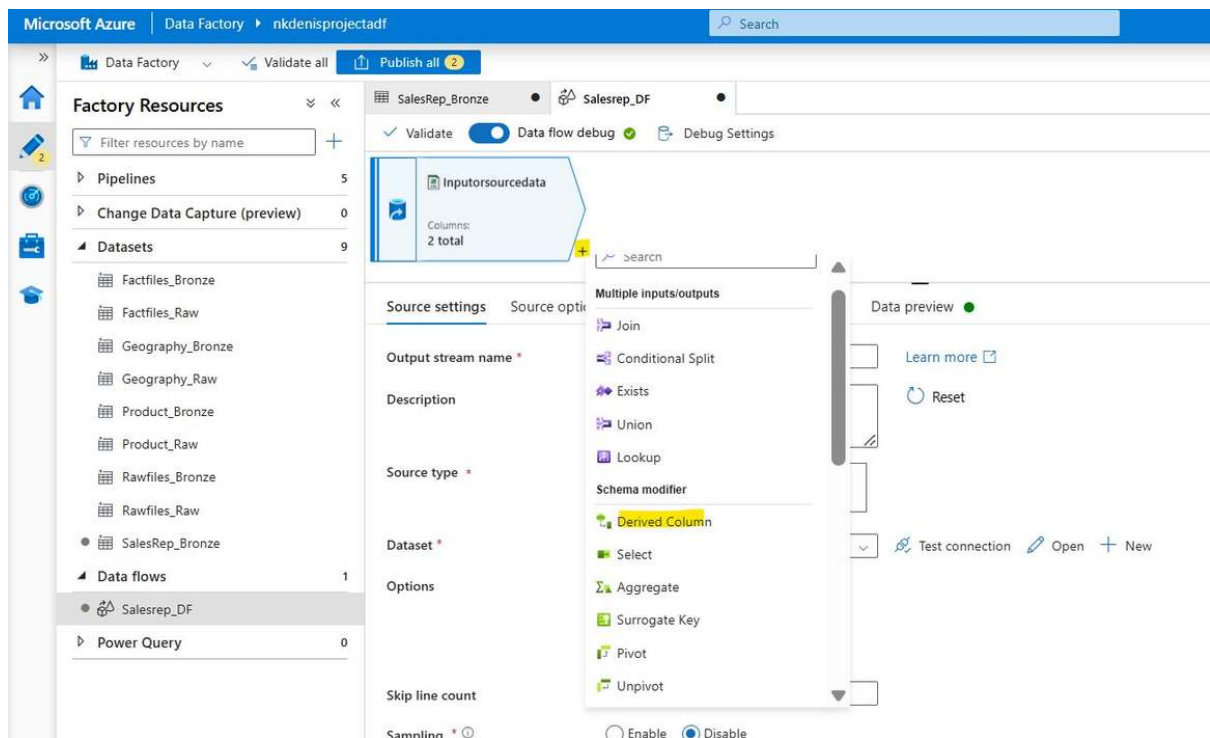
We can now preview the data



Requirement:

We need to remove ID – from the salesrepID(Need to transform)

Let's create a derived column



We will use same column and write the expression we need to get only numeric values:

Dataflow expression builder

derivedColumn1

Derived Columns

+ Create new

ANY 'SalesRepID'

Column name *

'SalesRepID'

Expression

`regexReplace(SalesRepID, '^[^0-9]*', '')`

Expression elements

All
Functions
Input schema
Parameters
Cached lookup
Data flow library functions
Locals

Expression values

Filter by keyword
+ Create new
abc SalesRepID
abc Sales Rep Name
123 abs(123 numeric_value)
123 acos(123 numeric_value)

Data preview Refresh

Save and finish

Cancel

Clear contents

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 9
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - SalesRep_Bronze
- Data flows: 1
 - Salesrep_DF
- Power Query: 0

SalesRep_DF

Validate | Data flow debug | Debug Settings

Reference: 1 | Columns: 2 total

derivedColumn1 | Columns: 3 total

Derived column's settings | Optimize | Inspect | Data preview

Output stream name: derivedColumn1

Description: Creating/updating the columns 'SalesRepID, Sales Rep Name, 'SalesRepID'

Incoming stream: Inputorsourcedata

Columns:

Column	Expression
'SalesRepID'	regexReplace(SalesRepID, '[0-9]', '')

Let's preview the data and check whether the requirement is satisfied or not

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 9
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - SalesRep_Bronze
- Data flows: 1
 - Salesrep_DF
- Power Query: 0

SalesRep_DF

Validate | Data flow debug | Debug Settings

Inputorsourcedata | Import data from SalesRep_Bronze

derivedColumn1 | Columns: 3 total

Derived column's settings | Optimize | Inspect | **Data preview**

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0 | ERROR 0 | TOTAL 7

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

	SalesRepID	Sales Rep Name	SalesRepID
+	ID - 6	Jan Novotny	6
+	ID - 7	John White	7
+	ID - 5	Ellen Woody	5
+	ID - 3	Mark Spancer	3
+	ID - 1	Ellie Gill	1
+	ID - 2	Bill Muray	2
+	ID - 4	El Bob	4

Hence requirement is satisfied but 2 columns became 3 columns ---so let's use the same column and transform

Data Factory | Validate all | Publish all 2

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 9
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - SalesRep_Bronze
 - Data flows: 1**
 - Salesrep_DF**
 - Power Query: 0

SalesRep_Bronze | **Salesrep_DF**

✓ Validate | Data flow debug | Debug Settings

Inputorsourcedata → derivedColumn1
Import data from SalesRep_Bronze | Columns: 2 total

Derived column's settings | Optimize | Inspect | **Data preview**

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

	SalesRepID	Sales Rep Name
+	6	Jan Novotny
+	7	John White
+	5	Ellen Woody
+	3	Mark Spencer
+	1	Ellie Gill
+	2	Bill Muray
+	4	El Bob

Now we need to dump into silver layer---need to configure in sink:

Data Factory | Validate all | Publish all 2

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 9
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - SalesRep_Bronze
 - Data flows: 1**
 - Salesrep_DF**
 - Power Query: 0

SalesRep_Bronze | **Salesrep_DF**

✓ Validate | Data flow debug | Debug Settings

Inputorsourcedata → derivedColumn1
Import data from SalesRep_Bronze | Columns: 2 total

Derived column's settings | Optimize | Inspect | **Data preview**

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

	SalesRepID	Sales Rep Name
+	6	Jan Novotny
+	7	John White
+	5	Ellen Woody
+	3	Mark Spencer
+	1	Ellie Gill
+	2	Bill Muray
+	4	El Bob

Cast

- Formatters
 - Flatten
 - Parse
 - Stringify
- Row modifier
 - Filter
 - Sort
 - Alter Row
 - Assert
- Flowlets
 - Flowlet
- Destination
 - Sink**

»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines5

Change Data Capture (preview)0

Datasets9

Factfiles_Bronze

Factfiles_Raw

Geography_Bronze

Geography_Raw

Product_Bronze

Product_Raw

Rawfiles_Bronze

Rawfiles_Raw

SalesRep_Bronze

Data flows1

SalesRep_DF

Power Query0

SalesRep_Bronze

SalesRep_DF

Validate

Data flow debug

Debug Settings

InputSourceData

Reference: 1

Columns: 2 total

Columns: 2 total

Columns: 2 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Add sink dataset

Reset

Incoming stream *

derivedColumn1

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

New

Options

Allow schema drift

Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Continue

Cancel

Select format

Choose the format type of your data


Avro


DelimitedText


JSON


ORC


Parquet


Binary

Continue

Back

Cancel

Set properties

Name

Salesrep_Silver

Linked service *

ADLS2adf_Is

File path

denis

/

Medallian/Silver

File name

▼

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

Data Factory | Validate all | Publish all 3

Factory Resources

Filter resources by name

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 10
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 1
 - Salesrep_DF
- Power Query: 0

SalesRep_Bronze | Salesrep_DF

✓ Validate | Data flow debug | Debug Settings

Input sourcedata → derivedColumn1 → sink1

Import data from SalesRep_Bronze | Creating/updating the columns 'SalesRepID, Sales Rep Name' | Columns: 2 total

Sink Settings

Output stream name: sink1 [Learn more](#)

Description: Export data to Salesrep_Silver [Reset](#)

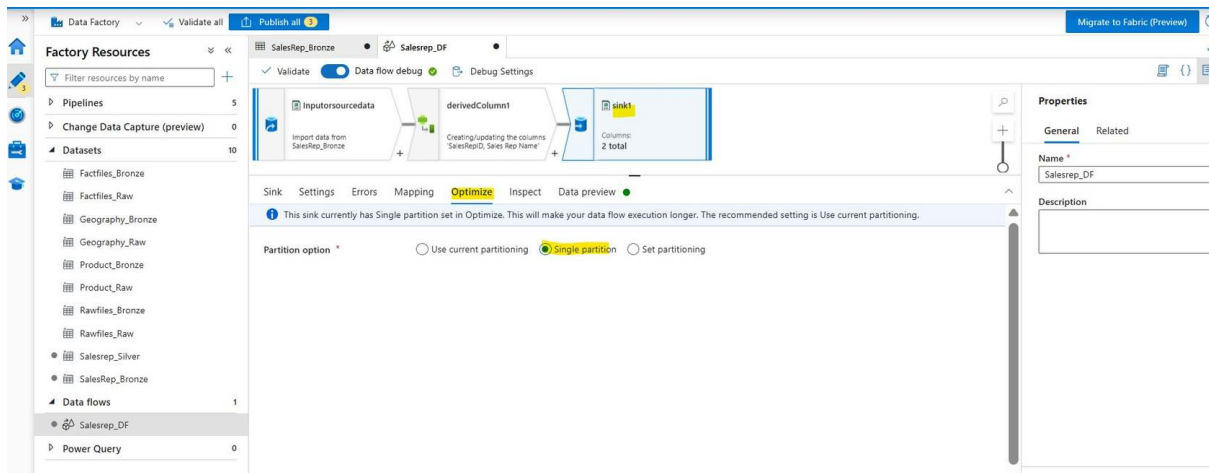
Incoming stream: derivedColumn1

Sink type: Dataset | Inline | Cache

Dataset: Salesrep_Silver [Test connection](#) [Open](#) [New](#)

Skip line count:

Options: ☒ Allow schema drift [?](#) ☐ Validate schema [?](#)

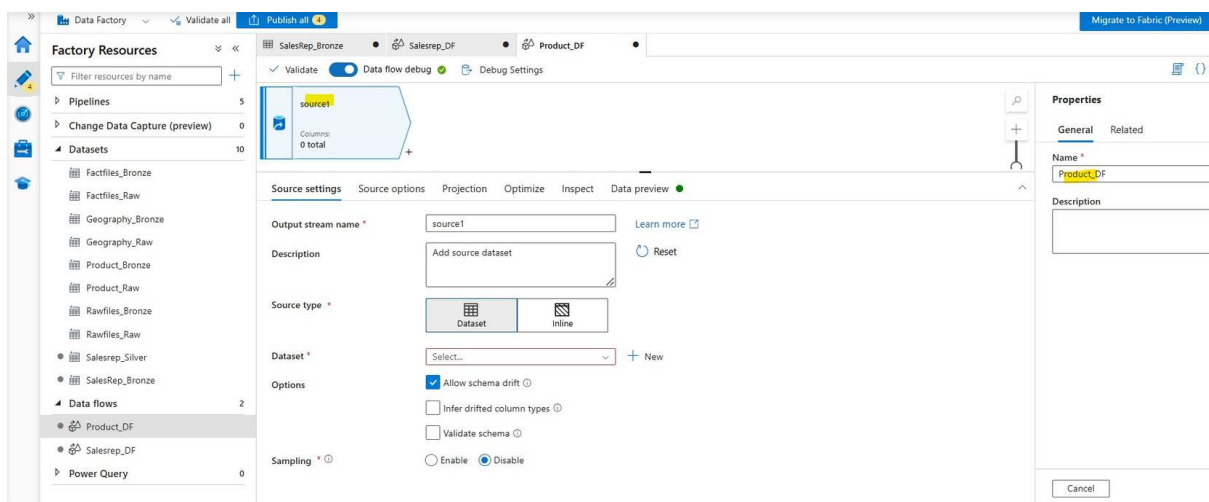


Successfully salesrepdata transformation has been done.

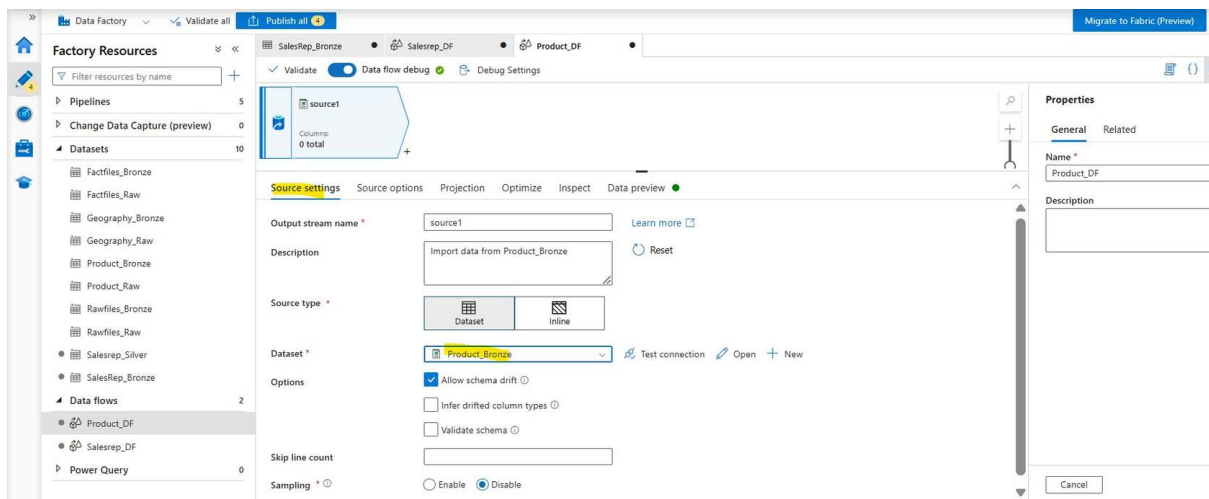
Publish all changes

2.Product dataflow:

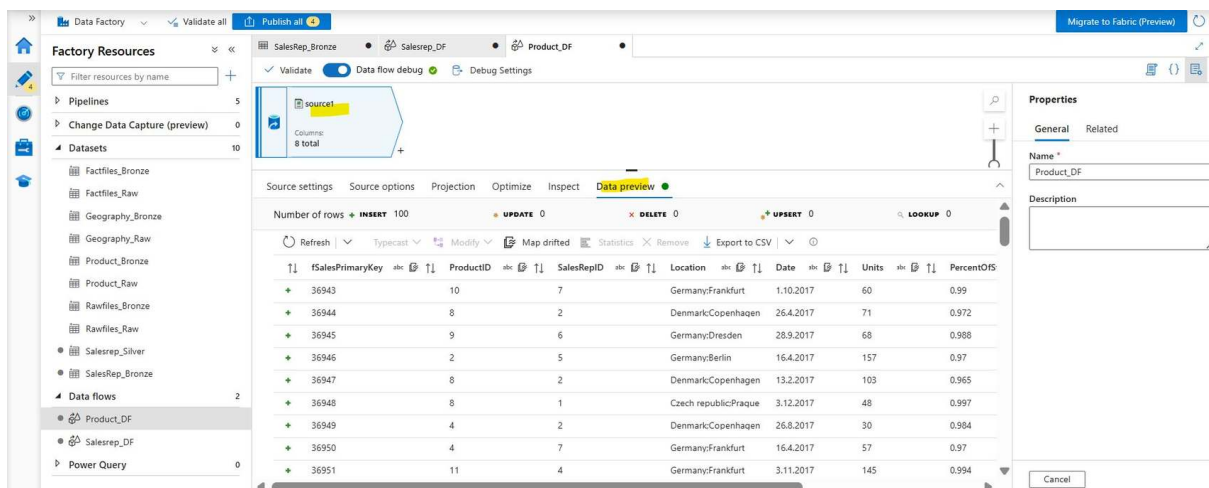
Create dataflow



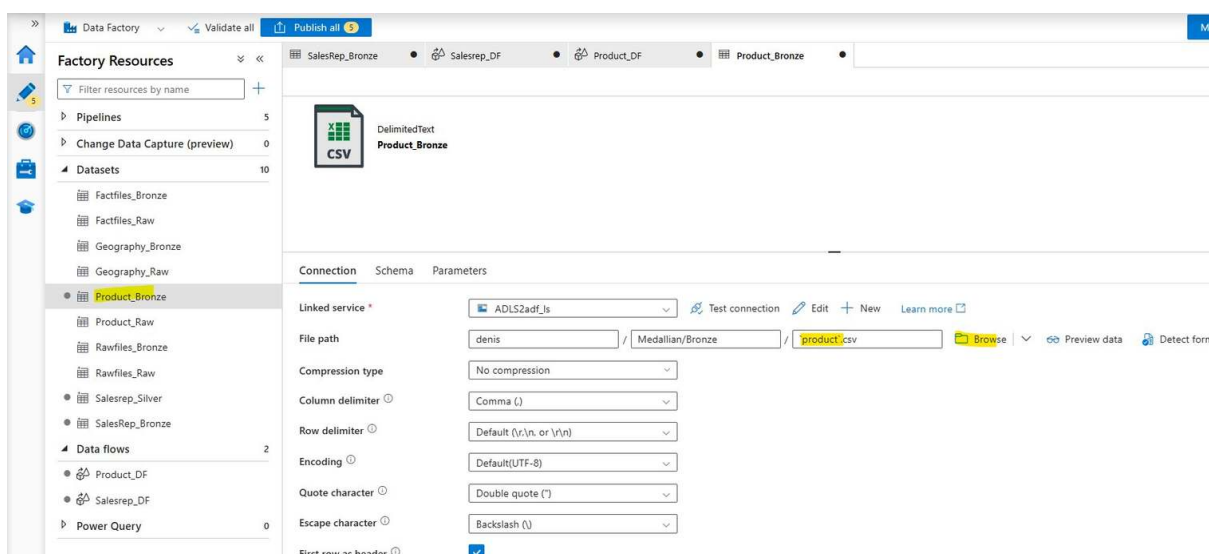
Already we have a data set product_bronze so we can use the data as input



**If the source data is showing 0 columns---go to projection---import projection--
-now you can see the columns in source**



While reading the data we need to give the file name



Factory Resources

Filter resources by name

Pipelines: 5

Change Data Capture (preview): 0

Datasets: 10

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Product_Bronze
- Product_Raw
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- SalesRep_DF
- Power Query

Data flows: 2

- Product_DF
- Salesrep_DF

Source settings

Source options

Projection

Optimize

Inspect

Data preview

ProductID	Sub Category Key	Color	ProductName	RetailPrice	StandardCost
1	3	Red	Alder	23.95	7.55
2	2	Blue	Linder	23.95	7.55
3	2	Green	Magnum	23.95	7.55
4	1	Red	Quad	43.95	13.75
5	1	Blue	Black Monk	43.95	13.75
6	4	Green	Quad	43.95	13.75
7	1	Red	Binq	26.95	8.25
8	3	Blue	VanHelen	26.95	8.25
9	1	Green	Magnum	26.95	8.25
10	1	Florescent Pink	Carlota	29.95	9.15
11	4	Florescent Blue	Carlota	29.95	9.15

Properties

General

Name: Product_DF

Description:

Requirement1: we have to remove the duplicate values

Factory Resources

Filter resources by name

Pipelines: 5

Change Data Capture (preview): 0

Datasets: 10

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Product_Bronze
- Product_Raw
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- SalesRep_DF
- Power Query

Data flows: 2

- Product_DF
- Salesrep_DF

Source settings

Source options

Projection

Optimize

Inspect

Data preview

Derived Column

- Select
- Aggregate
- Surrogate Key
- Pivot
- Unpivot
- Window
- Rank
- External Call
- Cast
- Formatters
- Flatten
- Parse
- Stringify

Please turn on debug mode

Window settings

Output stream name: window1

Description: Aggregates data based on a window and joins with original data

Incoming stream: source1

1. Over 2. Sort 3. Range by 4. Window columns

source1's column: 125 ProductID

Name as: ProductID

Window settings

Optimize

Inspect

Data preview

Output stream name *

window1

? Help

Learn more

Description

Aggregates data based on a window and joins with original data

Reset

Incoming stream *

source1

1. Over

2. Sort

3. Range by

4. Window columns

source1's column

Order

Nulls first

12s ProductID

Ascending

☒

+

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines 5

Change Data Capture (preview) 0

Datasets 10

Factfiles_Bronze

Factfiles_Raw

Geography_Bronze

Geography_Raw

Product_Bronze

Product_Raw

Rawfiles_Bronze

Rawfiles_Raw

Salesrep_Silver

Salesrep_Bronze

Data flows 2

Product_DF

Salesrep_DF

Power Query 0

SalesRep_Bronze

Salesrep_DF

Product_DF

Product_Bronze

Validate

Data flow debug

source1

Import data from Product_Bronze

+

window1

Columns: 7 total

+

Window settings

Optimize

Inspect

Data preview

Output stream name *

window1

? Help

Learn more

Description

Aggregates data based on a window and joins with original data

Reset

Incoming stream *

source1

1. Over

2. Sort

3. Range by

4. Window columns

Option *

Range by current row offset

Range by column value

Unbounded

☒

Dataflow expression builder

window1

Window Columns

+ Create new

ANY RN

Column name *

RN

Expression

rowNumber()

Expression elements

All

Functions

Input schema

Parameters

Cached lookup

Data flow library functions

Locals

Expression values

Filter by keyword

+ Create new

123 ProductID

123 Sub Category Key

abc Color

abc ProductName

123 RetailPrice

Save and finish

Cancel

Clear contents

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 10
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - Salesrep_Bronze
- Data flows: 2
 - Product_DF
 - Salesrep_DF
- Power Query: 0

Validate ☒ Data flow debug ☒

Reference: 1
Columns: 6 total

Window settings

Output stream name *: window1

Description: Aggregates data based on a window and joins with original data

Incoming stream *: source1

1. Over 2. Sort 3. Range by 4. Window columns

+ Add Clone Delete Open expression builder

Column	Expression
RN	rowNumber()

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 10
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - Salesrep_Bronze
- Data flows: 2
 - Product_DF
 - Salesrep_DF
- Power Query: 0

Validate ☒ Data flow debug ☒ Debug Settings

source1: Import data from Product_Bronze

window1: Columns: 7 total

Window settings: Optimize Inspect **Data preview**

1	3	Red	Alder	23.95	7.55	1
2	2	Blue	Linder	23.95	7.55	1
3	2	Green	Magnum	23.95	7.55	1
4	1	Red	Quad	43.95	13.75	1
5	1	Blue	Black Monk	43.95	13.75	1
6	4	Green	Quad	43.95	13.75	1
7	1	Red	Bing	26.95	8.25	1
7	1	Red	Bing	26.95	8.25	2
8	3	Blue	VanHelen	26.95	8.25	1
9	1	Green	Magnum	26.95	8.25	1
10	1	Florescent Pink	Carlota	29.95	9.15	1
10	1	Florescent Pink	Carlota	29.95	9.15	2
11	4	Florescent Blue	Carlota	29.95	9.15	1

We need to remove the duplicates which is aggregate 2

Using filter

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 10
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - Salesrep_Bronze
- Data flows: 2
 - Product_DF
 - Salesrep_DF
- Power Query: 0

Validate ☒ Data flow debug ☒ Debug Settings

source1: Import data from Product_Bronze

window1: Columns: 7 total

Window settings: Optimize Inspect **Data preview**

1	3	Red	Alder	23.95	7.55	1
2	2	Blue	Linder	23.95	7.55	1
3	2	Green	Magnum	23.95	7.55	1
4	1	Red	Quad	43.95	13.75	1
5	1	Blue	Black Monk	43.95	13.75	1
6	4	Green	Quad	43.95	13.75	1
7	1	Red	Bing	26.95	8.25	1
7	1	Red	Bing	26.95	8.25	2
8	3	Blue	VanHelen	26.95	8.25	1
9	1	Green	Magnum	26.95	8.25	1
10	1	Florescent Pink	Carlota	29.95	9.15	1
10	1	Florescent Pink	Carlota	29.95	9.15	2
11	4	Florescent Blue	Carlota	29.95	9.15	1

Rank, External Call, Cast, Formatters, Flatten, Parse, Stringify, Row modifier, Filter, Sort, Alter Row, Assert, Flowlets, Flowlet

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 10
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - Salesrep_Bronze
- Data flows: 2
 - Product_DF
 - Salesrep_DF
- Power Query: 0

Validate ☒ Data flow debug ☒ Debug Settings

source1: Import data from Product_Bronze

window1: Aggregates data based on a window and joins with original data

filter1: Columns: 7 total

Filter settings: Optimize Inspect **Data preview**

Output stream name: filter1

Description: Filtering rows using expressions on columns 'RN'

Incoming stream: window1

Filter on: RN = 1

Properties: General Related

Name: Product_DF

Description:

Cancel

Now we are able to see only the unique records

The screenshot shows the Data Factory interface with a data flow named 'Product_DF'. The flow consists of three steps: 'source1' (Import data from Product_Bronze), 'window1' (Aggregates data based on a window and joins with original data), and 'filter1' (Columns: 7 total). The 'filter1' step is selected, and its 'Data preview' tab is active. The preview shows 7 columns: ProductID, Sub Category Key, Color, ProductName, and two unnamed columns. The data is as follows:

ProductID	Sub Category Key	Color	ProductName		
1	3	Red	Alder		
2	2	Blue	Linder		
3	2	Green	Magnum		
4	1	Red	Quad		
5	1	Blue	Black Monk		
6	4	Green	Quad		
7	1	Red	Bing		
8	3	Blue	VanHelen		
9	1	Green	Magnum		
10	1	Florescent Pink	Carlota	29.95	9.15
11	4	Florescent Blue	Carlota	29.95	9.15

We will select the required columns

The screenshot shows the 'Select settings' tab for the 'filter1' step. The 'Incoming stream' is 'filter1'. The 'Options' section has 'Skip duplicate input columns' and 'Skip duplicate output columns' checked. The 'Input columns' section shows 6 mappings: 1 column(s) from the inputs left unmapped. The mappings are as follows:

filter1's column	Name as
12s ProductID	ProductID
12s Sub Category Key	SubCategory Key
abc Color	Color
abc ProductName	ProductName
1.2 RetailPrice	RetailPrice
1.2 StandardCost	StandardCost

The screenshot shows the Data Factory interface with a data flow named 'Product_DF'. The flow consists of four steps: 'source1' (Import data from Product_Bronze), 'window1' (Aggregates data based on a window and joins with original data), 'filter1' (Filtering rows using expressions on columns RN), and 'select1' (Columns: 6 total). The 'select1' step is selected, and its 'Data preview' tab is active. The preview shows 6 columns: ProductID, Sub Category Key, Color, ProductName, RetailPrice, and StandardCost. The data is as follows:

ProductID	Sub Category Key	Color	ProductName	RetailPrice	StandardCost
1	3	Red	Alder		
2	2	Blue	Linder		
3	2	Green	Magnum		
4	1	Red	Quad		
5	1	Blue	Black Monk		
6	4	Green	Quad		
7	1	Red	Bing		
8	3	Blue	VanHelen		
9	1	Green	Magnum		
10	1	Florescent Pink	Carlota	29.95	9.15
11	4	Florescent Blue	Carlota	29.95	9.15

»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines

Change Data Capture (preview)

Datasets

Factfiles_Bronze

Factfiles_Raw

Geography_Bronze

Geography_Raw

Product_Bronze

Product_Raw

Rawfiles_Bronze

Rawfiles_Raw

Salesrep_Silver

Salesrep_Bronze

Data flows

Product_DF

Salesrep_DF

Power Query

Salesrep_Bronze

Salesrep_DF

Product_DF

Product_Bronze

Validate

Data flow debug

Debug Settings

source1

window1

filter1

Reference 1

Columns: 6 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Add sink dataset

Incoming stream *

select1

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

New

Options

Allow schema drift

Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Continue

Cancel

Select format

Choose the format type of your data


Avro


DelimitedText


JSON


ORC


Parquet


Binary

Continue

Back

Cancel

Set properties

Name

Product_Silver

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Silver

/ File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

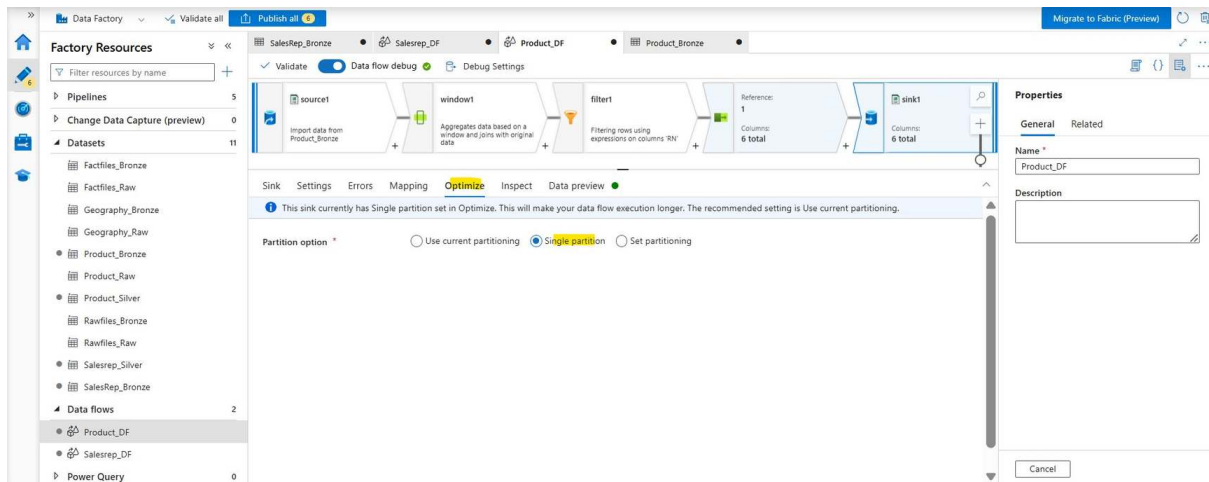
Back

Cancel

The screenshot displays the Azure Data Factory (ADF) console. On the left, the 'Factory Resources' pane shows a tree view of datasets, including 'Product_Silver'. The main workspace shows a data flow pipeline with a sink named 'sink1'. The 'Set properties' dialog is open for 'sink1', showing the following configuration:

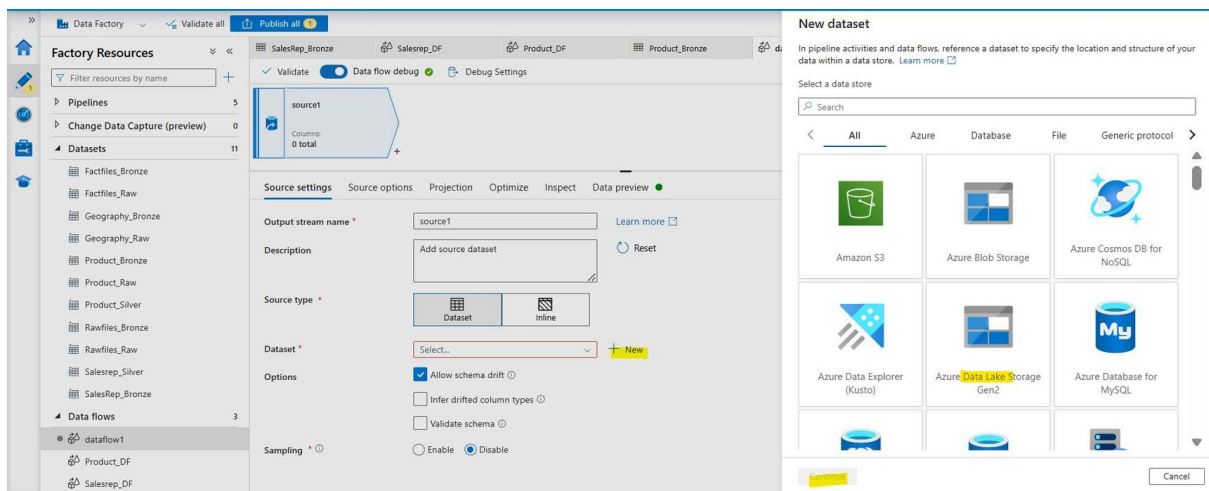
- Output stream name:** sink1
- Description:** Export data to Product_Silver
- Incoming stream:** select1
- Sink type:** Dataset
- Dataset:** Product_Silver
- Skip line count:** (empty)
- Options:**
 - ☒ Allow schema drift
 - ☐ Validate schema

The right-hand 'Properties' pane shows the 'General' tab with the 'Name' set to 'Product_DF'.



Successfully transformed

3.Category dataflow:



Hence the category data is present in rawfiles_bronze we need to create input data from bronze

Select excel because category is in excel format

Set properties

Name

Category_Bronze

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Bronze

/ Categories.xlsx

Worksheet mode

☒ Name ☐ Index

Sheet name * ⓘ

Categories

First row as header ☐

Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

The screenshot shows the 'Set properties' dialog for a Dataset in Azure Data Factory. The 'Name' is 'Category_Bronze', 'Linked service' is 'ADLS2adf_Is', 'File path' is 'denis / Medallian/Bronze / Categories.xlsx', 'Worksheet mode' is 'Name', 'Sheet name' is 'Categories', 'First row as header' is unchecked, and 'Import schema' is 'From connection/store'. The 'Advanced' section is expanded.

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 12
 - Category_Bronze
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 3
 - dataflow1
 - Product_DF

Source settings

Output stream name *: source1 [Learn more](#)

Description: Import data from Category_Bronze [Reset](#)

Source type *: ☒ Dataset ☐ Inline

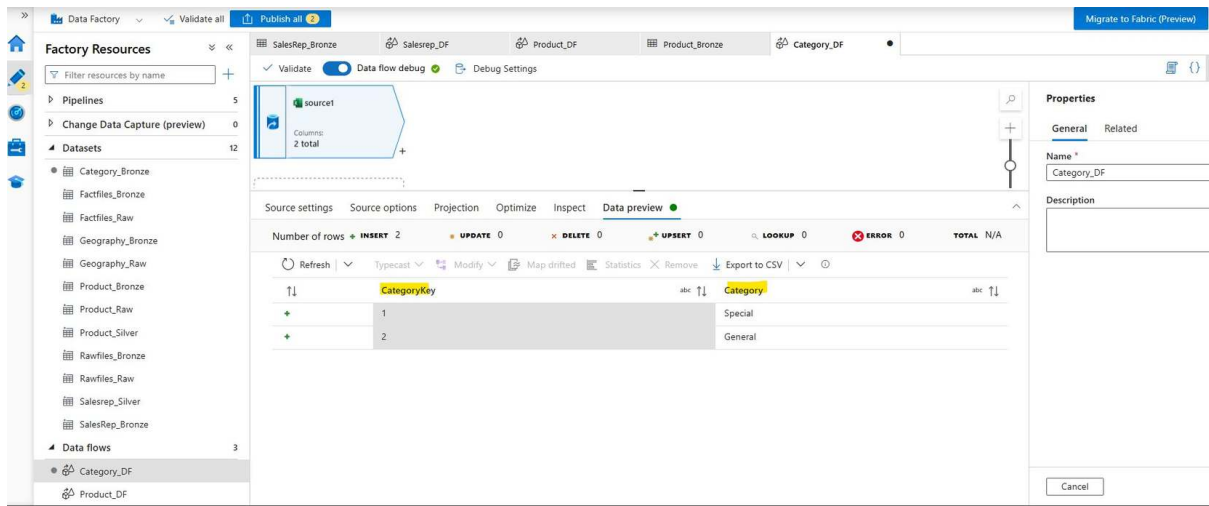
Dataset *: Category_Bronze [Test connection](#) [Open](#) [New](#)

Options

- ☒ Allow schema drift ⓘ
- ☐ Infer drifted column types ⓘ
- ☐ Validate schema ⓘ

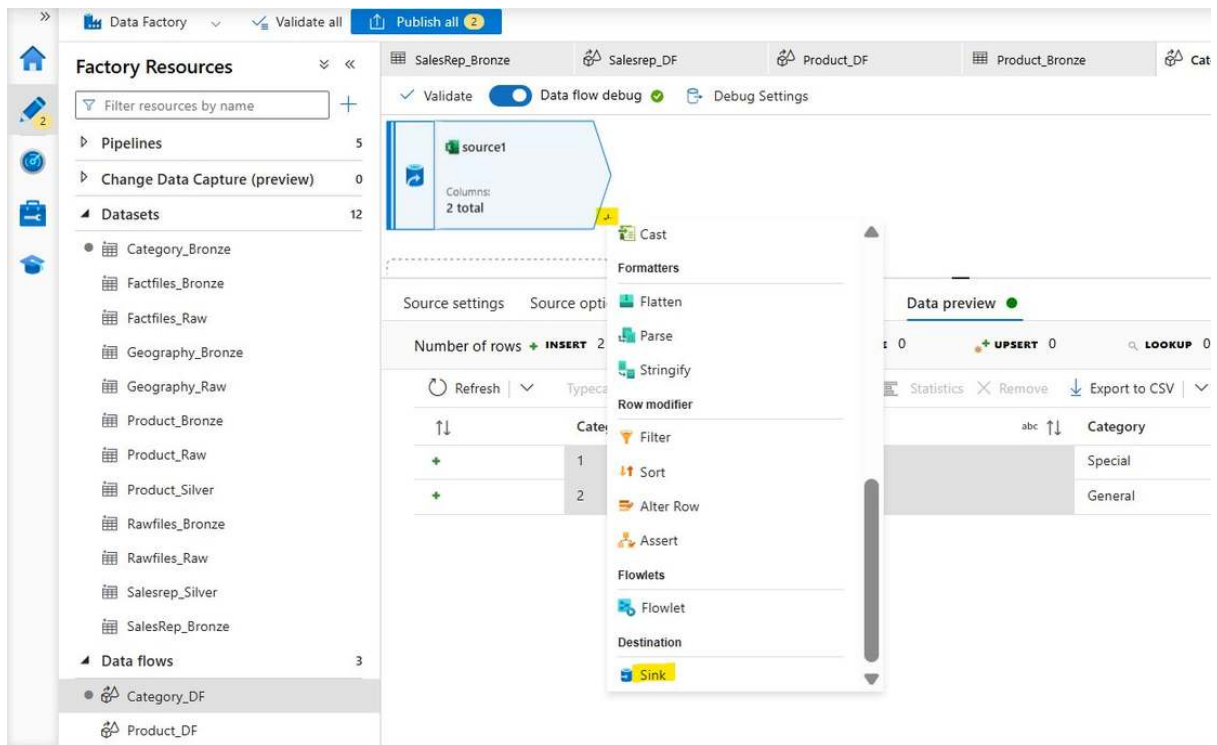
Sampling *: ☐ Enable ☒ Disable

Preview the data



There are no transformations required

Sink the data



»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines

Change Data Capture (preview)

Datasets

Category_Bronze

Factfiles_Bronze

Factfiles_Raw

Geography_Bronze

Geography_Raw

Product_Bronze

Product_Raw

Product_Silver

Rawfiles_Bronze

Rawfiles_Silver

Salesrep_Bronze

Salesrep_Silver

Data flows

Category_DF

Product_DF

SalesRep_Bronze

Salesrep_DF

Product_DF

Product_Bronze

Validate

Data flow debug

Debug Settings

source1

Import data from Category_Bronze

+

sink1

Columns: 2 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Add sink dataset

Reset

Incoming stream *

source1

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

New

Options

Allow schema drift

Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Continue

Cancel

Select format

Choose the format type of your data


Avro


DelimitedText


JSON


ORC


Parquet


Binary

Continue

Back

Cancel

Set properties

Name

Category_Silver

Linked service *

ADLS2adf_ls

File path

denis

/ Medallian/Silver

/ File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

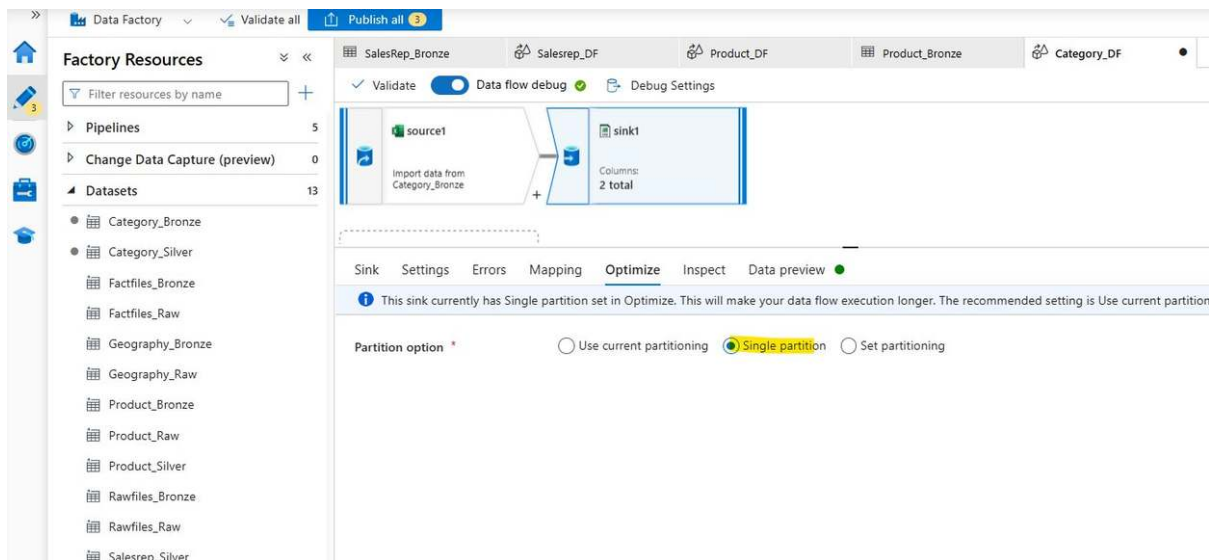
> Advanced

OK

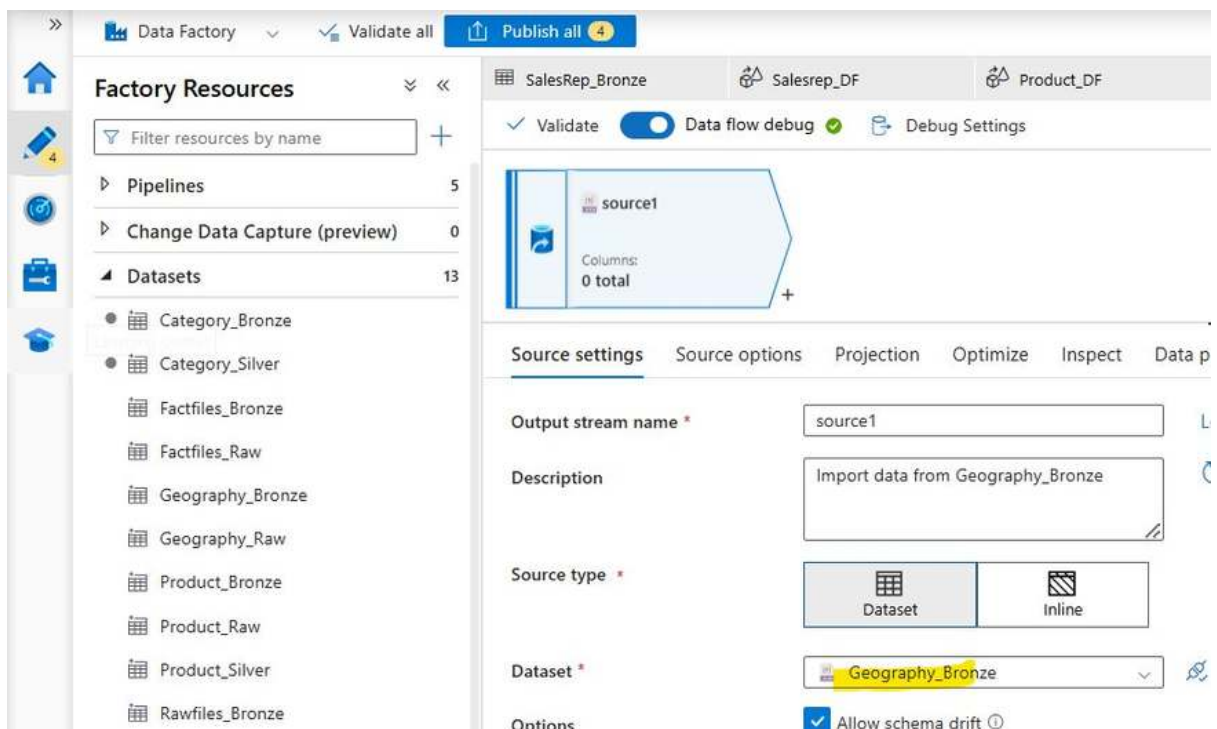
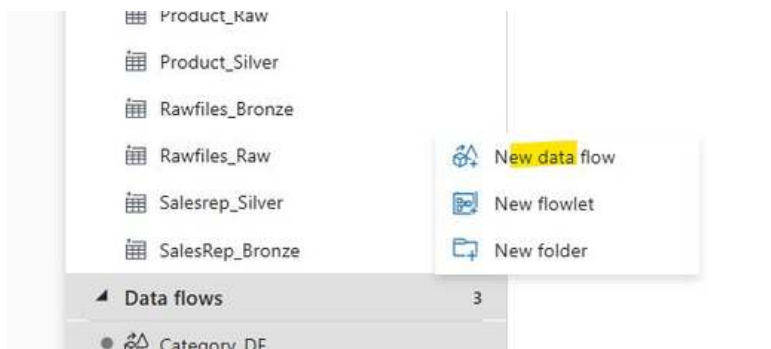
Back

Cancel

The screenshot shows the 'Set properties' dialog for a sink in Azure Data Factory. The 'Name' field is set to 'Category_Silver'. The 'Linked service' is 'ADLS2adf_ls'. The 'File path' is 'denis / Medallian/Silver / File name'. The 'First row as header' checkbox is checked. The 'Import schema' section has 'From connection/store' selected. The 'Advanced' section is expanded. Below the dialog, the 'Factory Resources' pane shows a list of datasets, including 'Category_Silver'. The 'Sink' settings pane shows the 'Output stream name' as 'sink1', the 'Description' as 'Export data to Category_Silver', the 'Incoming stream' as 'source1', the 'Sink type' as 'Dataset', the 'Dataset' as 'Category_Silver', and the 'Options' as 'Allow schema drift' checked and 'Validate schema' unchecked. The 'Properties' pane on the right shows the 'Name' as 'Category_DF' and the 'Description' as an empty text box.



4) Geography dataflow



Projection---import projection---preview data

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 4

SalesRep_Bronze | Salesrep_DF | Product_DF | Product_Bronze | Category_DF | Geography_DF

Validate | Data flow debug | Debug Settings

source1
Columns: 4 total

Source settings | Source options | Projection | Optimize | Inspect | **Data preview**

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0 | ERROR 0 | TOTAL N/A

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

Country	Town	Wikipedia	_id
Czech republic	Praque	https://en.wikipedia.org/wiki/Praque	6a55dd5ca916ab7bcfb8f7a1
Denmark	Copenhagen	https://en.wikipedia.org/wiki/Copenhagen	6a55dd5ca916ab7bcfb8f7a2
Germany	Frankfurt	https://en.wikipedia.org/wiki/Frankfurt	6a55dd5ca916ab7bcfb8f7a3
Germany	Berlin	https://en.wikipedia.org/wiki/Berlin	6a55dd5ca916ab7bcfb8f7a4
Germany	Dresden	https://en.wikipedia.org/wiki/Dresden	6a55dd5ca916ab7bcfb8f7a5
Czech republic	Brno	https://en.wikipedia.org/wiki/Brno	6a55dd5ca916ab7bcfb8f7a6
Czech republic	Ostrava	https://en.wikipedia.org/wiki/Ostrava	6a55dd5ca916ab7bcfb8f7a7

I want to remove oid columns

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 4
 - Category_DF

SalesRep_Bronze | Salesrep_DF | Product_DF

Validate | Data flow debug | Debug Settings

source1
Columns: 4 total

Source settings | Source options | Projection | Optimize | Inspect | **Data preview**

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0 | ERROR 0 | TOTAL N/A

Refresh | Typecast | Modify | Map drifted | Statistics | Remove | Export to CSV

Country	Town	Wikipedia	_id
Czech republic	Praque	https://en.wikipedia.org/wiki/Praque	6a55dd5ca916ab7bcfb8f7a1
Denmark	Copenhagen	https://en.wikipedia.org/wiki/Copenhagen	6a55dd5ca916ab7bcfb8f7a2
Germany	Frankfurt	https://en.wikipedia.org/wiki/Frankfurt	6a55dd5ca916ab7bcfb8f7a3
Germany	Berlin	https://en.wikipedia.org/wiki/Berlin	6a55dd5ca916ab7bcfb8f7a4
Germany	Dresden	https://en.wikipedia.org/wiki/Dresden	6a55dd5ca916ab7bcfb8f7a5
Czech republic	Brno	https://en.wikipedia.org/wiki/Brno	6a55dd5ca916ab7bcfb8f7a6
Czech republic	Ostrava	https://en.wikipedia.org/wiki/Ostrava	6a55dd5ca916ab7bcfb8f7a7

Multiple inputs/outputs

- Join
- Conditional Split
- Exists
- Union
- Lookup

Schema modifier

- Derived Column
- Select**
- Aggregate
- Surrogate Key
- Pivot
- Unpivot

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 4
 - Category_DF

Validate | Data flow debug | Debug Settings

source1: Import data from Geography_Bronze

select1: Columns: 2 total

Select settings | Optimize | Inspect | Data preview

Output stream name: select1

Description: Renaming source1 to select1 with columns 'Country, Town'

Incoming stream: source1

Options: ☒ Skip duplicate input columns, ☒ Skip duplicate output columns

Input columns: ☐ Auto mapping, ☐ Reset,

2 mappings: 2 column(s) from the inputs left unmapped

source1's column	Name as
abc Country	Country
abc Town	Town

Requirement is to add geokey

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 4
 - Category_DF

Validate | Data flow debug | Debug Settings

source1: Import data from Geography_Bronze

select1: Columns: 2 total

Select settings | Optimize | Inspect | Data preview

Output stream name: select1

Description: Renaming source1 to select1 with columns 'Country, Town'

Incoming stream: source1

Options: ☒ Skip duplicate input columns, ☒ Skip duplicate output columns

Input columns: ☐ Auto mapping, ☐ Reset,

2 mappings: 2 column(s) from the inputs left unmapped

source1's column	Name as
abc Country	Country
abc Town	Town

Search

Multiple inputs/outputs

- Join
- Conditional Split
- Exists
- Union
- Lookup

Schema modifier

- Derived Column
- Select
- Aggregate
- Surrogate Key
- Pivot
- Unpivot

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines
 - Change Data Capture (preview) 0
- Datasets 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows 4
 - Category_DF

SalesRep_Bronze | Salesrep_DF | Product_DF | Product_Bronze | Category

Validate | Data flow debug | Debug Settings

source1 | Reference: 1 | Columns: 2 total | **surrogateKey1** | Columns: 3 total

Surrogate key settings | Optimize | Inspect | Data preview

Output stream name: surrogateKey1 | Description: Adding new key Geokey starting from 1 with step 1 | Incoming stream: select1 | Key column: Geokey | Start value: 1 | Step value: 1

Preview data

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources

- Pipelines 5
 - Change Data Capture (preview) 0
- Datasets 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows 4
 - Category_DF

SalesRep_Bronze | Salesrep_DF | Product_DF | Product_Bronze | Category_DF | Geography_DF | Geography_Bronze

Validate | Data flow debug | Debug Settings

source1 | Reference: 1 | Columns: 2 total | **surrogateKey1** | Columns: 3 total

Surrogate key settings | Optimize | Inspect | Data preview

Number of rows: INSERT 7 | UPDATE 0 | DELETE 0 | UPSERT 0 | LOOKUP 0 | ERROR 0 | TOTAL 7

Country	Town	Geokey
Czech republic	Prague	1
Denmark	Copenhagen	2
Germany	Frankfurt	3
Germany	Berlin	4
Germany	Dresden	5
Czech republic	Brno	6
Czech republic	Ostrava	7

Properties

General | Related

Name: Geography_DF

Description:

Sink the data to target

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines 5
 - Change Data Capture (preview) 0
- Datasets 13
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows 4
 - Category_DF

SalesRep_Bronze | Salesrep_DF | Product_DF | Product_Bronze | Category_DF | Geography_DF | Geography_Bronze

Validate | Data flow debug | Debug Settings

source1 | Reference: 1 | Columns: 2 total | **select1** | Columns: 3 total | **sink1** | Columns: 3 total

Sink | Settings | Errors | Mapping | Optimize | Inspect | Data preview

Output stream name: sink1 | Description: Add sink dataset | Incoming stream: surrogateKey1 | Sink type: Dataset | Dataset: Select... | Options: Allow schema drift (checked) | Validate schema (unchecked)

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more

Select a data store

Search

All | Azure | Database | File | Generic protocol

Amazon S3 | Azure Blob Storage | Azure Cosmos DB for NoSQL | Azure Data Explorer (Kusto) | Azure Data Lake Storage Gen2 | Azure Database for MySQL

Continue | Cancel

Select format

Choose the format type of your data

 Avro	 DelimitedText	 JSON
 ORC	 Parquet	 Binary

Continue

Back

Cancel

Set properties

Name

Geography_Silver

Linked service *

ADLS2adf_ls

File path

denis

Medallian/Silver

File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines: 5
- Change Data Capture (preview): 0
- Datasets: 14
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Geography_Silver
 - Product_Bronze
 - Product_Raw
 - Product_Silver
 - Rawfiles_Bronze
 - Rawfiles_Raw
 - Salesrep_Silver
 - SalesRep_Bronze
- Data flows: 4

Data flow: Geography_DF

Validate | Data flow debug | Debug Settings

source1 (Import data from Geography_Bronze) → select1 (Renaming source1 to select1 with columns 'Country, Town') → Reference: 1, Columns: 3 total → sink1 (Columns: 3 total)

Sink Settings

Output stream name: sink1 [Learn more](#)

Description: Export data to Geography_Silver [Reset](#)

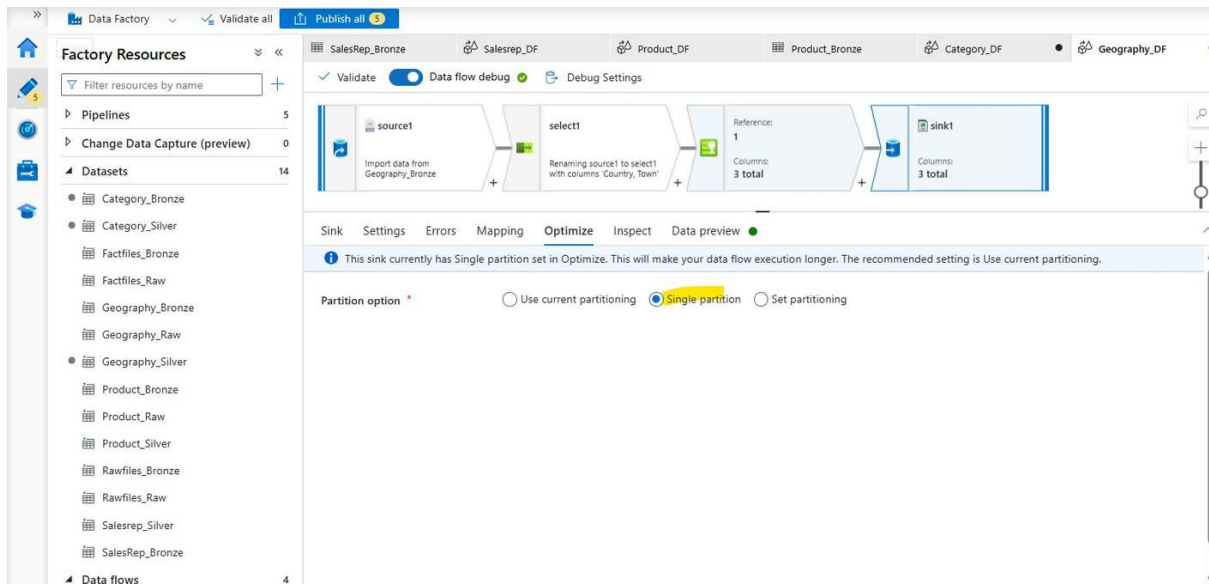
Incoming stream: surrogateKey1

Sink type: Dataset | Inline | Cache

Dataset: Geography_Silver [Test connection](#) [Open](#) [New](#)

Skip line count:

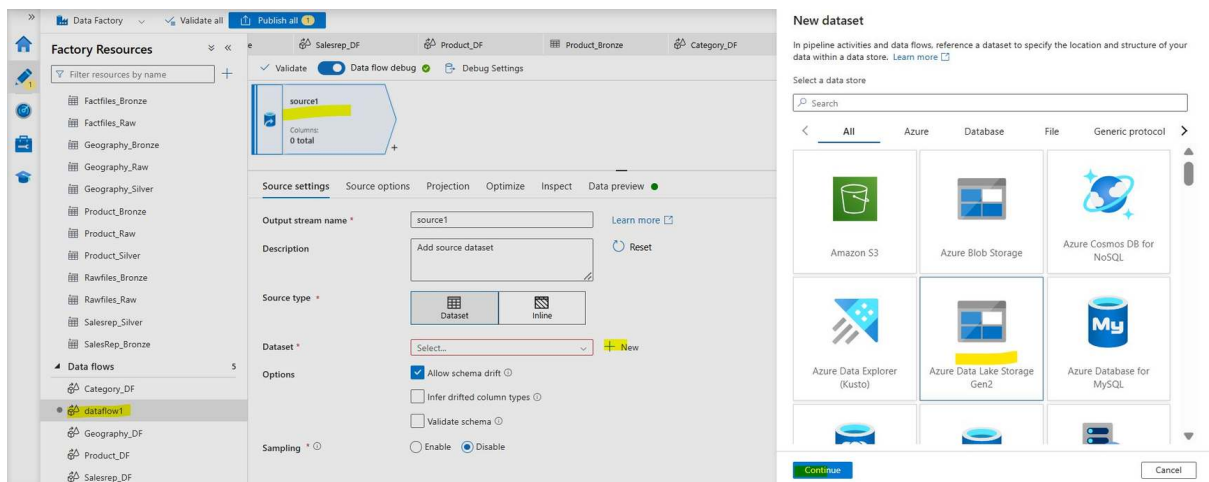
Options: ☒ Allow schema drift [?](#) ☐ Validate schema [?](#)



Publish all changes

5.Subcategory dataflow:

Create new pipeline----create inputdata



Select format

Choose the format type of your data

 Avro	 Delimited Text	 Excel
 JSON	 ORC	 Parquet
 XML	 Binary	

Continue

Back

Cancel

Set properties

Name

Subcategory_Bronze

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Bronze

/ SubCategories.csv

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

The screenshot shows the 'Set properties' dialog for a dataset named 'Subcategory_Bronze' in an Azure Data Factory environment. The dialog is part of a data flow named 'dataflow1'. The 'File path' is set to 'denis / Medallian/Bronze / SubCategories.csv'. The 'First row as header' checkbox is checked. The 'Import schema' section shows 'From connection/store' selected. The 'Advanced' section is expanded, showing 'Output stream name' as 'source1', 'Description' as 'Import data from Subcategory_Bronze', 'Source type' as 'Dataset', and 'Dataset' as 'Subcategory_Bronze'. The 'Options' section includes 'Allow schema drift' (checked), 'Infer drifted column types' (unchecked), and 'Validate schema' (unchecked). The 'Skip line count' is set to 0, and 'Sampling' is set to 'Disable'.

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- Subcategory_Bronze

Data flows

- Category_DF
- dataflow1
- Geography_DF
- Product_DF

Source settings

Output stream name * source1

Description Import data from Subcategory_Bronze

Source type * Dataset Inline

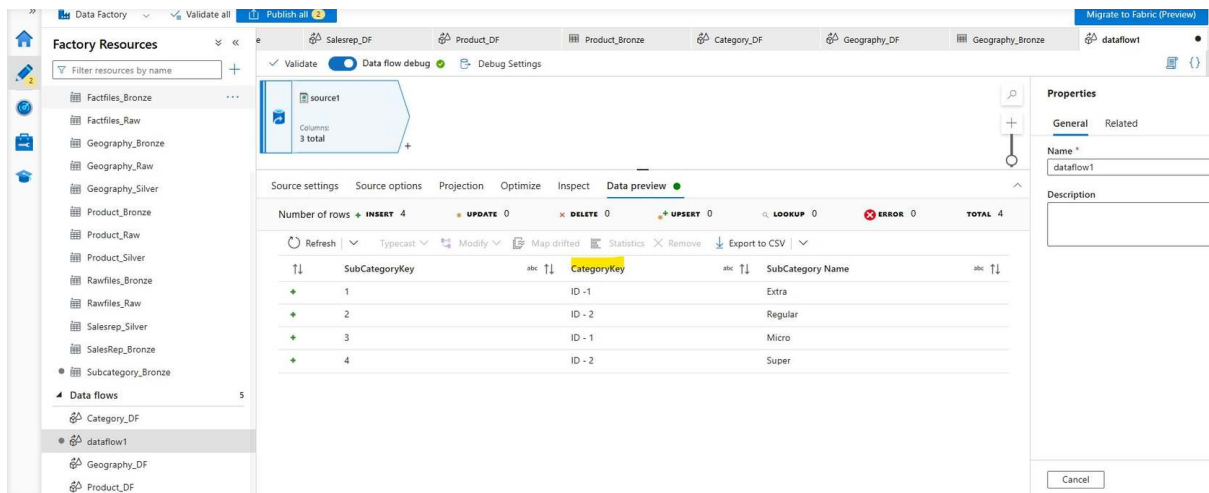
Dataset * Subcategory_Bronze

Options

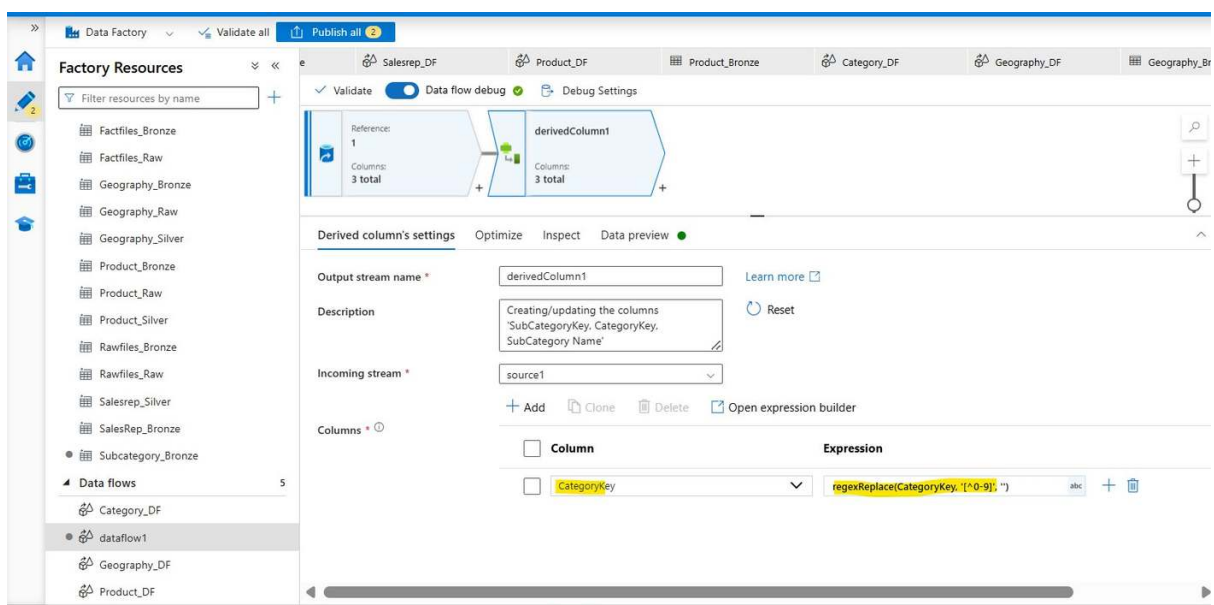
- ☒ Allow schema drift
- ☐ Infer drifted column types
- ☐ Validate schema

Skip line count

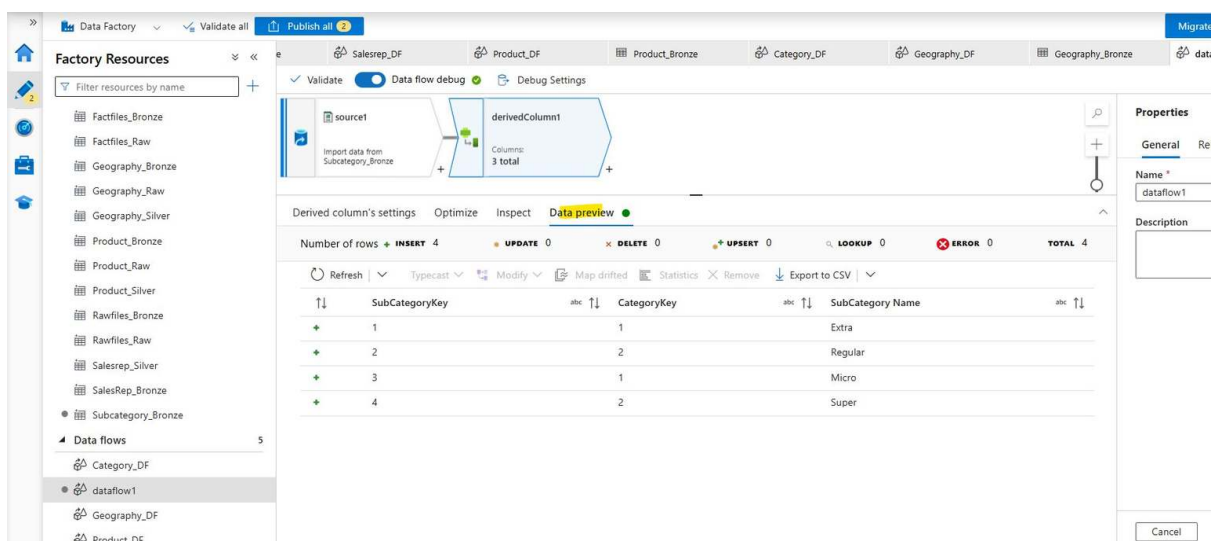
Sampling * ☐ Enable ☒ Disable



Requirement ---We need to split the id- from the category key



Preview data



Sink the data to target

»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

+

Factiles_Bronze

Factiles_Raw

Geography_Bronze

Geography_Raw

Geography_Silver

Product_Bronze

Product_Raw

Product_Silver

Rawfiles_Bronze

Rawfiles_Raw

Salesrep_Silver

SalesRep_Bronze

Subcategory_Bronze

Data flows

Category_DF

dataflow1

Geography_DF

Product_DF

Validate

Data flow debug

Debug Settings

source1

Import data from Subcategory_Bronze

Reference: 1

Columns: 3 total

+

sink1

Columns: 3 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Add sink dataset

Reset

Incoming stream *

derivedColumn1

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

New

Options

☒ Allow schema drift

☐ Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Continue

Cancel

Select format

Choose the format type of your data


Avro


DelimitedText


JSON


ORC


Parquet


Binary

Continue

Back

Cancel

Set properties

Name

Subcategory_Silver

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Silver

File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- Subcategory_Bronze
- Subcategory_Silver
- Data flows
- Category_DF
- Subcategory_DF
- Geography_DF

Data Factory

Validate all

Publish all

Salesrep_DF

Product_DF

Product_Bronze

Category_DF

Validate

Data flow debug

Debug Settings

source1

Import data from Subcategory_Bronze

Reference: 1

Columns: 3 total

sink1

Columns: 3 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Export data to Subcategory_Silver

Reset

Incoming stream *

derivedColumn1

Sink type *

Dataset

Inline

Cache

Dataset *

Subcategory_Silver

Test connection

Open

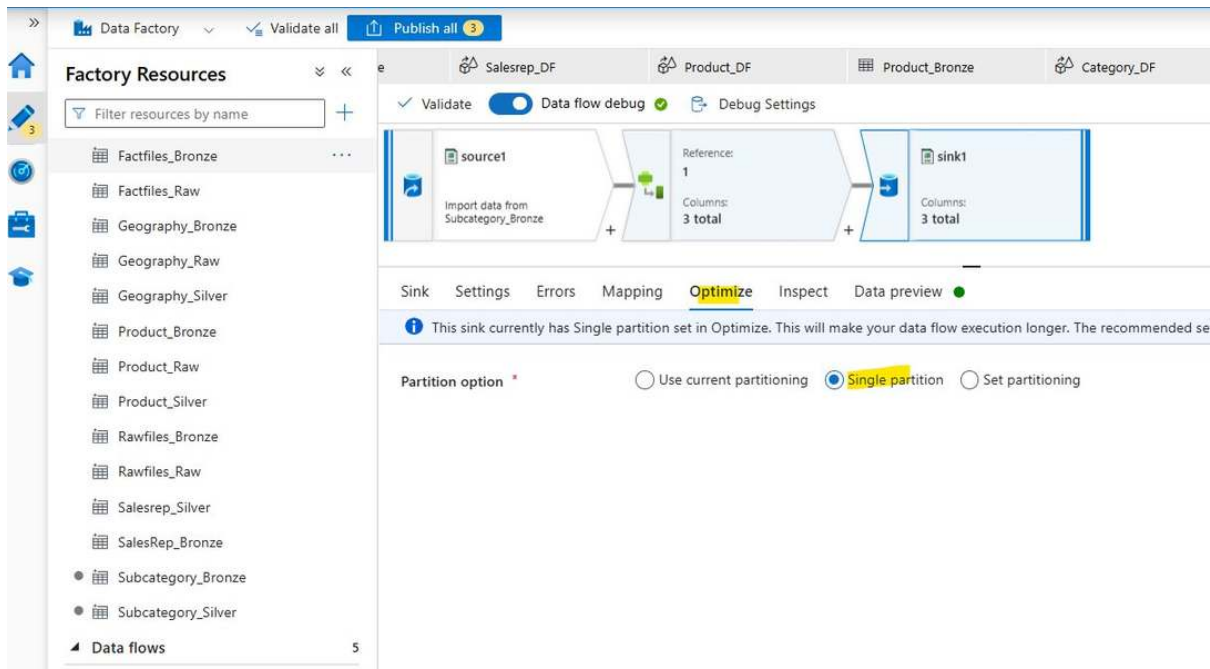
New

Skip line count

Options

☒ Allow schema drift

☐ Validate schema

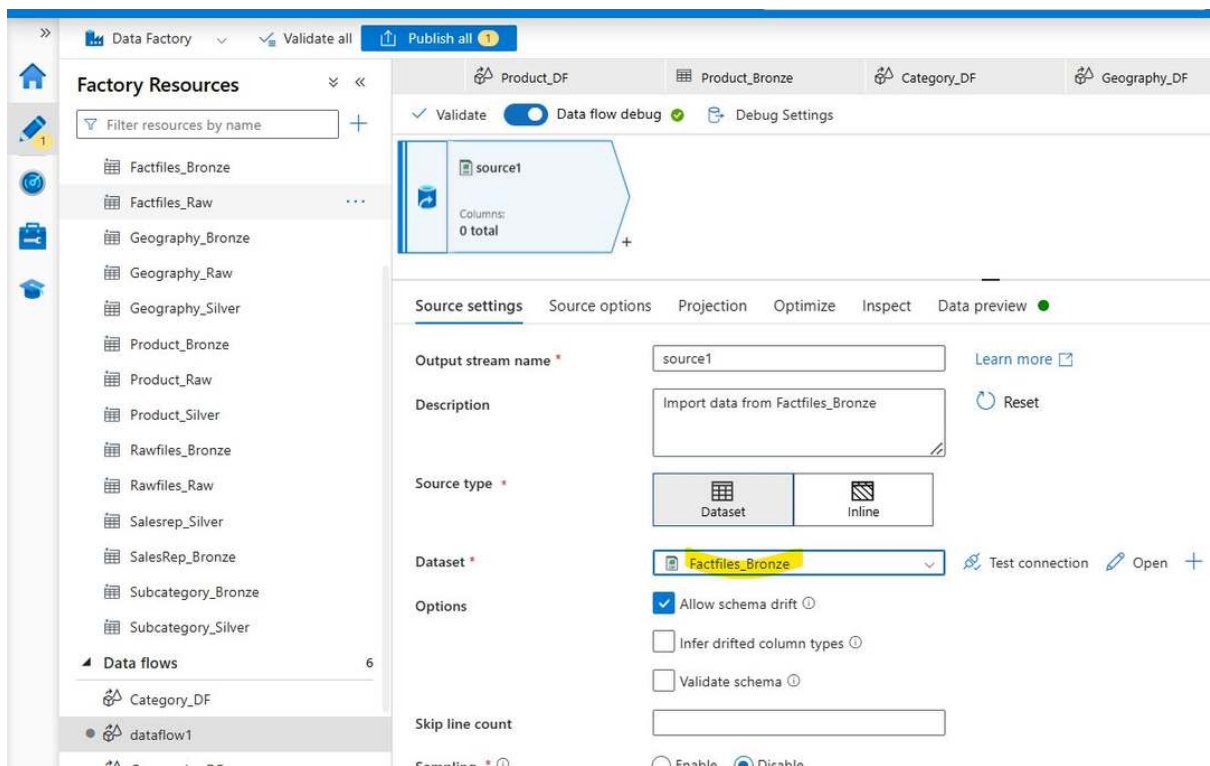


Publish all changes

6.Facttables(sales) dataflow:

Create pipeline

We have all the sales files in factfiles_bronze---all the files are in same format--it will merge automatically



If source is showing 0 columns---projection---import projection

Factory Resources

Filter resources by name

Factfiles_Bronze, Factfiles_Raw, Geography_Bronze, Geography_Raw, Geography_Silver, Product_Bronze, Product_Raw, Product_Silver, Rawfiles_Bronze, Rawfiles_Raw, Salesrep_Silver, SalesRep_Bronze, SalesRep_Silver, Subcategory_Bronze, Subcategory_Silver

Data flows

Category_DF, dataflow1, Geography_DF

Product_DF, Product_Bronze, Category_DF, Geography_DF, Geography_Bronze, Subcategory_DF, dataflow1

Validate, Data flow debug, Debug Settings

source1

Columns: 8 total

Source settings, Source options, Projection, Optimize, Inspect, Data preview

Refresh, Typecast, Modify, Map drifted, Statistics, Remove, Export to CSV

Number of rows: INSERT 100, UPDATE 0, DELETE 0, UPSERT 0, LOOKUP 0, ERROR 0

tsSalesPrimaryKey	ProductID	SalesRepID	Location	Date	Units	PercentOfStandardCost	RevenueDiscount
36943	10	7	Germany:Frankfurt	1.10.2017	60	0.99	0.35
36944	8	2	Denmark:Copenhagen	26.4.2017	71	0.972	0.35
36945	9	6	Germany:Dresden	28.9.2017	68	0.988	0.35
36946	2	5	Germany:Berlin	16.4.2017	157	0.97	0.55
36947	8	2	Denmark:Copenhagen	13.2.2017	103	0.965	0.5
36948	8	1	Czech republic:Prague	3.12.2017	48	0.997	0.25
36949	4	2	Denmark:Copenhagen	26.8.2017	30	0.984	0.25
36950	4	7	Germany:Frankfurt	16.4.2017	57	0.97	0.35
36951	11	4	Germany:Frankfurt	3.11.2017	145	0.994	0.5

Properties

General, Related

Name: dataflow1

Description

Cancel

Here in this data we have country and town same in one column called location

Factory Resources

Filter resources by name

Factfiles_Bronze, Factfiles_Raw, Geography_Bronze, Geography_Raw, Geography_Silver, Product_Bronze, Product_Raw, Product_Silver, Rawfiles_Bronze, Rawfiles_Raw, Salesrep_Silver, SalesRep_Bronze, SalesRep_Silver, Subcategory_Bronze, Subcategory_Silver

Data flows

Category_DF, Sales_DF, Geography_DF

Product_DF, Product_Bronze, Category_DF, Geography_DF, Geography_Bronze, Subcategory_DF, Sales_DF

Validate, Data flow debug, Debug Settings

source1

Import data from Factfiles_Bronze

Columns: 10 total

Derived column's settings, Optimize, Inspect, Data preview

Output stream name: derivedColumn1

Description: Creating/updating the columns 'SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, PercentOfStandardCost, RevenueDiscount'

Incoming stream: source1

Columns

Column	Expression
Country	split(Location,':')[1]
Town	split(Location,':')[2]

Properties

General, Related

Name: Sales_DF

Description

Cancel

Preview data

Factory Resources

Filter resources by name

Factfiles_Bronze, Factfiles_Raw, Geography_Bronze, Geography_Raw, Geography_Silver, Product_Bronze, Product_Raw, Product_Silver, Rawfiles_Bronze, Rawfiles_Raw, Salesrep_Silver, SalesRep_Bronze, SalesRep_Silver, Subcategory_Bronze, Subcategory_Silver

Data flows

Category_DF, Sales_DF, Geography_DF

Product_DF, Product_Bronze, Category_DF, Geography_DF, Geography_Bronze, Subcategory_DF, Sales_DF

Validate, Data flow debug, Debug Settings

source1

Import data from Factfiles_Bronze

Columns: 10 total

Derived column's settings, Optimize, Inspect, Data preview

Number of rows: DELETE 0, UPSERT 0, LOOKUP 0, ERROR 0, TOTAL 1000

Map drifted, Statistics, Remove, Export to CSV

SalesRepID	Location	Date	Units	PercentOfStandardCost	RevenueDiscount	Country	Town
36943	Germany:Frankfurt	1.10.2017	60	0.99	0.35	Germany	Frankfurt
36944	Denmark:Copenhagen	26.4.2017	71	0.972	0.35	Denmark	Copenhagen
36945	Germany:Dresden	28.9.2017	68	0.988	0.35	Germany	Dresden
36946	Germany:Berlin	16.4.2017	157	0.97	0.55	Germany	Berlin
36947	Denmark:Copenhagen	13.2.2017	103	0.965	0.5	Denmark	Copenhagen
36948	Czech republic:Prague	3.12.2017	48	0.997	0.25	Czech republic	Prague
36949	Denmark:Copenhagen	26.8.2017	30	0.984	0.25	Denmark	Copenhagen
36950	Germany:Frankfurt	16.4.2017	57	0.97	0.35	Germany	Frankfurt
36951	Germany:Frankfurt	3.11.2017	145	0.994	0.5	Germany	Frankfurt

Properties

General, Related

Name: Sales_DF

Description

Cancel

Requirement 2: date column format

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- Subcategory_Bronze
- Subcategory_Silver

Data flows

- Category_DF
- Sales_DF
- Geography_DF

Product_DF

Validate Data flow debug Debug Settings

source1 Import data from Factfiles_Bronze

derivedColumn1 Columns: 10 total

Derived column's settings Optimize Inspect Data preview

Output stream name * derivedColumn1 [Learn more](#)

Description Creating/updating the columns 'SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, Units' [Reset](#)

Incoming stream * source1

[Add](#) [Clone](#) [Delete](#) [Open expression builder](#)

Column	Expression
Country	split(Location,')[1]
Town	split(Location,')[2]
Date	toDate(Date,'dd.mm.yyyy')

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- SalesRep_Bronze
- Subcategory_Bronze
- Subcategory_Silver

Data flows

- Category_DF
- Sales_DF
- Geography_DF

Product_DF

Validate Data flow debug Debug Settings

source1 Import data from Factfiles_Bronze

derivedColumn1 Columns: 10 total

Derived column's settings Optimize Inspect Data preview

Number of rows + INSERT 100 + UPDATE 0 + DELETE 0 + UPSERT 0 LOOKUP 0

Refresh Typecast Modify Map drifted Statistics Remove Export to CSV

fSalesPrimaryKey	ProductID	SalesRepID	Location	Date	Units	PercentOfStandardCost
36943	10	7	Germany:Frankfurt	2017-01-01	60	0.99
36944	8	2	Denmark:Copenhagen	2017-01-26	71	0.972
36945	9	6	Germany:Dresden	2017-01-28	68	0.988
36946	2	5	Germany:Berlin	2017-01-16	157	0.97
36947	8	2	Denmark:Copenhagen	2017-01-13	103	0.965
36948	8	1	Czech republic:Prague	2017-01-03	48	0.997
36949	4	2	Denmark:Copenhagen	2017-01-26	30	0.984
36950	4	7	Germany:Frankfurt	2017-01-16	57	0.97
36951	11	4	Germany:Frankfurt	2017-01-03	145	0.994

Let's remove unnecessary column like location

Factory Resources

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- Salesrep_Bronze
- Subcategory_Bronze
- Subcategory_Silver

Data flows

- Category_DF
- Sales_DF**
- Geography_DF

Product_DF

Validate ☒ Data flow debug ☒ Debug Settings

source1: Import data from Factfiles_Bronze

derivedColumn1: Creating/updating the columns: SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, PercentOfStandardCost.

select1: Columns: 9 total

sink1: Add sink dataset

Select settings Optimize Inspect **Data preview**

UPDATE 0 DELETE 0 UPSERT 0 LOOKUP 0 ERROR 0 TOTAL 1000

Modify Map drifted Statistics Remove Export to CSV

ProductID	SalesRepID	Date	Units	PercentOfStandardCost	RevenueDiscount	Country	Town
10	7	2017-01-01	60	0.99	0.35	Germany	Frankfurt
8	2	2017-01-26	71	0.972	0.35	Denmark	Copenhagen
9	6	2017-01-28	68	0.988	0.35	Germany	Dresden
2	5	2017-01-16	157	0.97	0.55	Germany	Berlin
8	2	2017-01-13	103	0.965	0.5	Denmark	Copenhagen
8	1	2017-01-03	48	0.997	0.25	Czech republic	Prague
4	2	2017-01-26	30	0.984	0.25	Denmark	Copenhagen
4	7	2017-01-16	57	0.97	0.35	Germany	Frankfurt
11	4	2017-01-03	145	0.994	0.5	Germany	Frankfurt

Sink the data

Factory Resources

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Product_Bronze
- Product_Raw
- Product_Silver
- Rawfiles_Bronze
- Rawfiles_Raw
- Salesrep_Silver
- Salesrep_Bronze
- Subcategory_Bronze
- Subcategory_Silver

Data flows

- Category_DF
- Sales_DF**
- Geography_DF

Product_DF

Validate ☒ Data flow debug ☒ Debug Settings

source1: Import data from Factfiles_Bronze

Reference: 1 Columns: 11 total

sink1: Columns: 11 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name: sink1 [Learn more](#)

Description: Add sink dataset [Reset](#)

Incoming stream: derivedColumn1

Sink type: Dataset Inline Cache

Dataset: Select... [+ New](#)

Options: ☒ Allow schema drift ☐ Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All Azure Database File Generic protocol

Amazon S3 Azure Blob Storage Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto) Azure Data Lake Storage Gen2 Azure Database for MySQL

[Continue](#) [Cancel](#)

Select format

Choose the format type of your data



Avro



DelimitedText



JSON



ORC



Parquet



Binary

Set properties

Name

Sales_Silver

Linked service *

ADLS2adf_Is

File path

denis

/ Medallian/Silver

/ File name

First row as header



Import schema

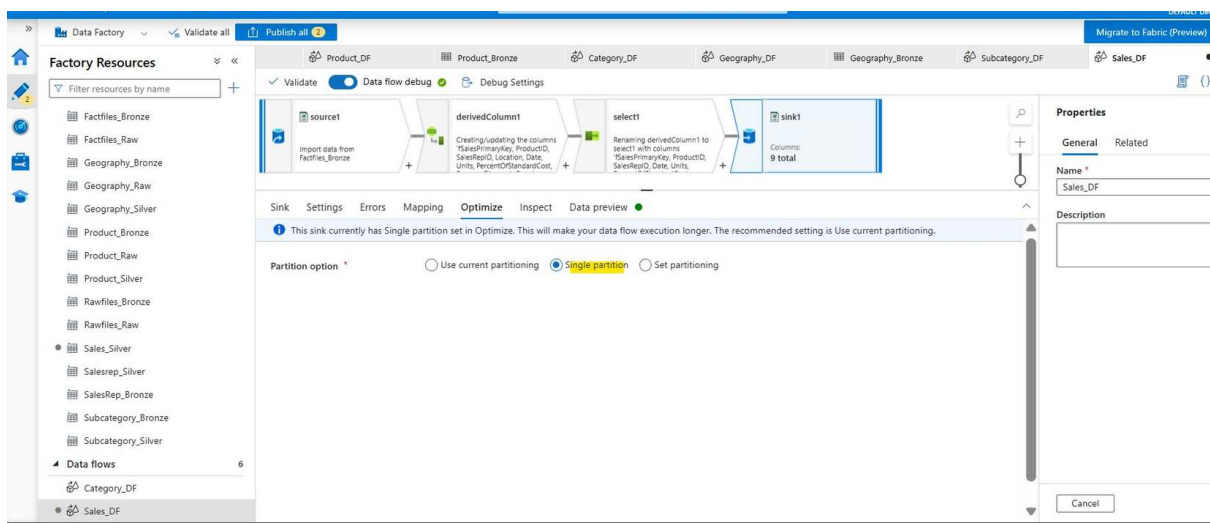
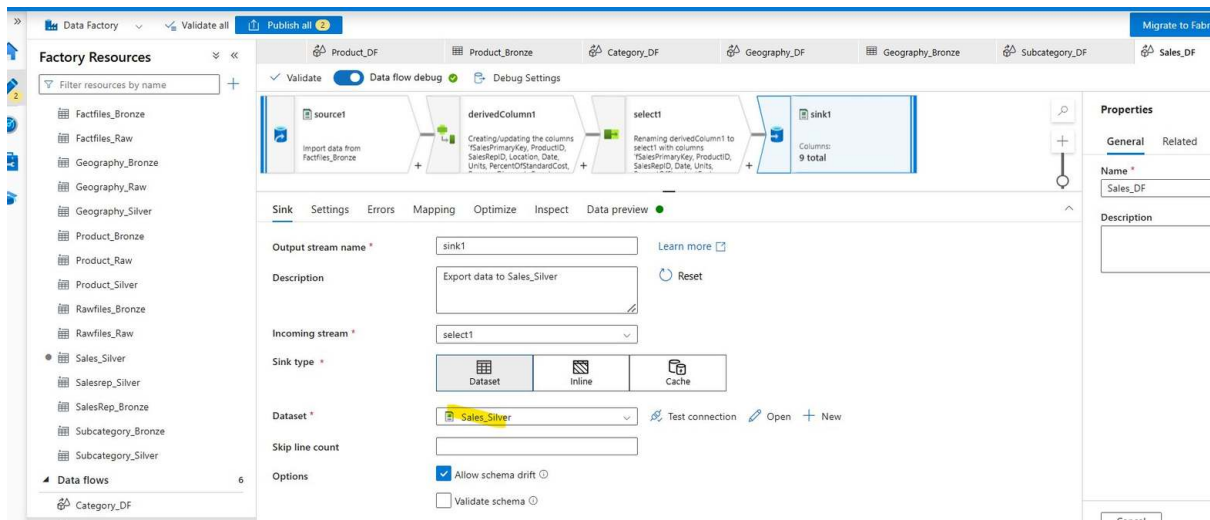
☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

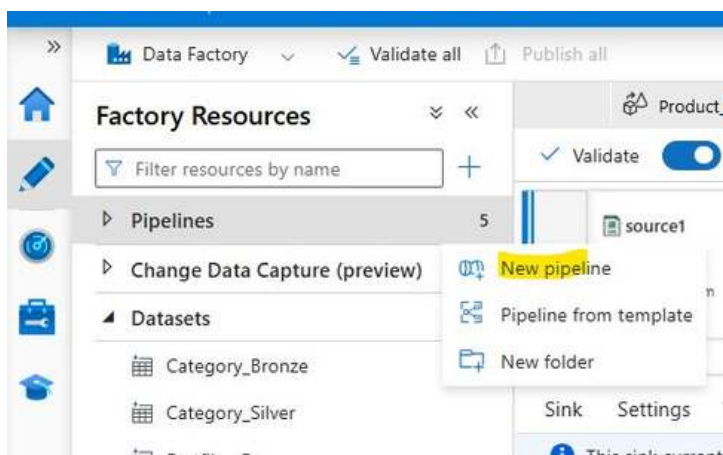
Back

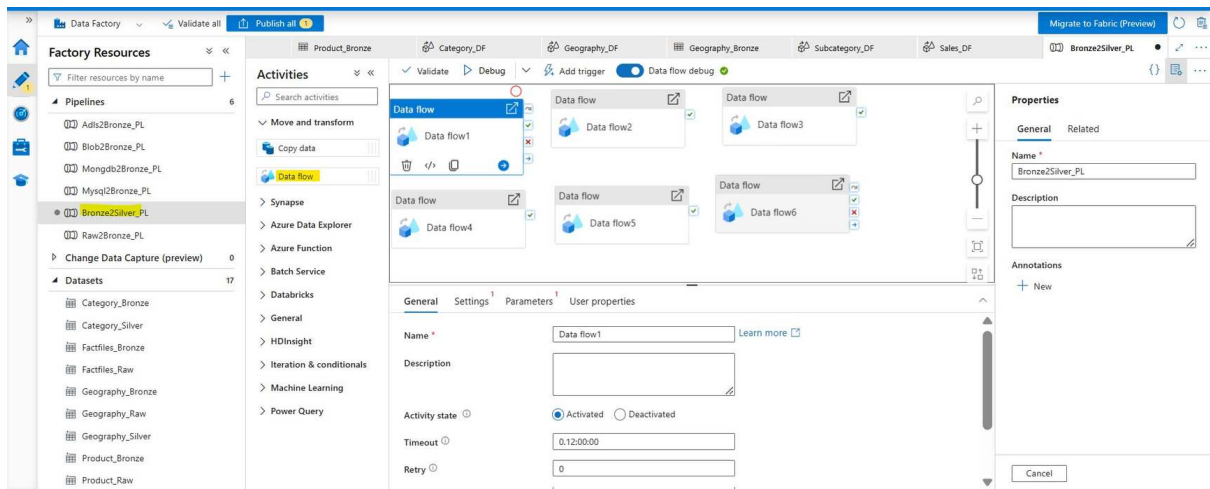
Cancel



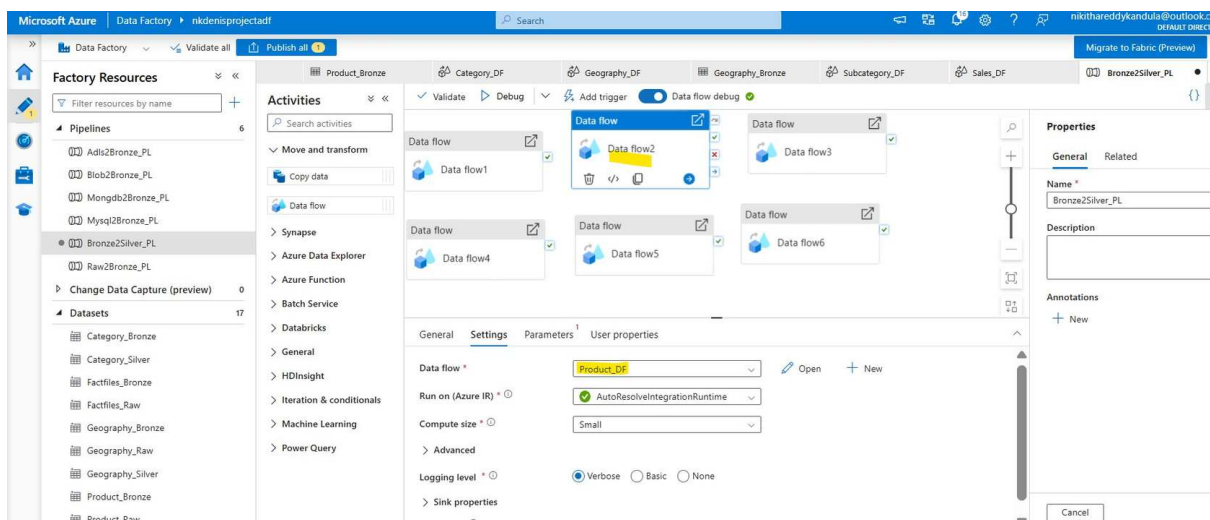
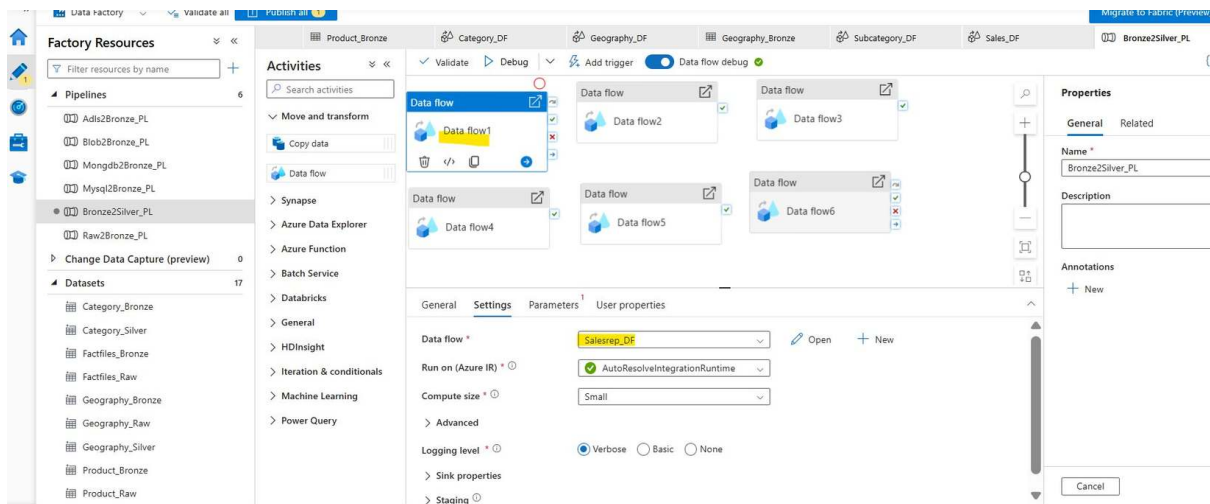
Publish all changes

Create a pipeline and run all the dataflows





For every dataflow give one dataflow



Microsoft Azure Data Factory interface showing the configuration of a Data flow activity.

Factory Resources:

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Bronze2Silver_PL (selected)
 - Raw2Bronze_PL
- Datasets: 17
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Geography_Silver
 - Product_Bronze
 - Product_Raw

Activities:

- Move and transform
 - Copy data
 - Data flow (selected)
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
 - General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

Data flow configuration (Data flow3):

- General
 - Data flow: Category_DF
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose
- Settings
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose

Properties:

- Name: Bronze2Silver_PL
- Description:

Microsoft Azure Data Factory interface showing the configuration of a Data flow activity.

Factory Resources:

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Bronze2Silver_PL (selected)
 - Raw2Bronze_PL
- Datasets: 17
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Geography_Silver
 - Product_Bronze
 - Product_Raw

Activities:

- Move and transform
 - Copy data
 - Data flow (selected)
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
 - General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

Data flow configuration (Data flow4):

- General
 - Data flow: Geography_DF
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose
- Settings
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose

Properties:

- Name: Bronze2Silver_PL
- Description:

Microsoft Azure Data Factory interface showing the configuration of a Data flow activity.

Factory Resources:

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Bronze2Silver_PL (selected)
 - Raw2Bronze_PL
- Datasets: 17
 - Category_Bronze
 - Category_Silver
 - Factfiles_Bronze
 - Factfiles_Raw
 - Geography_Bronze
 - Geography_Raw
 - Geography_Silver
 - Product_Bronze
 - Product_Raw

Activities:

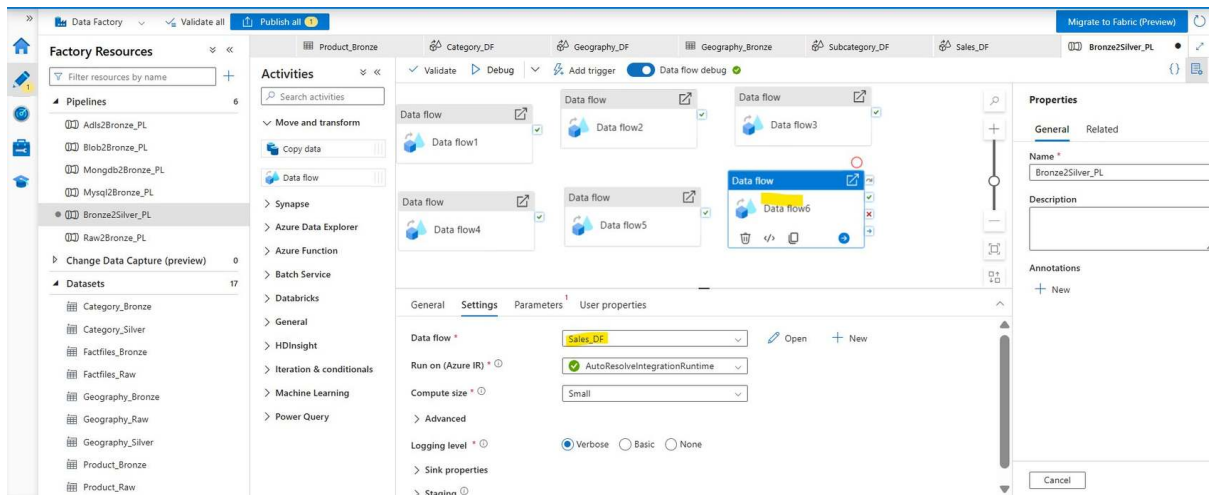
- Move and transform
 - Copy data
 - Data flow (selected)
- Synapse
 - Azure Data Explorer
 - Azure Function
 - Batch Service
 - Databricks
 - General
 - HDInsight
 - Iteration & conditionals
 - Machine Learning
 - Power Query

Data flow configuration (Data flow5):

- General
 - Data flow: Subcategory_DF
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose
- Settings
 - Run on (Azure IR): AutoResolveIntegrationRuntime
 - Compute size: Small
 - Logging level: Verbose

Properties:

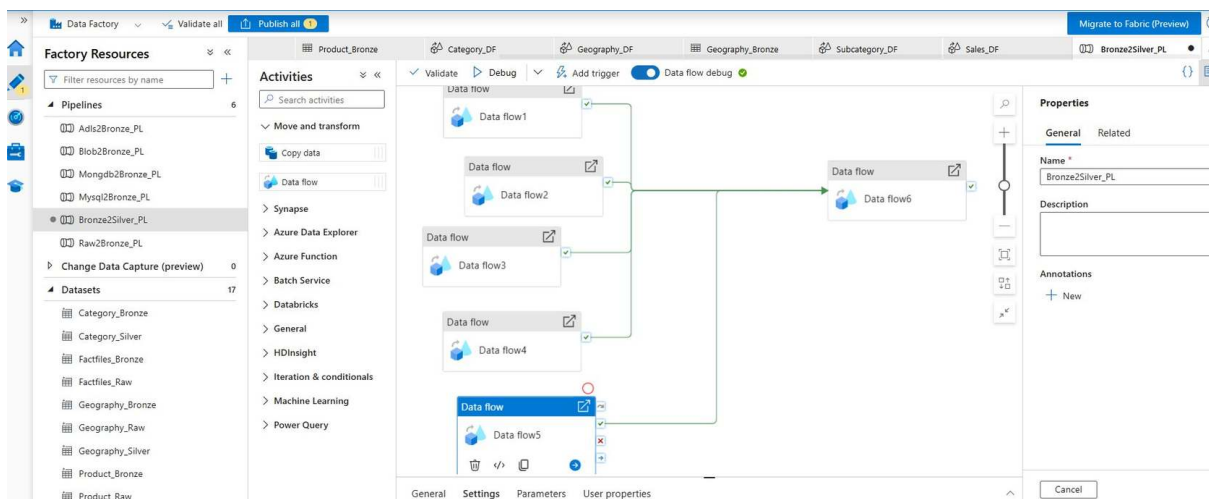
- Name: Bronze2Silver_PL
- Description:



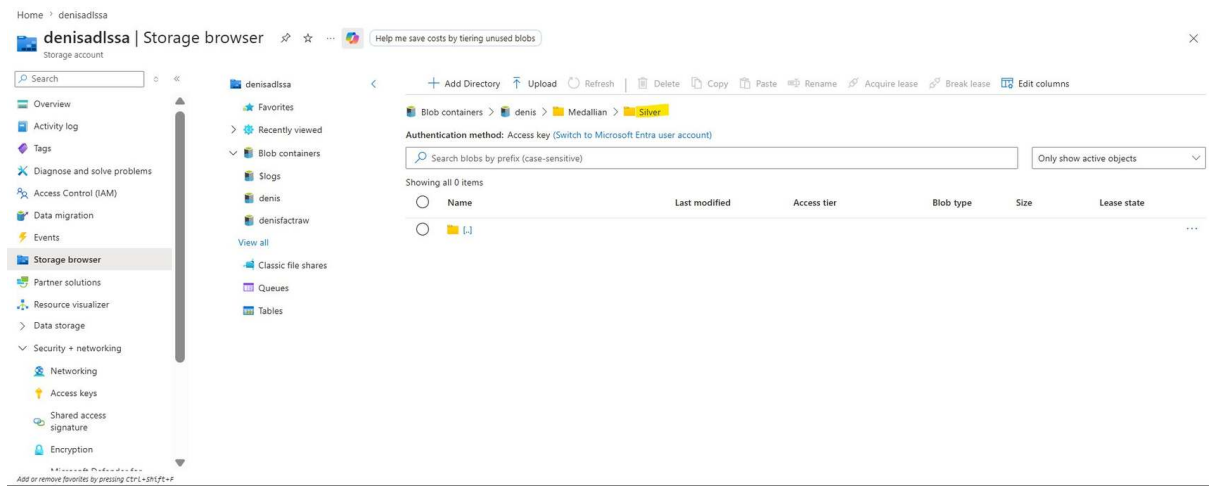
Publish all changes

*******Note:**

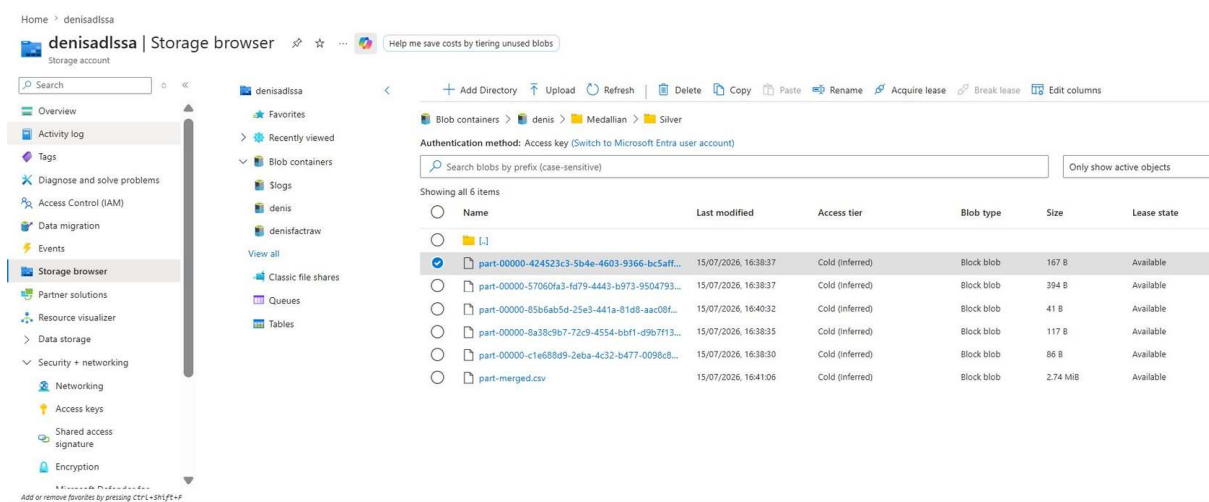
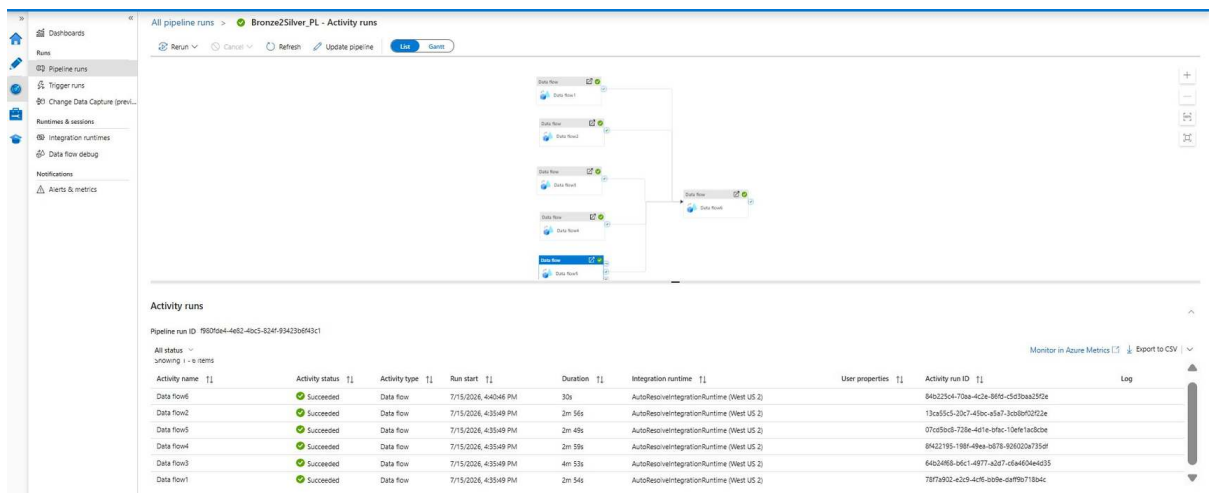
First all the dimtables dataflow should run then only facttable dataflow should execute



Before running the pipeline let's check the silver layer is empty or not



Publish all---add trigger---trigger now



Successfully transformed the data into the silver layer

*****Iam not happy with the file name---it is randomly giving some orbitaly format---let's fix this.

---Every data flow in the sink---it is taking the format as default

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 17
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Category_DF

Validate Data flow debug Debug Settings

source1 Import data from Category_Bronze

sink1 Columns: 2 total

Sink Settings Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is...

Clear the folder ☐

File name option * Default

Quote All ☐

Headers

Umask ☐ R ☐ W ☐ X
☐ R ☒ W ☐ X
☐ R ☒ W ☐ X

Octal 022

Pre/post commands

Let's change to pattern

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 17
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Category_DF

Validate Data flow debug Debug Settings

source1 Import data from Category_Bronze

sink1 Columns: 2 total

Sink Settings Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is...

Clear the folder ☐

File name option * Pattern

Pattern *

Quote All ☐

Headers

Umask ☐ R ☐ W ☐ X
☐ R ☒ W ☐ X
☐ R ☒ W ☐ X

Octal 022

For Geography we need to change

» Data Factory Validate all Publish all

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 17
- Data flows: 6
 - Category_DF
 - Geography_DF**
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Category_DF • **Geography_DF** •

✓ Validate Data flow debug Debug Settings

source1 Import data from Geography_Bronze + select1 Renaming source1 to select1 with columns 'Country, Town' + surrogateKey1 Adding new key Geokey starting from 1 with step 1 + sink1 Columns: 3 total

Sink **Settings** Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is Use current partitioning.

Clear the folder ☐

File name option *

Pattern *

Quote All ☐

Headers ANY

Umask ☐

Owner ☐ R ☐ W ☐ X
Group ☐ R ☒ W ☐ X
Others ☐ R ☒ W ☐ X
Octal

Pre/post commands

» Data Factory Validate all Publish all Migrate to

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 17
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF**
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Category_DF • **Geography_DF** • **Product_DF** •

✓ Validate Data flow debug Debug Settings

source1 Import data from Product_Bronze + window1 Aggregates data based on a window and joins with original data + filter1 Filtering rows using expressions on columns 'SN' + select1 Renaming filter1 to select1 with columns ProductID, SubCategory Key, Color, ProductName, RetailPrice. + sink1 Columns: 6 total

Sink **Settings** Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is Use current partitioning.

Clear the folder ☐

File name option *

Pattern *

Quote All ☐

Headers ANY

Umask ☐

Owner ☐ R ☐ W ☐ X
Group ☐ R ☒ W ☐ X
Others ☐ R ☒ W ☐ X
Octal

Pre/post commands

File pre command

» Data Factory Validate all Publish all

Factory Resources

- Pipelines 6
- Change Data Capture (preview) 0
- Datasets 17
- Data flows 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
 - Power Query 0

Category_DF

Validate Data flow debug Debug Settings

source1 Import data from Factfiles_Bronze

derivedColumn1 Creating/updating the columns 'SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, PercentOfStandardCost,

select1 Renaming derivedColumn1 to select1 with columns 'SalesPrimaryKey, ProductID, SalesRepID, Date, Units,

sink1 Columns: 9 total

Sink Settings Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is Use current partitioning.

Clear the folder ☐

File name option * Pattern

Pattern * Sales_Silver.csv

Quote All ☐

Headers Enter expression... ANY

Umask

Owner ☐ R ☐ W ☐ X

Group ☐ R ☒ W ☐ X

Others ☐ R ☒ W ☐ X

Octal 022

Pre/post commands

» Data Factory Validate all Publish all Migrate to Fabric (Preview)

Factory Resources

- Pipelines 6
- Change Data Capture (preview) 0
- Datasets 17
- Data flows 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
 - Power Query 0

Category_DF

Validate Data flow debug Debug Settings

InputSourceData Import data from Salesrep_Bronze

derivedColumn1 Creating/updating the columns 'SalesRepID, Sales Rep Name,

sink1 Columns: 2 total

Sink Settings Errors Mapping Optimize Inspect Data preview

This sink currently has Single partition set in Optimize. This will make your data flow execution longer. The recommended setting is Use current partitioning.

Clear the folder ☐

File name option * Pattern

Pattern * Salesrep_Silver.csv

Quote All ☐

Headers Enter expression... ANY

Umask

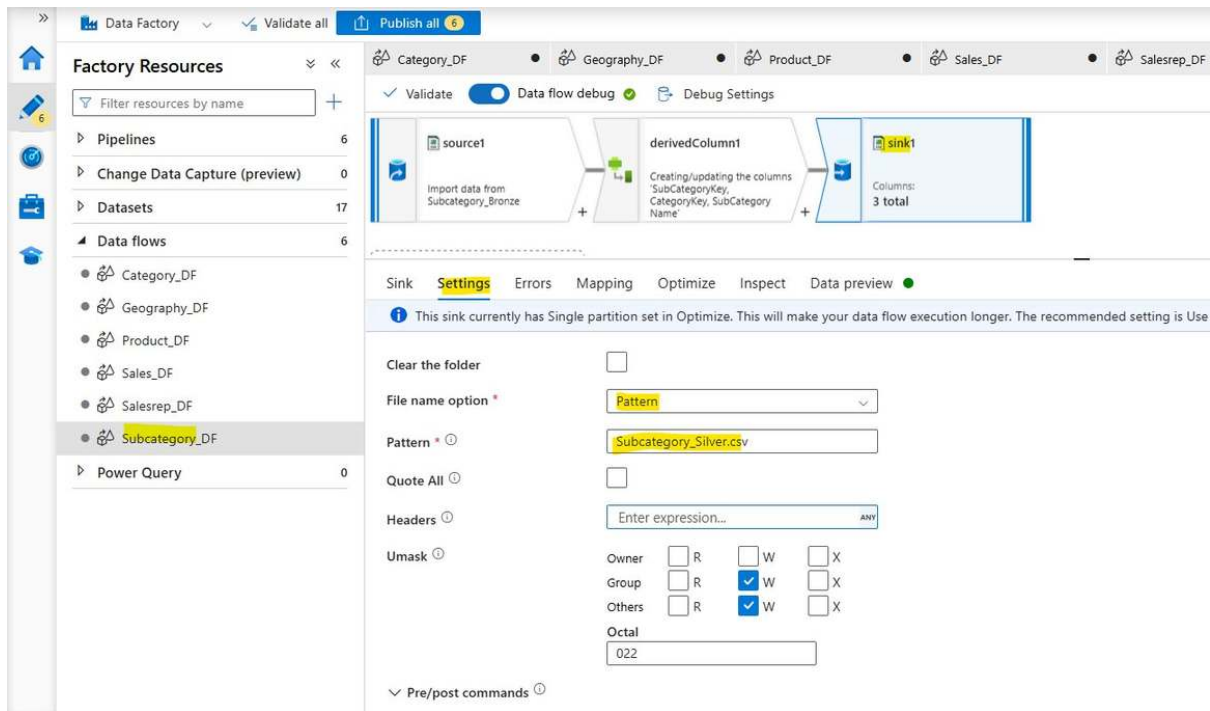
Owner ☐ R ☐ W ☐ X

Group ☐ R ☒ W ☐ X

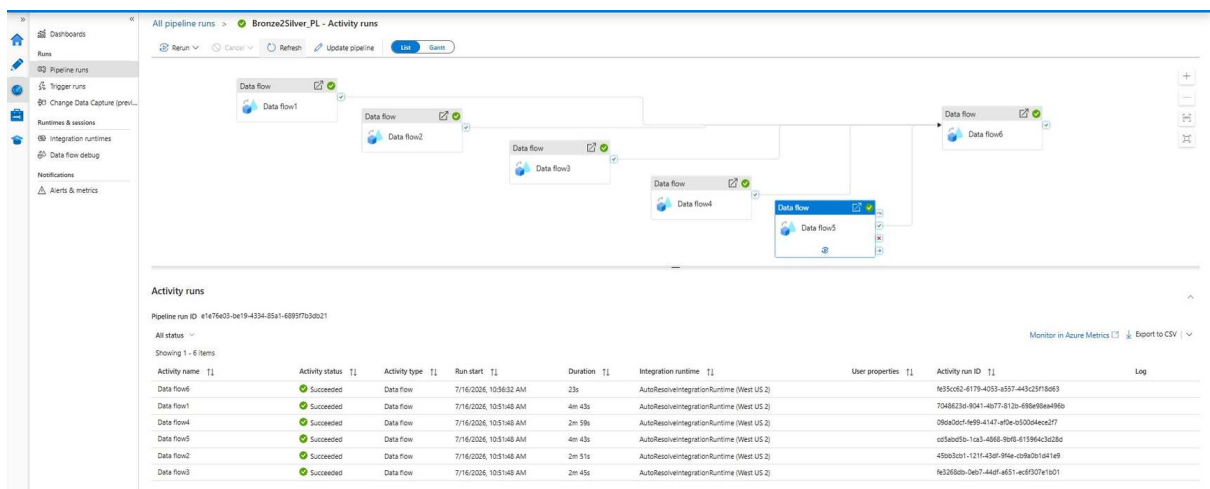
Others ☐ R ☒ W ☐ X

Octal 022

Pre/post commands

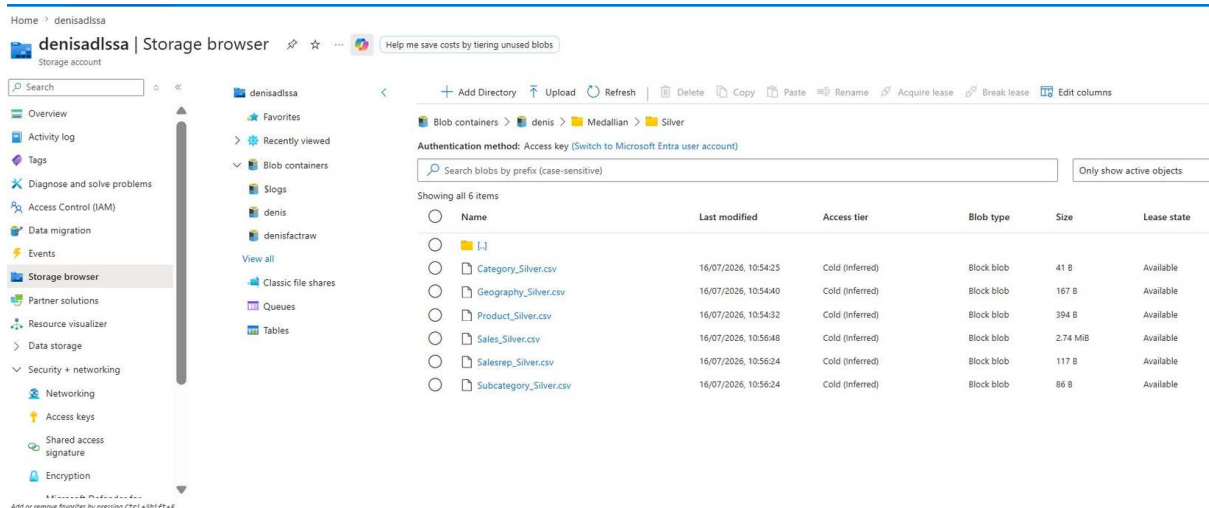


**Publish all the changes----select --Bronze2Silver_PL--add trigger----
triggernow—run pipeline**



Successfully pipeline has executed

Now let's check in the silver layer



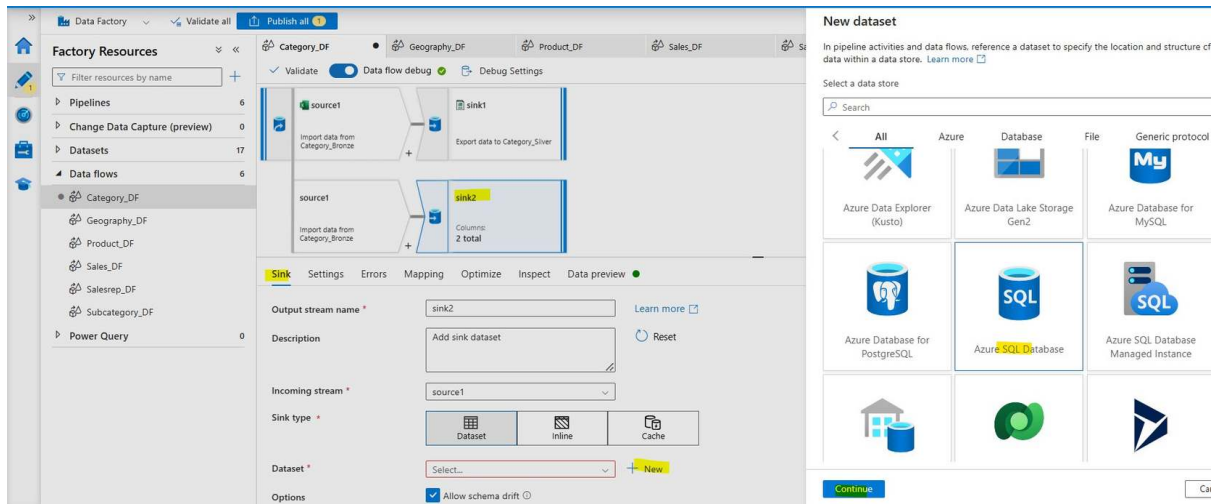
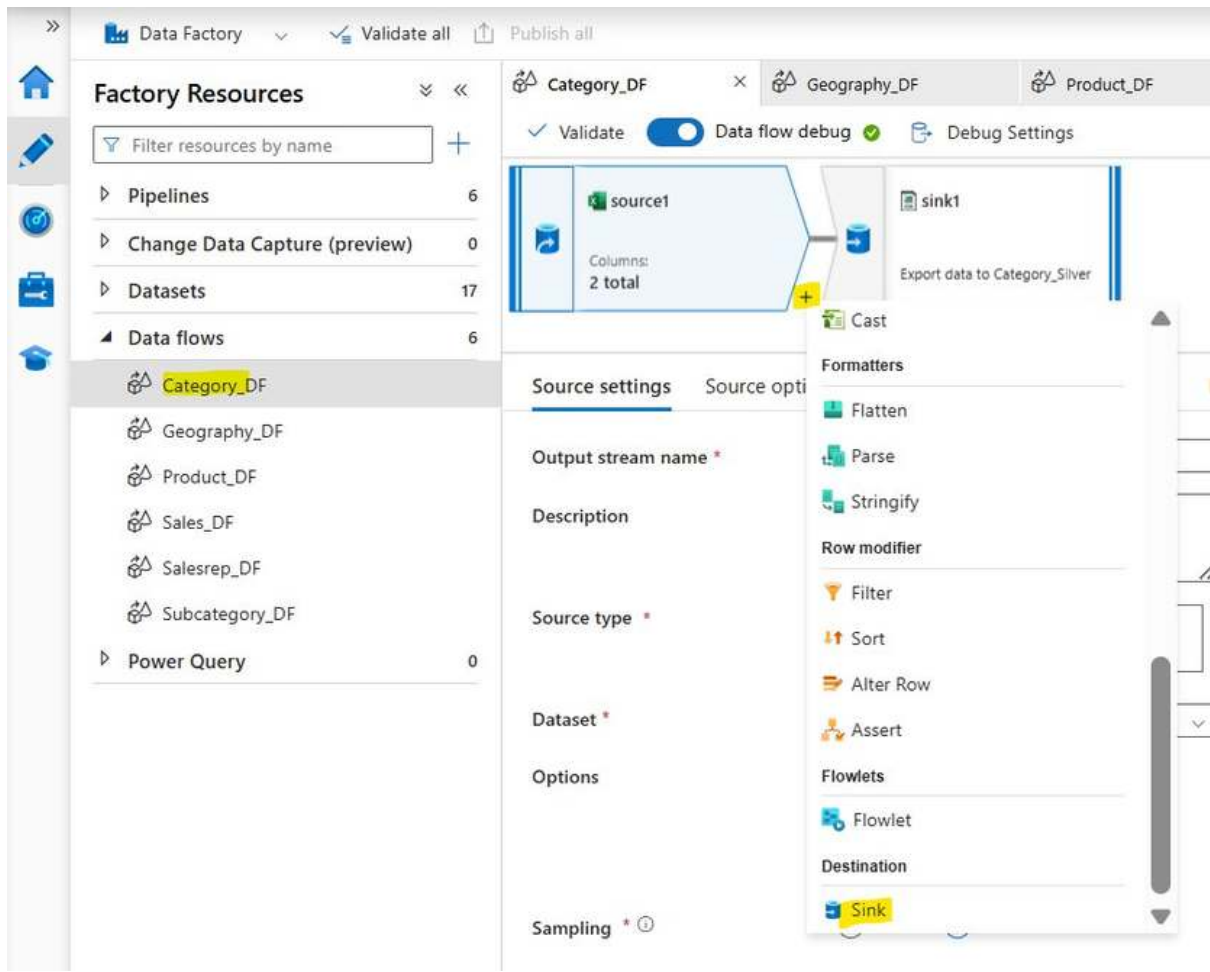
Now the format is fixed as per the requirement

Requirement: Normalized table in the database

if client don't want the file based and need as the normalized table format ---already we have the normalized file format in the silver ---then we need to dump into the database as table format for OLTP

Implementation:

Go to dataflow---add another sink and dump into the target(db)



Set properties

Name

Linked service *

☐ Select from existing table ☒ New table

Schema and table name
Schema name

> Advanced

Data Factory | Validate all | Publish all 2

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 18
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Category_DF | Geography_DF | Product_DF | Sales_DF | Salesrep_DF

✓ Validate | Data flow debug | Debug Settings

source1: Import data from Category_Bronze

sink1: Export data to Category_Silver

source1: Import data from Category_Bronze

sink2: Columns: 2 total

Sink | Settings | Errors | Mapping | Optimize | Inspect | Data preview

Output stream name *: sink2 [Learn more](#)

Description: Export data to Category_SS [Reset](#)

Incoming stream *: source1

Sink type *: ☒ Dataset ☐ Inline ☐ Cache

Dataset *: Category_SS [Test connection](#) [Open](#) [New](#)

Options: ☒ Allow schema drift

For Geography:

Data Factory | Validate all | Publish all 2

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 18
- Data flows: 6
 - Category_DF
 - Geography_DF**
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Geography_DF

source1: Import data from Geography_Bronze

select1: Renaming source1 to select1 with columns 'Country, Town'

surrogateKey1: Columns: 3 total

sink1: Export data to Geography Silver

Surrogate key settings | Optimize | Inspect | Data preview

Output stream name: surrogateKey1

Description: Adding new key Geokey starting from 1 with step 1

Incoming stream: select1

Key column: Geokey

Start value: 1

Formatters

- Flatten
- Parse
- Stringify

Row modifier

- Filter
- Sort
- Alter Row
- Assert

Flowlets

- Flowlet

Destination

- Sink

Data Factory | Validate all | Publish all 2

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 18
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Geography_DF

source1: Import data from Geography_Bronze

select1: Renaming source1 to select1 with columns 'Country, Town'

surrogateKey1: Adding new key Geokey starting from 1 with step 1

surrogateKey1: Adding new key Geokey starting from 1 with step 1

sink1: Export data to Geography Silver

sink2: Columns: 3 total

Sink | Settings | Errors | Mapping | Optimize | Inspect | Data preview

Output stream name: sink2

Description: Add sink dataset

Incoming stream: surrogateKey1

Sink type: Dataset

Dataset: Select...

Options: ☒ Allow schema drift

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All | Azure | Database | File | Generic protocol

- Azure Data Explorer (Kusto)
- Azure Data Lake Storage Gen2
- Azure Database for MySQL
- Azure Database for PostgreSQL
- Azure SQL Database
- Azure SQL Database Managed Instance

Continue | Cancel

Set properties

Name

Geography_SS

Linked service *

AzureSqlDatabase2adf_ls

☐ Select from existing table ☒ New table

Schema and table name

Schema name

Geography_TB

> Advanced

OK

Back

Cancel

The screenshot shows the 'Set properties' dialog for a sink in Azure Data Factory. The 'Output stream name' is 'sink2'. The 'Description' is 'Export data to Geography_SS'. The 'Incoming stream' is 'surrogateKey1'. The 'Sink type' is 'Dataset'. The 'Dataset' is 'Geography_SS'. The 'Options' section has 'Allow schema drift' checked.

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 19
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Data flow diagram:

- source1: Import data from Geography_Bronze
- select1: Renaming source1 to select1 with columns 'Country, Town'
- surrogateKey1: Adding new key Geokey starting from 1 with step 1
- sink1: Export data to Geography_Silver
- surrogateKey1: Adding new key Geokey starting from 1 with step 1
- sink2: Columns: 3 total

Sink Settings:

- Output stream name: sink2
- Description: Export data to Geography_SS
- Incoming stream: surrogateKey1
- Sink type: Dataset
- Dataset: Geography_SS
- Options: ☒ Allow schema drift

For Product:

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 19
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF**
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Product_DF

source1: Import data from Product_Bronze

window1: Aggregates data based on a window and joins with original data

filter1: Filtering rows using expressions on columns 'RN'

select1: Columns: 6 total

sink1: Export data to Product_Silver

Cast

- Formatters
 - Flatten
 - Parse
 - Stringify
- Row modifier
 - Filter
 - Sort
 - Alter Row
 - Assert
- Flowlets
 - Flowlet
- Destination
 - Sink**

Select settings | Optimize | Inspect | Data preview

Output stream name *: select1 [Learn more](#)

Description: Renaming filter1 to select1 with columns 'ProductID, SubCategory Key, Color, ProductName, RetailPrice, StandardCost'

Incoming stream *: filter1

Options

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 19
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF**
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Product_DF

source1: Import data from Product_Bronze

window1: Aggregates data based on a window and joins with original data

filter1: Filtering rows using expressions on columns 'RN'

select1: Renaming filter1 with columns 'ProductID, SubCategory Key, Color, ProductName, RetailPrice, StandardCost'

select1: Renaming filter1 with columns 'ProductID, SubCategory Key, Color, ProductName, RetailPrice, StandardCost'

Sink | Settings | Errors | Mapping | Optimize | Inspect | Data preview

Output stream name *: sink2 [Learn more](#)

Description: Add sink dataset

Incoming stream *: select1

Sink type *: ☒ Dataset ☐ Inline ☐ Cache

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

Search

All | Azure | Database | File | Generic protocol

- Azure Data Explorer (Kusto)
- Azure Data Lake Storage Gen2
- Azure Database for MySQL
- Azure Database for PostgreSQL
- Azure SQL Database
- Azure SQL Database Managed Instance

[Continue](#) [Cancel](#)

Set properties

Name

Product_SS

Linked service *

AzureSqlDatabase2adf_Is

☐ Select from existing table

☒ New table

Schema and table name

Schema name

Product_TB

> Advanced

OK

Back

Cancel

Data Factory

Validate all

Publish all

Migrate to Fabric (Preview)

Factory Resources

Pipelines

6

Change Data Capture (preview)

0

Datasets

20

Data flows

6

Category_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Subcategory_DF

Power Query

0

Category_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Subcategory_DF

Bronze2Silver_PL

Validate

Data flow debug

Debug Settings

source1

Aggregates data based on a window and joins with original data

filter1

Filtering rows using expressions on columns 'N'

select1

Renaming filter1 to select1 with columns: ProductID, SubCategory Key, Color, ProductName, RetailPrice

sink1

Export data to Product_Silver

select1

Renaming filter1 to select1 with columns: ProductID, SubCategory Key, Color, ProductName, RetailPrice

sink2

Columns: 6 total

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink2

Learn more

Description

Export data to Product_SS

Reset

Incoming stream *

select1

Sink type *

Dataset

Inline

Cache

Dataset *

Product_SS

Test connection

Open

New

Options

Allow schema drift

For Sales

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 20
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Data flow debug

source1: Import data from Factories_Bronze

derivedColumn1: Creating/updating the columns 'SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, PercentOfStandardCost'

select1: Renaming derivedColumn1 to select1 with columns 'SalesPrimaryKey, ProductID, SalesRepID, Date, Units'

sink1: Export data

sink2: Columns: 9 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name: sink2

Description: Add sink dataset

Incoming stream: select1

Sink type: Dataset

Dataset: Select...

Set properties

Name: Sales_SS

Linked service: AzureSqlDatabase2adf_js

Select from existing table ☒ New Table

Schema and table name: Schema name: Sales_TB

Advanced

OK Back Cancel

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 21
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Data flow debug

source1: Import data from Factories_Bronze

derivedColumn1: Creating/updating the columns 'SalesPrimaryKey, ProductID, SalesRepID, Location, Date, Units, PercentOfStandardCost'

select1: Renaming derivedColumn1 to select1 with columns 'SalesPrimaryKey, ProductID, SalesRepID, Date, Units'

sink1: Export data to Sales_Silver

sink2: Columns: 9 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name: sink2

Description: Export data to Sales_SS

Incoming stream: select1

Sink type: Dataset

Dataset: Sales_SS

Test connection Open New

5.Salesrep

Factory Resources

- Pipelines: 6
- Change Data Capture (preview): 0
- Datasets: 21
- Data flows: 6
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Data flow debug

Inputsourcedata: Import data from Salesrep_Bronze

derivedColumn1: Creating/updating the columns 'SalesRepID, Sales Rep Name'

sink1: Export data to Salesrep_Silver

derivedColumn1: Creating/updating the columns 'SalesRepID, Sales Rep Name'

sink2: Columns: 2 total

Sink Settings Errors Mapping Optimize Inspect Data preview

Output stream name: sink2

Description: Add sink dataset

Incoming stream: derivedColumn1

Sink type: Dataset

Dataset: Select...

Options: ☒ Allow schema drift

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more

Select a data store

Search

All Azure Database File Generic protocol

Azure Database for PostgreSQL Azure SQL Database Azure SQL Database Managed Instance

Azure Synapse Analytics Datawarehouse (Common Data Service for Apps) Dynamics 365

Continue Cancel

Set properties

Name

Salesrep_SS

Linked service *

AzureSqlDatabase2adf_Is

☐ Select from existing table ☒ New table

Schema and table name

Schema name

Sales_rep_TB

> Advanced

OK

Back

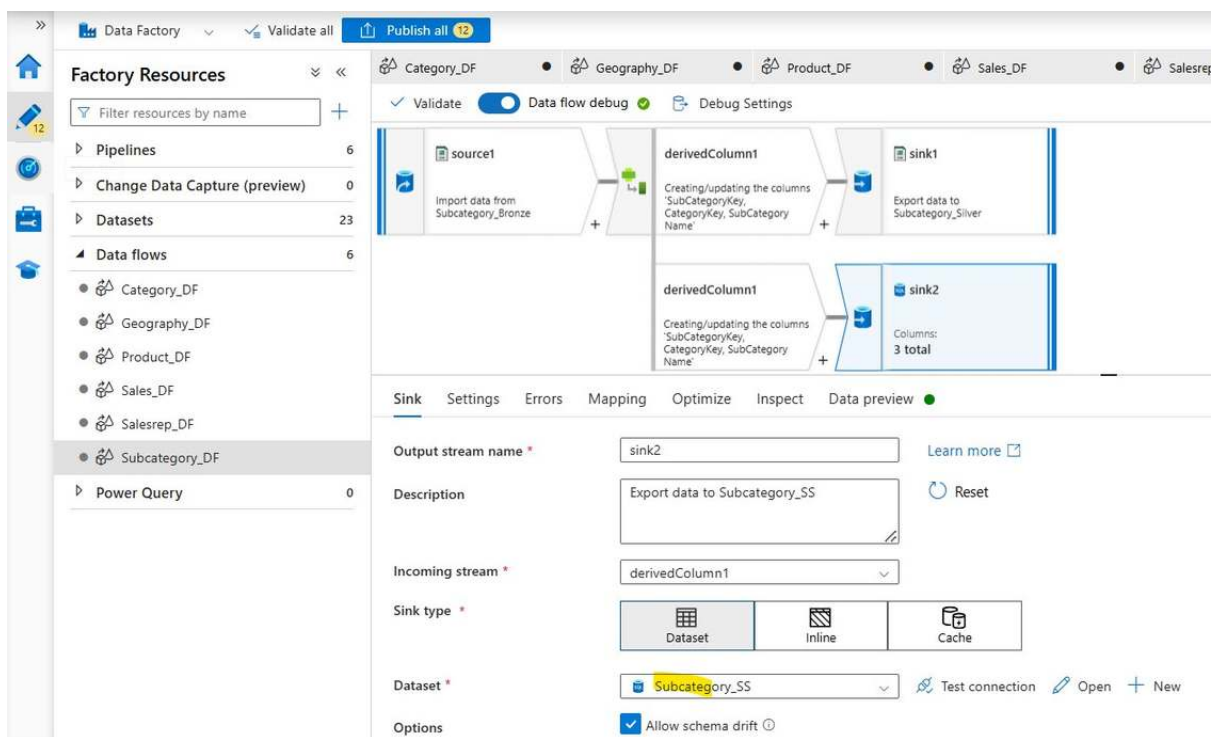
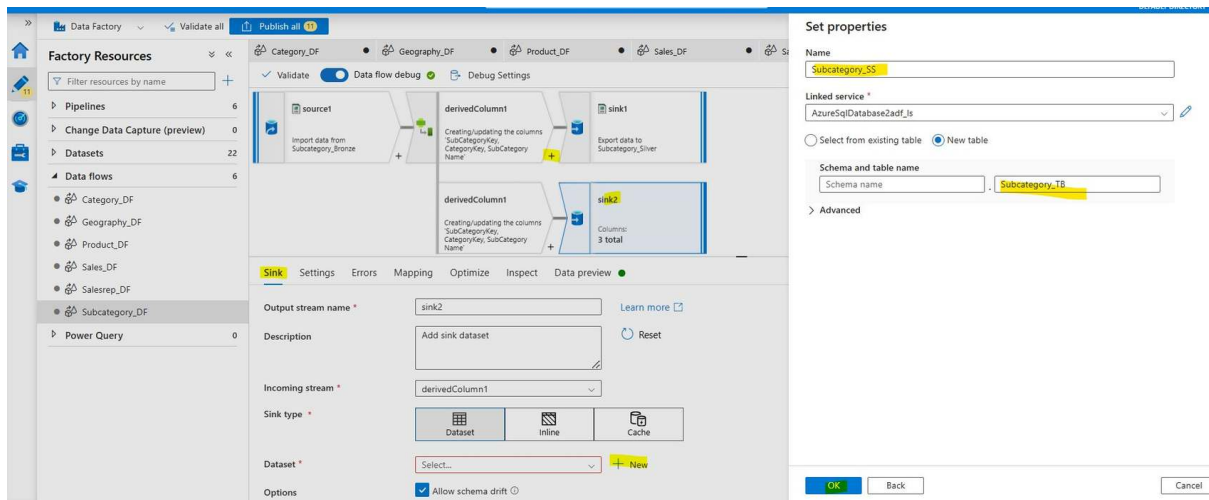
Cancel

The screenshot shows the 'Set properties' dialog for a sink in Azure Data Factory. The 'Factory Resources' pane on the left lists various resources, with 'Salesrep_DF' selected. The main area displays a data flow diagram with two sinks: 'sink1' and 'sink2'. The 'sink2' settings are expanded, showing the following configuration:

- Output stream name ***: sink2
- Description**: Export data to Salesrep_SS
- Incoming stream ***: derivedColumn1
- Sink type ***: Dataset
- Dataset ***: Salesrep_SS
- Options**: ☒ Allow schema drift

Buttons for 'Test connection', 'Open', and 'New' are visible next to the dataset selection.

6.For Subcategory



Publish all the changes in adf

Before running the pipeline ---let's check the Azure sql db---no tables are created

Home > denisazuresqladb (denisazuresqlserver/denisazuresqladb)

denisazuresqladb (denisazuresqlserver/denisazuresqladb) | Query editor (preview) ☆ ...

SQL database

Search

New query Templates Open in Feedback Sign out Classic experience

Overview
Activity log
Tags
Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
Settings
Data management
Integrations
Power Platform
Security
Intelligent performance
Monitoring
Automation
Help

Explorer

Search

denisazuresqladb

dbo

Tables

Views

Stored Procedures

Functions

Queries

Start exp

Write and run SQL queries, create databases

No tables are in the azure sql db

Now in adf—select bronze2silver_DF---add trigger---triggernow—runpipeline

All pipeline runs > Bronze2Silver_PL - Activity runs

Run Cancel Refresh Update pipeline List Gantt

Data flow

Data flow1

Data flow2

Data flow3

Data flow4

Data flow5

Data flow6

Activity runs

Pipeline run ID: 8697460-c0b0-4c93-829f-dc39d0873ee

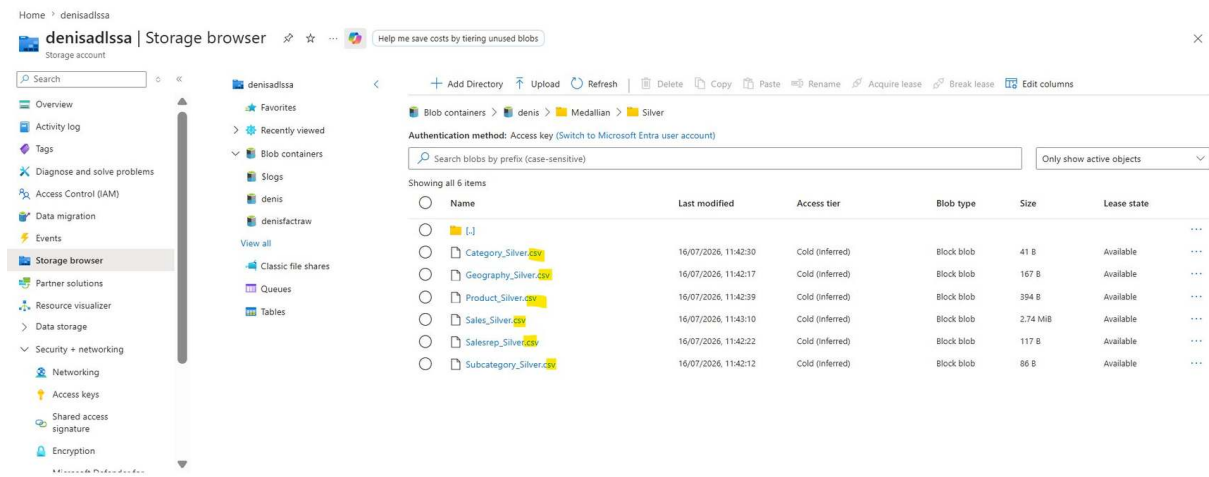
All status

Showing 1 - 6 items

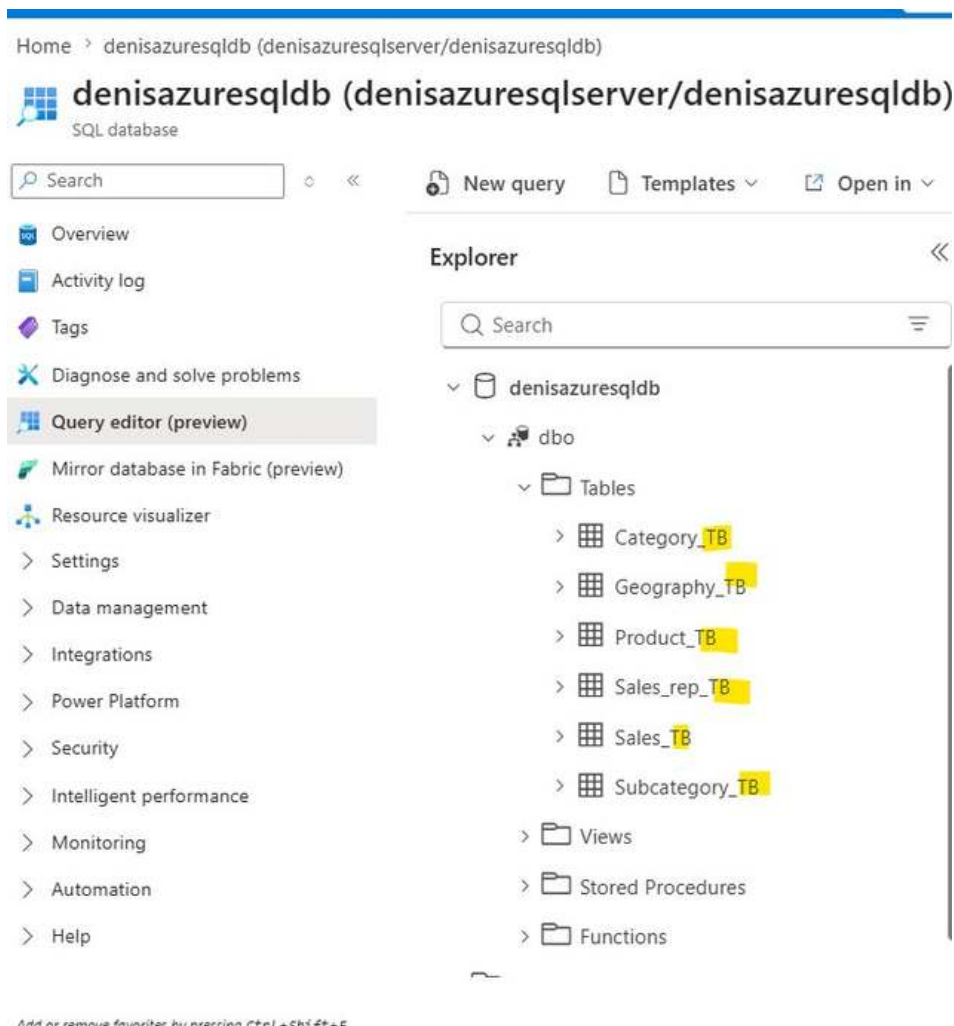
Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
Data flow6	Succeeded	Data flow	7/16/2026, 11:42:49 AM	28s	AutoResolveIntegrationRuntime (West US 2)		4a7854f1-e377-4c46-be57-20c3410b6709	
Data flow4	Succeeded	Data flow	7/16/2026, 11:37:29 AM	4m 55s	AutoResolveIntegrationRuntime (West US 2)		62c0219e-866d-4c91-86db-b1b0850e2b90	
Data flow1	Succeeded	Data flow	7/16/2026, 11:37:29 AM	5m 0s	AutoResolveIntegrationRuntime (West US 2)		4ae83360-0b64-4709-a089-2175340b101c	
Data flow3	Succeeded	Data flow	7/16/2026, 11:37:29 AM	5m 0s	AutoResolveIntegrationRuntime (West US 2)		320c0f13-b114-4135-9e54-a4ed17415318	
Data flow2	Succeeded	Data flow	7/16/2026, 11:37:29 AM	5m 19s	AutoResolveIntegrationRuntime (West US 2)		e9f89910-bc7f-4bdc-9f9b-cf8a3a494310	
Data flow5	Succeeded	Data flow	7/16/2026, 11:37:29 AM	4m 55s	AutoResolveIntegrationRuntime (West US 2)		73b76ca6-c9e5-459f-8404-b7a682e37e84	

Monitor in Azure Metrics

Let's see in the adls silver layer should be file format----in azure sqldb should be in tb format



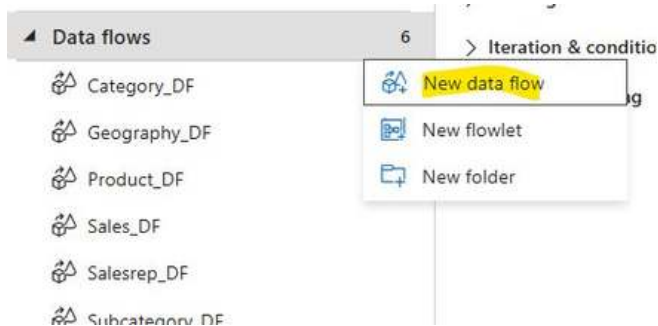
In ads silver layer---succesfully got the file format normalized



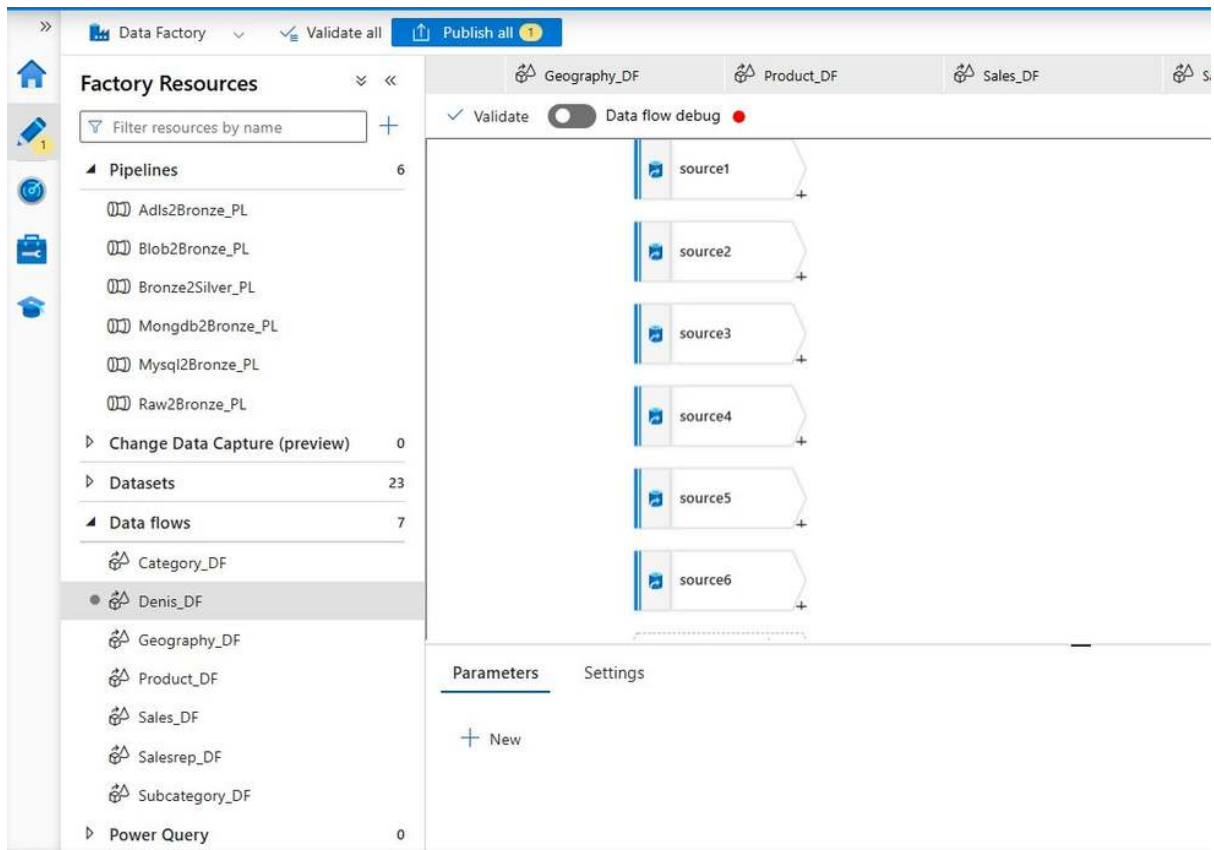
In azure sq db---we dumped successfully as the normalized tables

Requirement: Denormalized tables---we need to dump into the gold layer

Create dataflo:



Add 6 sources:



Data Factory | Validate all | Publish all 1

Factory Resources

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Raw2Bronze_PL
- Change Data Capture (preview): 0
- Datasets: 23
- Data flows: 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Geography_DF | Product_DF | Sales_DF | Salesrep_DF | Subcategory_DF | Bronze2

✓ Validate | Data flow debug

Source settings | Source options | Projection | Optimize | Inspect | Data preview

Output stream name: **source1** | Learn more

Description: Import data from Category_Silver | Reset

Source type: Dataset | Inline

Dataset: **Category_Silver** | Test connection | Open | New

Options:

- ☒ Allow schema drift
- ☐ Infer drifted column types
- ☐ Validate schema

Skip line count:

Data Factory | Validate all | Publish all 1

Factory Resources

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Raw2Bronze_PL
- Change Data Capture (preview): 0
- Datasets: 23
- Data flows: 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Geography_DF | Product_DF | Sales_DF | Salesrep_DF

✓ Validate | Data flow debug

Source settings | Source options | Projection | Optimize | Inspect | Data preview

Output stream name: **source2** | Learn more

Description: Import data from Geography_Silver | Reset

Source type: Dataset | Inline

Dataset: **Geography_Silver** | Test connection | Open | New

Options:

- ☒ Allow schema drift
- ☐ Infer drifted column types
- ☐ Validate schema

Skip line count:

» Data Factory Validate all Publish all 1

Factory Resources

Filter resources by name +

Pipelines 6

- Adls2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview) 0

Datasets 23

Data flows 7

- Category_DF
- Denis_DF
- Geography_DF
- Product_DF
- Sales_DF
- Salesrep_DF
- Subcategory_DF

Power Query 0

Geography_DF Product_DF Sales_DF Salesrep_DF Subcategory_DF

Validate Data flow debug

source1 source2

Source settings Source options Projection Optimize Inspect Data preview

Output stream name * source3 Learn more

Description Import data from Product_Silver Reset

Source type * Dataset Inline

Dataset * Product_Silver Test connection Open + New

Options

- ☒ Allow schema drift
- ☐ Infer drifted column types
- ☐ Validate schema

Skip line count

» Data Factory Validate all Publish all 1

Factory Resources

Filter resources by name +

Pipelines 6

- Adls2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview) 0

Datasets 23

Data flows 7

- Category_DF
- Denis_DF
- Geography_DF
- Product_DF
- Sales_DF
- Salesrep_DF
- Subcategory_DF

Geography_DF Product_DF Sales_DF Salesrep_DF

Validate Data flow debug

source1

Source settings Source options Projection Optimize Inspect Data preview

Output stream name * source4 Learn more

Description Import data from Sales_Silver Reset

Source type * Dataset Inline

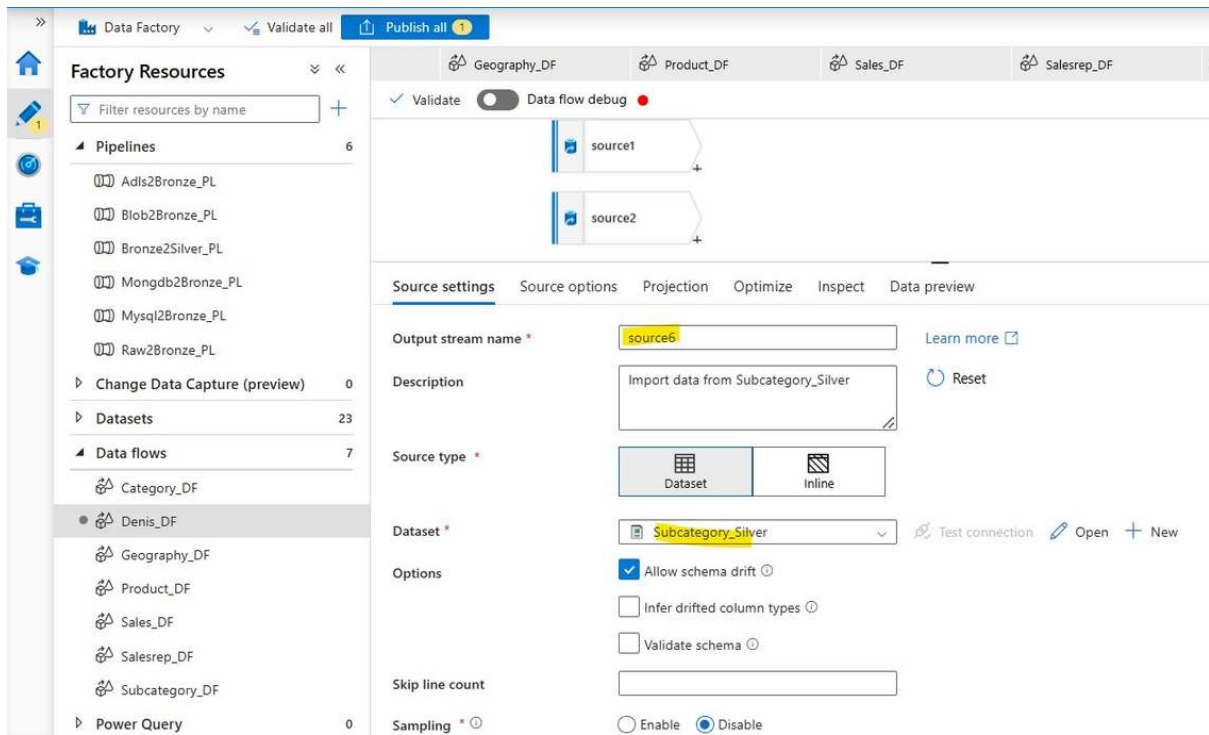
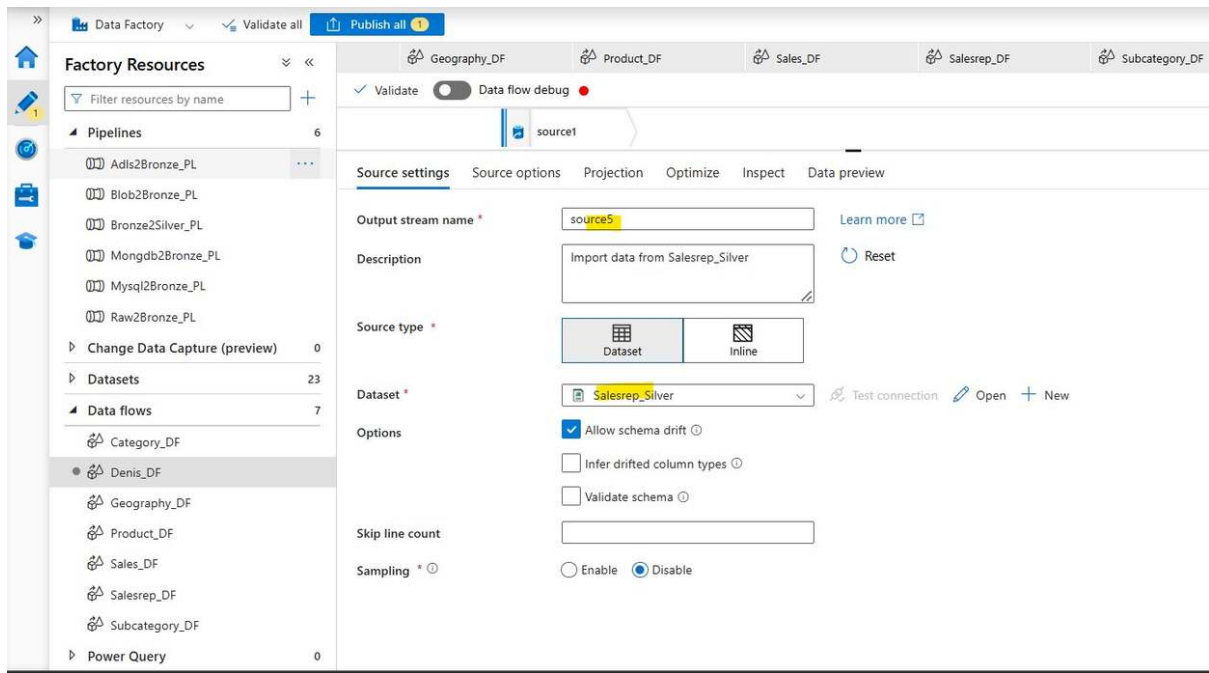
Dataset * Sales_Silver Test connection Open + New

Options

- ☒ Allow schema drift
- ☐ Infer drifted column types
- ☐ Validate schema

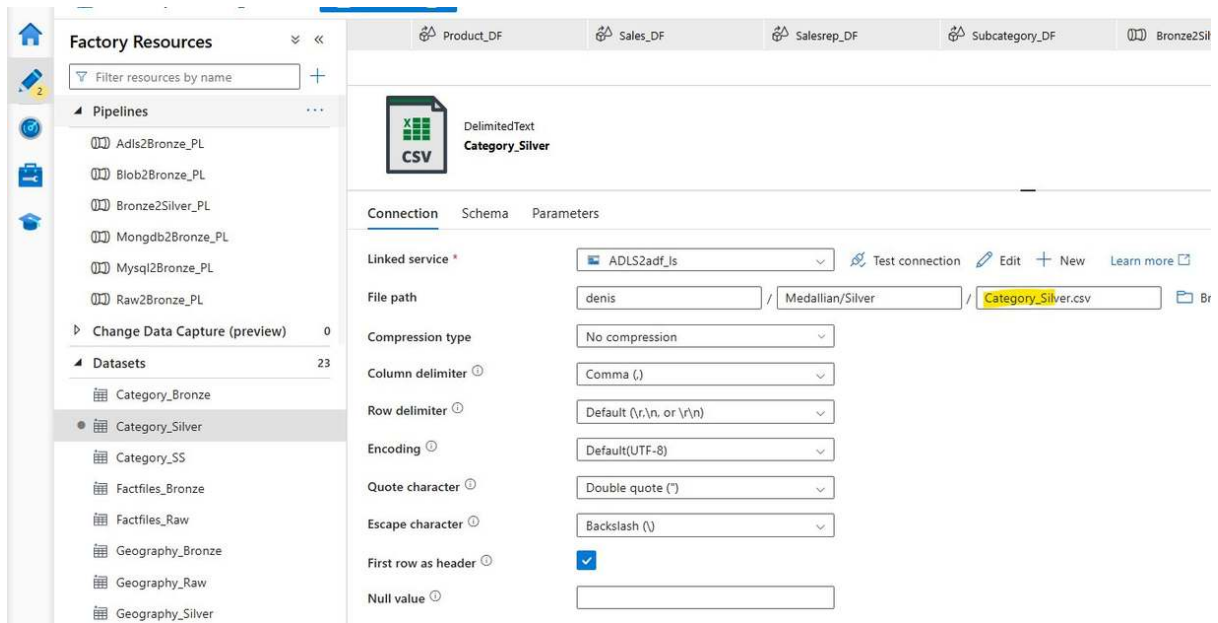
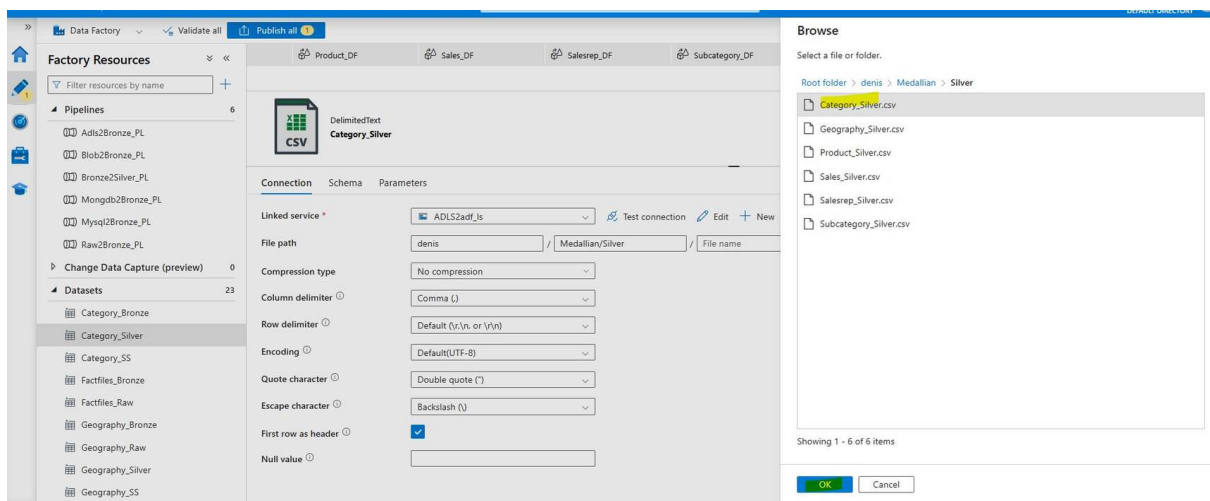
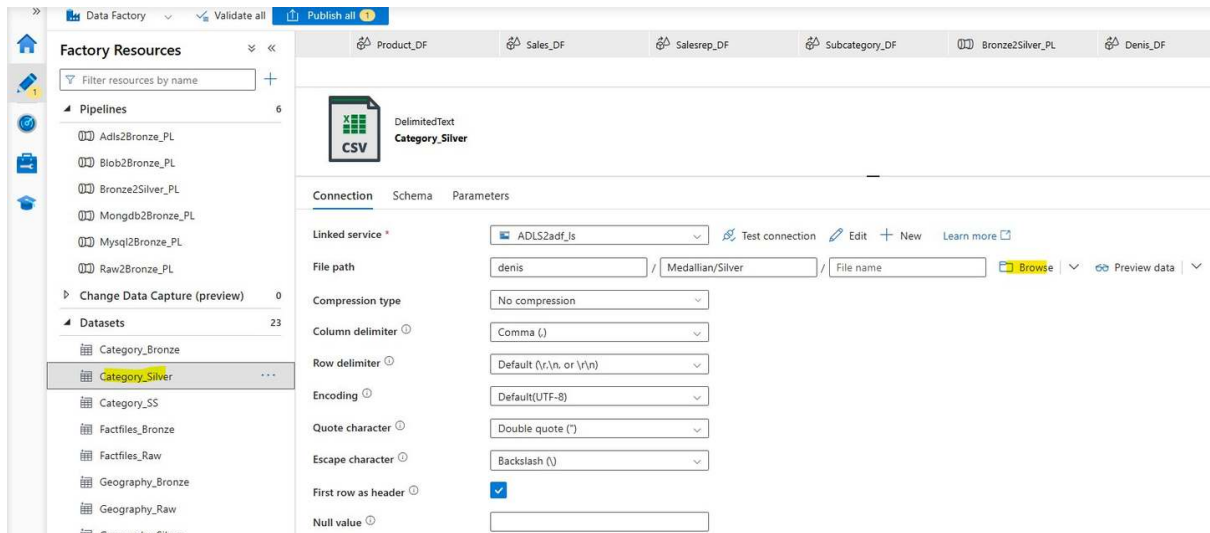
Skip line count

Sampling * ☐ Enable ☒ Disable



For all the sources added the input data from the silver layer

Now the problem is we are not able to see the columns in the source---when ever we are reading the file we gave only the path to store---now let's give the dataset aswell



For geography

» Data Factory Validate all Publish all 2

Factory Resources

Filter resources by name

Pipelines 6

- Adls2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview) 0

Datasets 23

- Category_Bronze
- Category_Silver
- Category_SS
- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver

DelimitedText Geography_Silver

CSV

Connection Schema Parameters

Linked service * ADLS2adf_ls Test connection Edit + New Learn more

File path denis / Medallian/Silver / File name Browse

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r\n, or \n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

» Data Factory Validate all Publish all 3

Factory Resources

Filter resources by name

Pipelines 6

- Adls2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview) 0

Datasets 23

- Category_Bronze
- Category_Silver
- Category_SS
- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver

DelimitedText Geography_Silver

CSV

Connection Schema Parameters

Linked service * ADLS2adf_ls Test connection Edit + New Learn more

File path denis / Medallian/Silver / Geography_Silver.csv Browse

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r\n, or \n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

For product

Data Factory | Validate all | Publish all

Factory Resources

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Geography_SS
- Product_Bronze
- Product_Raw
- Product_Silver**
- Product_SS
- Rawfiles_Bronze
- Rawfiles_Raw
- Sales_Silver
- Sales_SS
- Salesrep_Silver
- Salesrep_SS
- SalesRep_Bronze
- Subcategory_Bronze

DelimitedText Product_Silver

Connection | Schema | Parameters

Linked service * | ADLS2adf_Is | Test connection | Edit | + New | Learn more

File path | denis | / Medallian/Silver | File name | Browse

Compression type | No compression

Column delimiter | Comma (,)

Row delimiter | Default (\r\n or \n\n)

Encoding | Default(UTF-8)

Quote character | Double quote (")

Escape character | Backslash (\)

First row as header | ☒

Data Factory | Validate all | Publish all

Factory Resources

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Geography_SS
- Product_Bronze
- Product_Raw
- Product_Silver**
- Product_SS
- Rawfiles_Bronze
- Rawfiles_Raw
- Sales_Silver
- Sales_SS
- Salesrep_Silver
- Salesrep_SS
- SalesRep_Bronze
- Subcategory_Bronze

DelimitedText Product_Silver

Connection | Schema | Parameters

Linked service * | ADLS2adf_Is | Test connection | Edit | + New | Learn more

File path | denis | / Medallian/Silver | Product_Silver.csv | Browse

Compression type | No compression

Column delimiter | Comma (,)

Row delimiter | Default (\r\n or \n\n)

Encoding | Default(UTF-8)

Quote character | Double quote (")

Escape character | Backslash (\)

First row as header | ☒

For sales:

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

Factory Resources

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Geography_SS
- Product_Bronze
- Product_Raw
- Product_Silver
- Product_SS
- Rawfiles_Bronze
- Rawfiles_Raw
- Sales_Silver**
- Sales_SS
- Salesrep_Silver
- Salesrep_SS
- SalesRep_Bronze
- Subcategory_Bronze

DelimitedText Sales_Silver

Connection | Schema | Parameters

Linked service * | ADLS2adf_Is | Test connection | Edit | + New | Learn more

File path | denis | / Medallian/Silver | File name | Browse | Preview data | Detect format

Compression type | No compression

Column delimiter | Comma (,)

Row delimiter | Default (\r\n or \n\n)

Encoding | Default(UTF-8)

Quote character | Double quote (")

Escape character | Backslash (\)

First row as header | ☒

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Geography_SS
- Product_Bronze
- Product_Raw
- Product_Silver
- Product_SS
- Rawfiles_Bronze
- Rawfiles_Raw
- Sales_Silver**
- Sales_SS
- Salesrep_Silver
- Salesrep_SS
- SalesRep_Bronze
- Subcategory_Bronze

DelimitedText Sales_Silver

CSV

Connection Schema Parameters

Linked service * ADLS2adf_Is Test connection Edit + New Learn more

File path denis / Medallian/Silver / **Sales_Silver.csv**

Compression type No compression

Column delimiter Comma (,)

Row delimiter Default (\r,\n, or \r\n)

Encoding Default(UTF-8)

Quote character Double quote (")

Escape character Backslash (\)

First row as header ☒

For salesrep:

Factory Resources

Filter resources by name

- Factfiles_Bronze
- Factfiles_Raw
- Geography_Bronze
- Geography_Raw
- Geography_Silver
- Geography_SS
- Product_Bronze
- Product_Raw
- Product_Silver
- Product_SS
- Rawfiles_Bronze
- Rawfiles_Raw
- Sales_Silver
- Sales_SS
- Salesrep_Silver**
- Salesrep_SS
- SalesRep_Bronze
- Subcategory_Bronze

DelimitedText Salesrep_Silver

CSV

Connection Schema Parameters

Linked service * ADLS2adf_Is Test connection Edit + New Learn more

File path denis / Medallian/Silver / File name Browse

Compression type No compression

Column delimiter Comma (,)

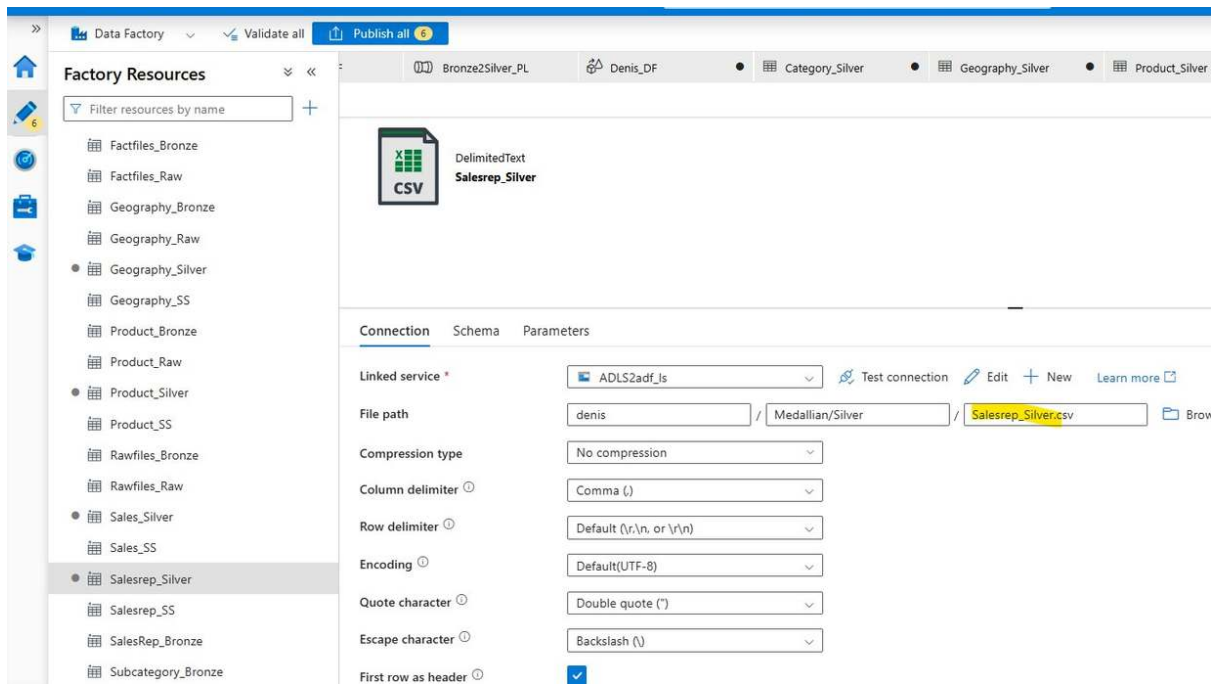
Row delimiter Default (\r,\n, or \r\n)

Encoding Default(UTF-8)

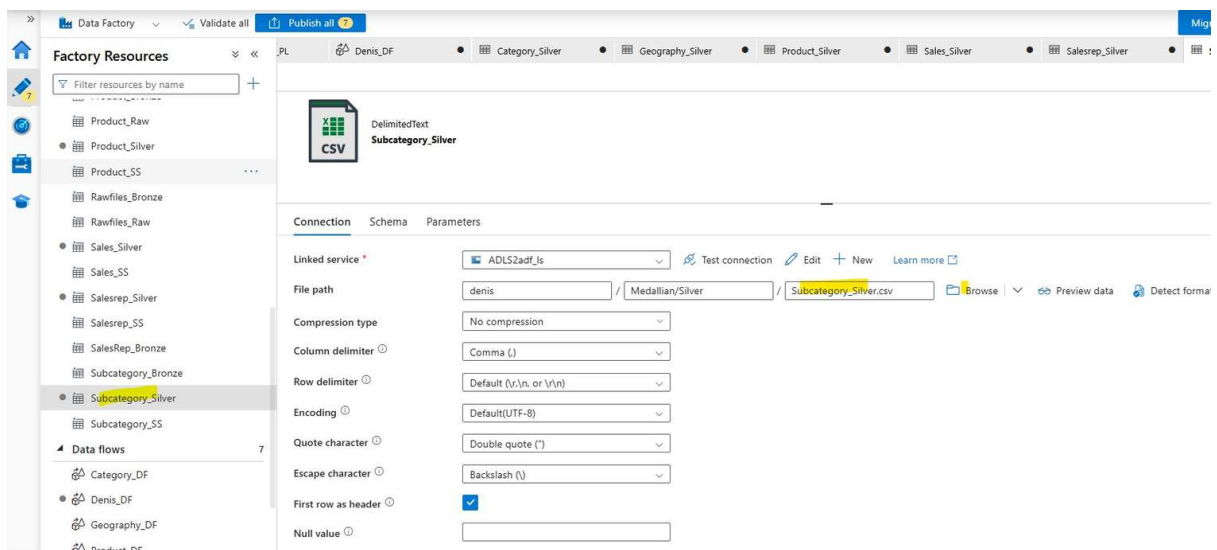
Quote character Double quote (")

Escape character Backslash (\)

First row as header ☒

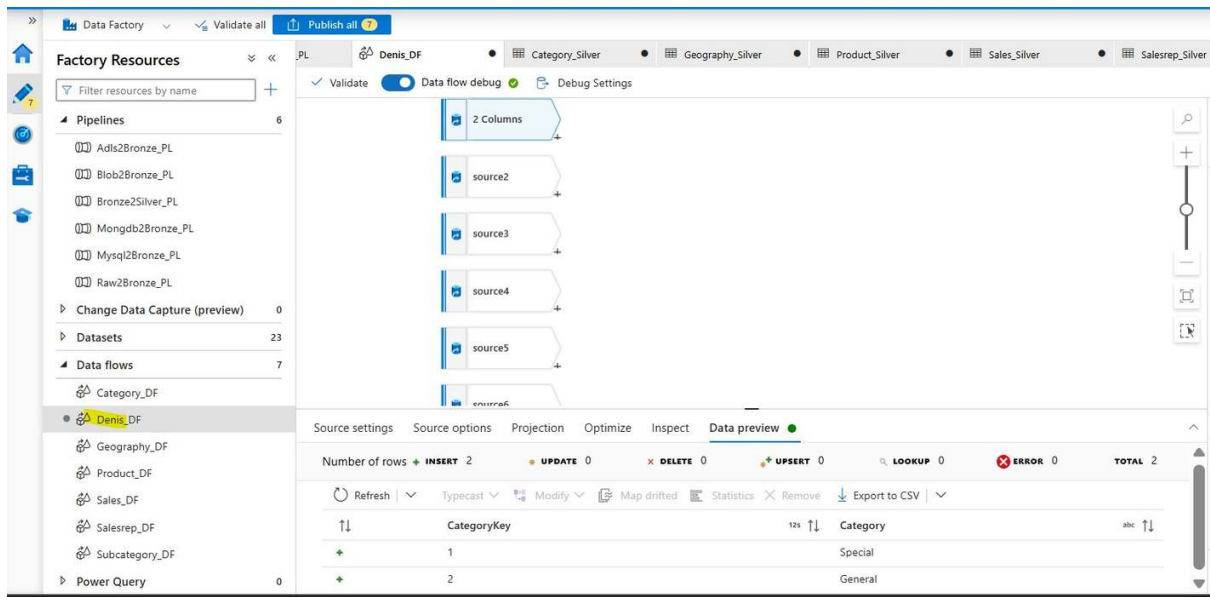


For subcategory

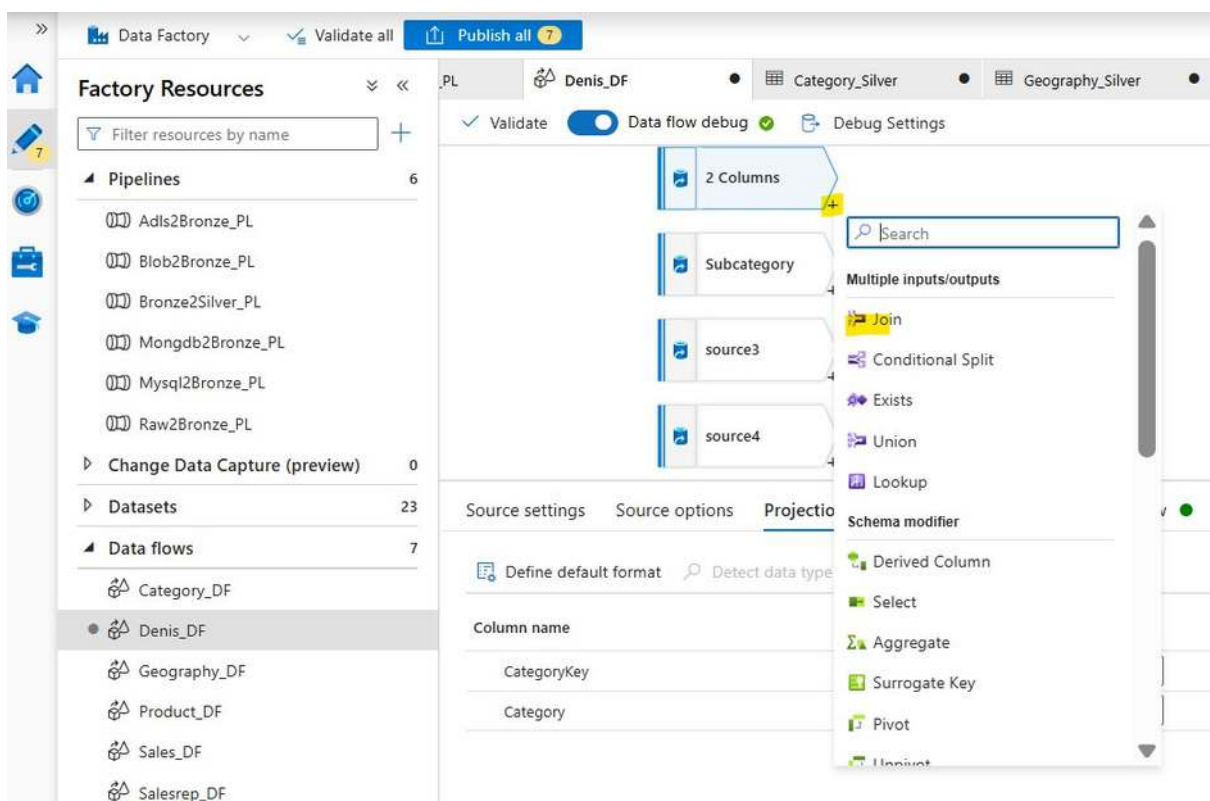


Go to denis_DF---for each and every source---projection---import projection

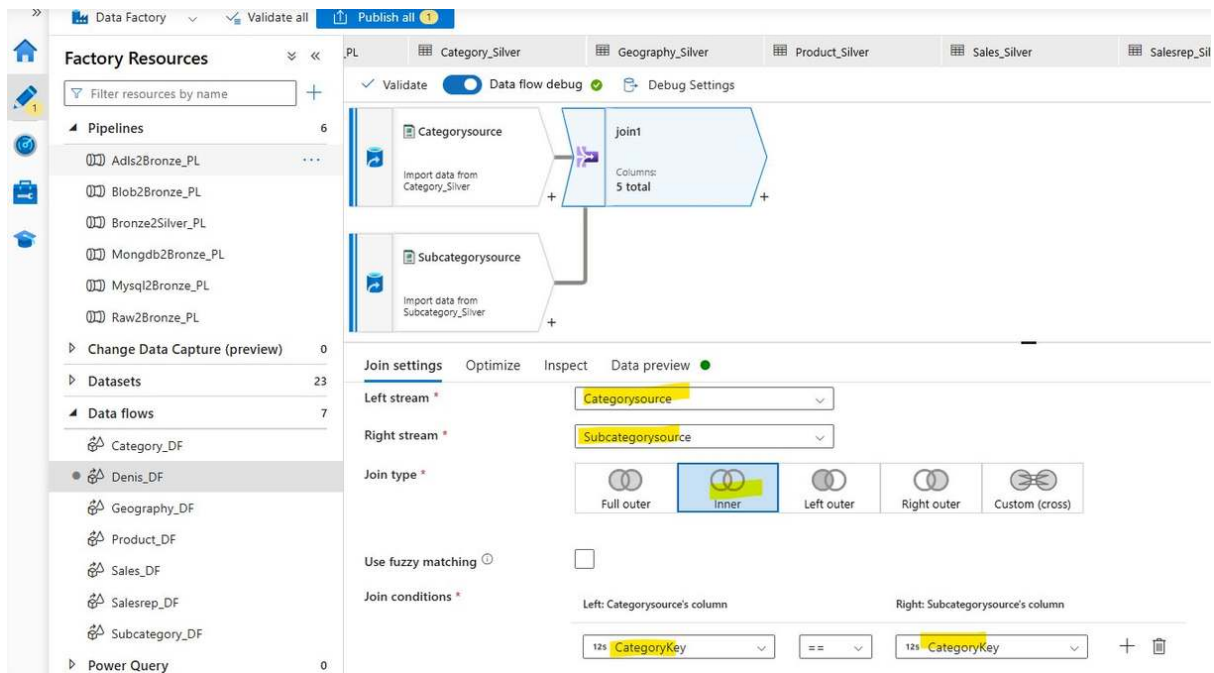
Succesfully we have the data in the source now



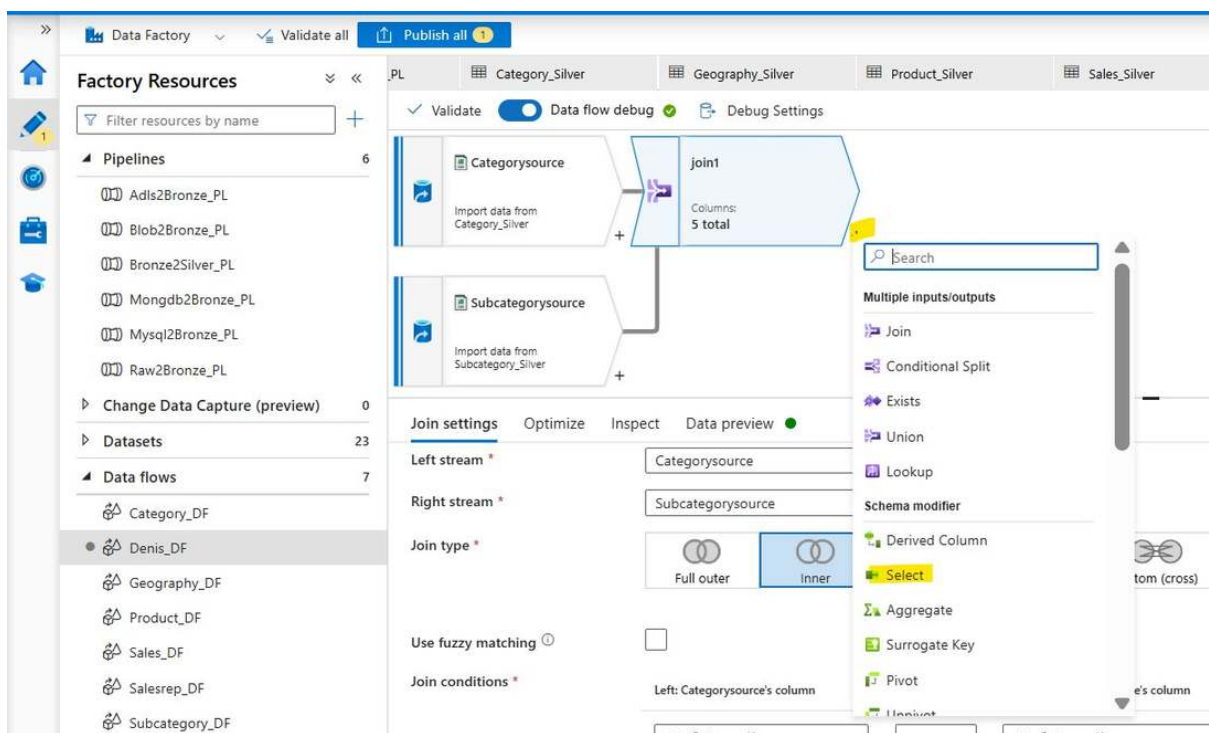
Now in source 1 we have category data---source 2 we have subcategory data---let's join the two tables-



Joining 2 tables with common column called categorykey from both tables



We will select the required columns



Now we have the same subcategorykey in the product table as well we need to join them

Factory Resources

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Raw2Bronze_PL
- Change Data Capture (preview): 0
- Datasets: 23
- Data flows: 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Join settings

Output stream name: join2

Description: Inner join on 'Categoryandsubcategory' and 'Productsource'

Left stream: Categoryandsubcategory

Right stream: Productsource

Join type: Inner

Use fuzzy matching: ☐

Join conditions:

- Left: Categoryandsubcategory's column: 12s SubCategoryKey
- Operator: ==
- Right: Productsource's column: 12s SubCategoryKey

Let's remove unnecessary columns

Factory Resources

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Raw2Bronze_PL
- Change Data Capture (preview): 0
- Datasets: 23
- Data flows: 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Select settings

Output stream name: select1

Description: Renaming join2 to select1 with columns 'CategoryKey, Category, SubCategory Name, ProductID, Color, ProductName, RetailPrice, StandardCost'

Incoming stream: join2

Options:

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns

Input columns: 12 columns

Output columns: 10 columns

Factory Resources

- Pipelines: 6
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Raw2Bronze_PL
- Change Data Capture (preview): 0
- Datasets: 23
- Data flows: 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF
 - Subcategory_DF
- Power Query: 0

Select settings

Output stream name: select1

Description: Renaming join2 to select1 with columns 'CategoryKey, Category, SubCategory Name, ProductID, Color, ProductName, RetailPrice, StandardCost'

Incoming stream: join2

Options:

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns

Input columns: 12 columns

Output columns: 10 columns

8 mappings: 2 column(s) from the inputs left unmapped

join2's column	Name as
12s CategoryKey	CategoryKey
12s Category	Category
12s SubCategory Name	SubCategory Name
12s ProductID	ProductID
12s Color	Color
12s ProductName	ProductName
12s RetailPrice	RetailPrice
12s StandardCost	StandardCost

- **Successfully joined 3 tables---category with subcategory---with product**

- The geography table doesnot have any common column to join

But the salesrep and sales data has common column called salesrepid so we can join them

The top screenshot shows the Microsoft Azure Data Factory interface. On the left, the 'Factory Resources' pane lists various pipelines and data flows. The main canvas displays a data flow with a 'Join' task. A dropdown menu is open, showing various tasks like 'Join', 'Conditional Split', 'Exists', 'Union', 'Lookup', 'Schema modifier', 'Derived Column', 'Select', 'Aggregate', 'Surrogate Key', 'Pivot', and 'Unpivot'. The 'Join' task is selected.

The bottom screenshot shows the 'Join settings' tab for a join task named 'join3'. The 'Output stream name' is 'join3'. The 'Description' is 'Inner join on 'Salesrepsource' and 'Salesource''. The 'Left stream' is 'Salesrepsource' and the 'Right stream' is 'Salesource'. The 'Join type' is set to 'Inner'. The 'Use fuzzy matching' checkbox is unchecked. The 'Join conditions' are defined as 'SalesRepID' from the left stream and 'SalesRepID' from the right stream, with an equals sign operator.

Let's remove the unnecessary columns

The top screenshot shows the Azure Data Factory interface with a data flow named 'Category_Silver'. The flow includes a 'Join' node that combines data from 'Categorysource', 'Subcategorysource', 'Productsource', 'Geographysource', 'Salesrepsource', and 'Salessource'. The 'Data preview' tab is active, showing a table with 11 columns. The bottom screenshot shows the 'Data preview' settings for the 'join3' stream. The 'Incoming stream' is 'join3'. The 'Options' section has 'Skip duplicate input columns' and 'Skip duplicate output columns' checked. The 'Input columns' section shows a table with columns and their corresponding names in the output stream.

Join3's column	Name as
125 Salesrepsource@SalesRepID	SalesRepID
abc Sales Rep Name	Sales Rep Name
123 fSalesPrimarykey	fSalesPrimarykey
125 ProductID	ProductID
abc Date	Date
125 Units	Units
1.2 PercentOfStandardCost	PercentOfStandardCost
1.2 RevenueDiscount	RevenueDiscount
abc Country	Country
abc Town	Town

**In the main datasets
categorysubcategoryproduct,salesrepsalesgeography we have a
common column called product id**

Factory Resources

Filter resources by name

Pipelines

- Adis2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview)

Datasets

Data flows

- Category_DF
- Denis_DF
- Geography_DF
- Product_DF
- Sales_DF
- Salesrep_DF
- Subcategory_DF

Power Query

Category_Silver

Geography_Silver

Product_Silver

Sales_Silver

Salesrep_Silver

Subcategory_Silver

Validate

Data flow debug

Debug Settings

Join settings

Output stream name *

Description

Inner join on 'catesubcatprod' and 'salesrepandsalesandgeography'

Incoming stream *

Join2

7 Columns

Search

Multiple inputs/outputs

- Join
- Conditional Split
- Exists
- Union
- Lookup

Schema modifier

- Derived Column
- Select
- Aggregate
- Surrogate Key
- Pivot
- Unpivot

Join settings

Output stream name *

join4

Learn more

Description

Inner join on 'catesubcatprod' and 'salesrepandsalesandgeography'

Reset

Left stream *

catesubcatprod

Right stream *

salesrepandsalesandgeography

Join type *

Full outer

Inner

Left outer

Right outer

Custom (cross)

Use fuzzy matching

Join conditions *

Left: catesubcatprod's column

Right: salesrepandsalesandgeography's column

12s ProductID

==

12s ProductID

+

-

Select the columns

Factory Resources

Filter resources by name

Pipelines

- Adis2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Raw2Bronze_PL

Change Data Capture (preview)

Datasets

Data flows

- Category_DF
- Denis_DF
- Geography_DF
- Product_DF
- Sales_DF
- Salesrep_DF
- Subcategory_DF

Power Query

Category_Silver

Geography_Silver

Product_Silver

Sales_Silver

Salesrep_Silver

Subcategory_Silver

Validate

Data flow debug

Debug Settings

Join settings

Output stream name *

join4

Learn more

Description

Inner join on 'catesubcatprod' and 'salesrepandsalesandgeography'

Reset

Left stream *

catesubcatprod

Right stream *

salesrepandsalesandgeography

Join type *

Full outer

Inner

Left outer

Right outer

Custom (cross)

Search

Multiple inputs/outputs

- Join
- Conditional Split
- Exists
- Union
- Lookup

Schema modifier

- Derived Column
- Select
- Aggregate
- Surrogate Key
- Pivot
- Unpivot

Join4's column	Name as
123 SalesPrimaryKey	Transactionid
123 catesubcatprod@ProductID	ProductID
123 SalesRepID	SalesRepID
abc Sales Rep Name	Sales Rep Name
abc ProductName	ProductName
abc Category	Category
abc SubCategory Name	SubCategory Name
abc Color	Color
1.2 RetailPrice	RetailPrice
1.2 StandardCost	StandardCost
Date	Date
123 Units	Units
1.2 PercentOfStandardCost	PercentOfStandardCost
1.2 RevenueDiscount	RevenueDiscount
abc Country	Country
abc Town	Town

Finally we have the denormalized data(with all the table info)

Requirement :we need to add 3 new columns

1)TotalRevenue: RetailPrice * Units

select1

Description

Renaming join4 to select1 with columns Transactionid, ProductID, SalesRepID, Sales Rep Name,

Incoming stream

join4

Options

- ☒ Skip duplicate input columns
- ☒ Skip duplicate output columns

2)Totalcost: StandardCost * Units

Dataflow expression builder

derivedColumn1

Derived Columns

+ Create new

123 TotalRevenue

ANY TotalCost

Column name *

TotalCost

Expression

StandardCost * Units

Expression elements

All
Functions
Input schema
Parameters
Cached lookup
Data flow library functions
Locals

Expression values

Filter by keyword

+ Create new

123 StandardCost

Date

123 Units

123 PercentOfStandardCost

Data preview Refresh

Save and finish

Cancel

Clear contents

Factory Resources

Filter resources by name

- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Denis_PL
- Raw2Bronze_PL
- Silver2Gold_PL
- Change Data Capture (preview) 0
- Datasets 24
- Data flows 7
 - Category_DF
 - Denis_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF

Derived column's settings

Output stream name * derivedColumn1

Description Creating/updating the columns Transactionid, ProductID, SalesRepID, Sales Rep Name, ProductName.

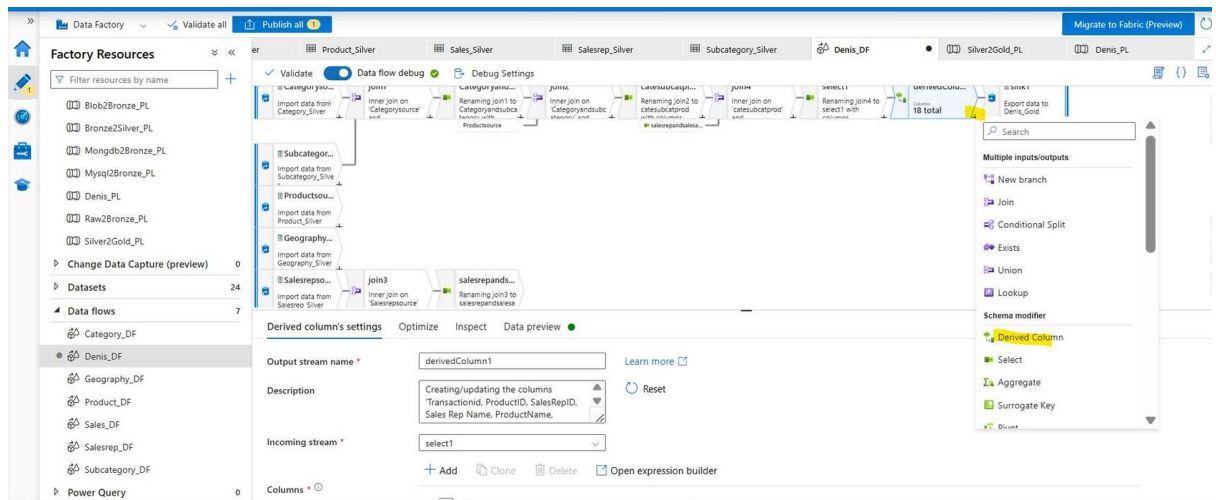
Incoming stream * select1

Columns *

Column	Expression
TotalRevenue	RetailPrice * Units
TotalCost	StandardCost * Units

3)Gross profit—TotalRevenue-TotalCost

But this cannot add in this because the 2 columns we derived here it's not possible to derive from those column---add derived column



Dataflow expression builder

derivedColumn2

Derived Columns

+ Create new

ANY GrossProfit

Column name *

GrossProfit

Expression

TotalRevenue - TotalCost

Expression elements

All

Functions

Input schema

Parameters

Cached loc No results found

Data flow library functions

Locals

Expression values

Filter by keyword

+ Create new

abc Town

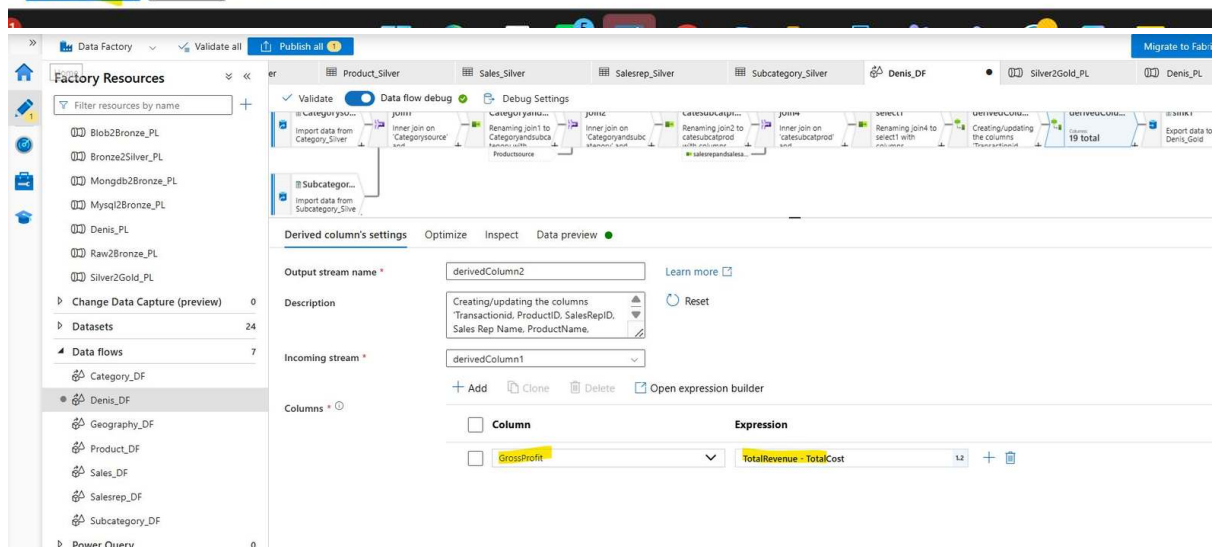
123 TotalRevenue

123 abs(123 numeric_value)

123 acos(123 numeric_value)

Data preview Refresh

Save and finish Cancel Clear contents



Preview the data:

Factory Resources

Filter resources by name

Change Data Capture (preview)

Datasets

Data flows

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Subcategory_DF

Power Query

Validate

Data flow debug

Debug Settings

Derived column's settings

Optimize

Inspect

Data preview

UPSET 0

LOOKUP 0

ERROR 0

TOTAL 1000

StandardCost	Date	Units	PercentOfStandardCost	RevenueDiscount	Country	Town	TotalRevenue	TotalCost	GrossProfit
7.55	2017-01-07	76	0.977	0.4	Germany	Berlin	1820.2	573.8	1246.4
7.55	2017-01-27	134	0.989	0.5	Czech republic	Prague	3209.2999999999997	1011.6999999999999	2197.6
7.55	2017-01-09	75	0.992	0.4	Germany	Berlin	1796.25	566.25	1230.0
7.55	2017-01-23	85	0.993	0.4	Germany	Frankfurt	2035.75	641.75	1394.0
7.55	2017-01-05	35	0.988	0.25	Denmark	Copenhagen	838.25	264.25	574.0
7.55	2017-01-12	172	0.987	0.55	Germany	Frankfurt	4119.4	1298.6	2820.7999999999997
7.55	2017-01-21	8	0.995	0.0	Czech republic	Prague	191.6	60.4	131.2
7.55	2017-01-26	7	0.996	0.0	Germany	Frankfurt	167.65	52.85	114.80000000000001

3 new column has added according to the business requirement

Let's add sink and dump into the gold

Factory Resources

Filter resources by name

Change Data Capture (preview)

Datasets

Data flows

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Subcategory_DF

Power Query

Validate

Data flow debug

Debug Settings

Select settings

Optimize

Inspect

Data preview

Join4's column

123 fSalesPrimaryKey

123 catesubcatprod@ProductID

123 SalesRepID

abc Sales Rep Name

abc ProductName

Name as

Transactionid

ProductID

SalesRepID

Sales Rep Name

ProductName

Sink

Factory Resources

Filter resources by name

Change Data Capture (preview)

Datasets

Data flows

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Subcategory_DF

Power Query

Validate

Data flow debug

Debug Settings

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

Link1

Description

Add sink dataset

Incoming stream *

select1

Sink type *

Dataset

Dataset *

Options

Allow schema drift

Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. Learn more

Select a data store

Search

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Continue

Cancel

Select format

Choose the format type of your data

 Avro	 DelimitedText	 JSON
 ORC	 Parquet	 Binary

Continue

Back

Cancel

Set properties

Name

Denis_Gold

Linked service *

ADLS2adf_ls

File path

denis

/ Medallian/Gold

/ File name

First row as header



Import schema

☒ From connection/store ☐ From sample file ☐ None

> Advanced

OK

Back

Cancel

Factory Resources

Filter resources by name

Pipelines

6

Adls2Bronze_PL

Blob2Bronze_PL

Bronze2Silver_PL

Mongodb2Bronze_PL

Mysql2Bronze_PL

Raw2Bronze_PL

Change Data Capture (preview)

0

Datasets

24

Data flows

7

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Category_Silver

Geography_Silver

Product_Silver

Sales_Silver

Salesrep_Silver

Subcategory_Sil

Validate

Data flow debug

Debug Settings

Categorysource

join1

Categoryandsu...

join2

catesubcatprod

join4

select1

16 Co

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink1

Learn more

Description

Export data to Denis_Gold

Reset

Incoming stream *

select1

Sink type *

Dataset

Inline

Cache

Dataset *

Denis_Gold

Test connection

Open

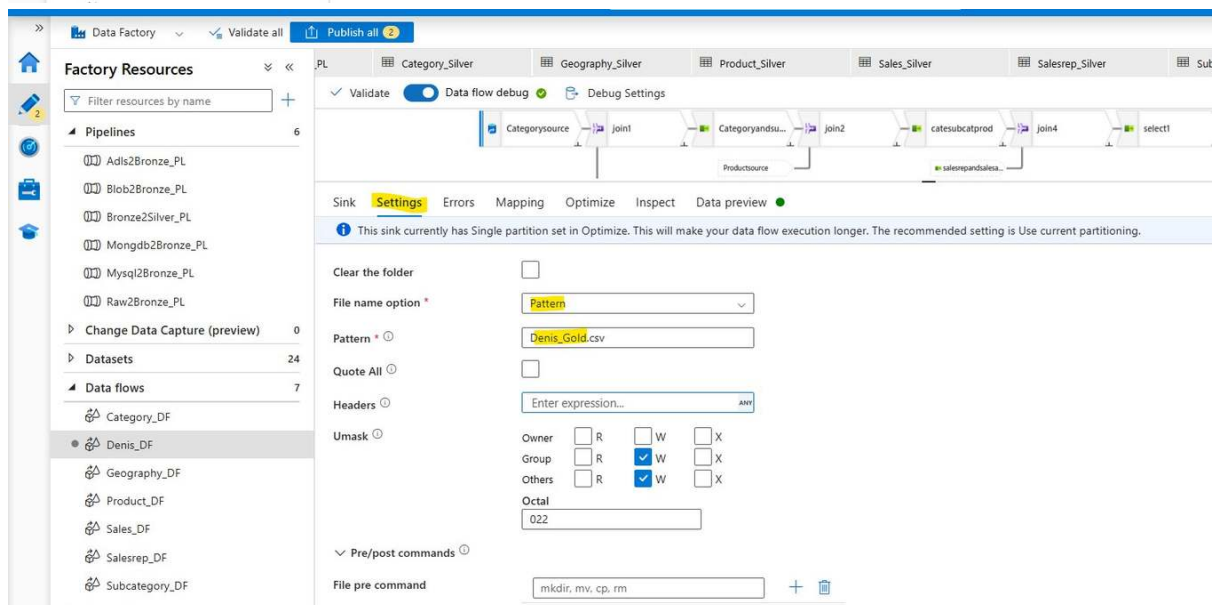
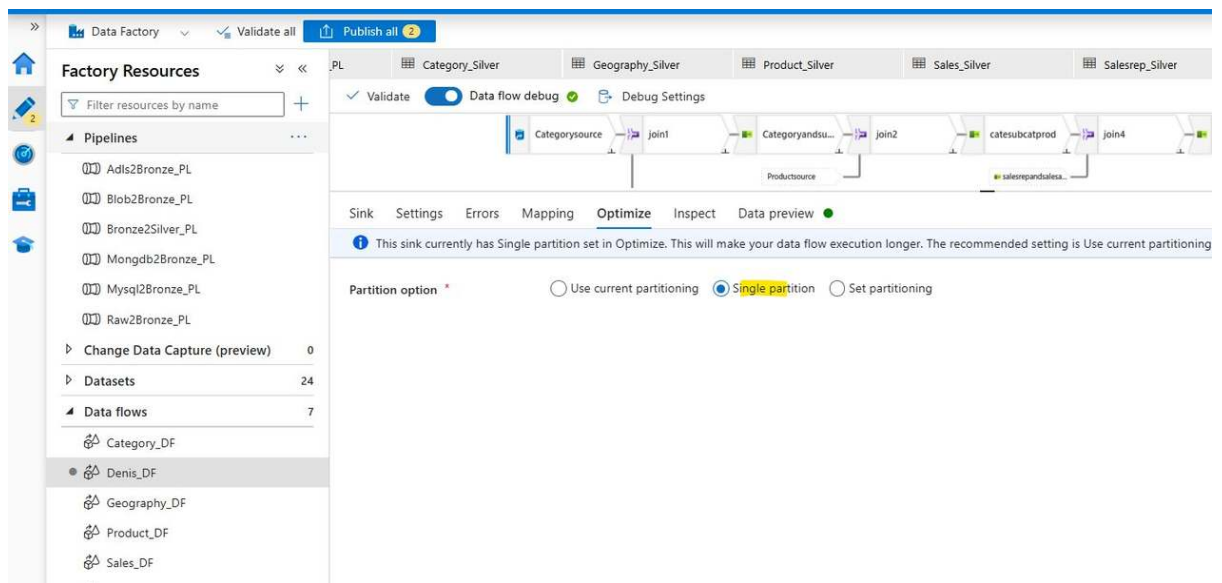
New

Skip line count

Options

☒ Allow schema drift

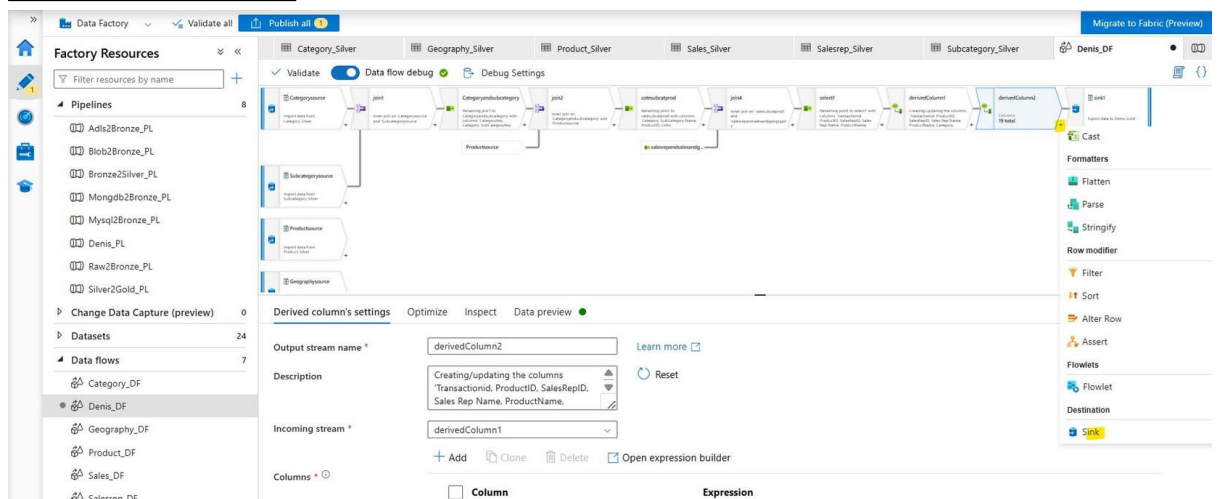
☐ Validate schema



Publish all changes in adf ---but this is for the file base format

*****Requirement:If I need it in thetable format:**

Add one more sink



»

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines

Adis2Bronze_PL

Blob2Bronze_PL

Bronze2Silver_PL

Mongodb2Bronze_PL

Mysql2Bronze_PL

Denis_PL

Raw2Bronze_PL

Silver2Gold_PL

Change Data Capture (preview)

Datasets

Data flows

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Category_Silver

Geography_Silver

Product_Silver

Sales_Silver

Validate

Data flow debug

Debug Settings

Import data from Category_Silver

Inner join on Category_Silver

Renaming join1

Inner join on Category_Silver

Renaming join2

Inner join on Category_Silver

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink2

Learn more

Description

Add sink dataset

Reset

Incoming stream *

derivedColumn2

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

New

Options

Allow schema drift

Validate schema

New dataset

In pipeline activities and data flows, reference a dataset to specify the location and structure of your data within a data store. [Learn more](#)

Select a data store

All

Azure

Database

File

Generic protocol

Amazon S3

Azure Blob Storage

Azure Cosmos DB for NoSQL

Azure Data Explorer (Kusto)

Azure Data Lake Storage Gen2

Azure Database for MySQL

Azure Database for PostgreSQL

Azure SQL Database

Azure SQL Database Managed Instance

Continue

Cancel

Set properties

Name

Denis_GG

Linked service *

AzureSqlDatabase2adf_Is

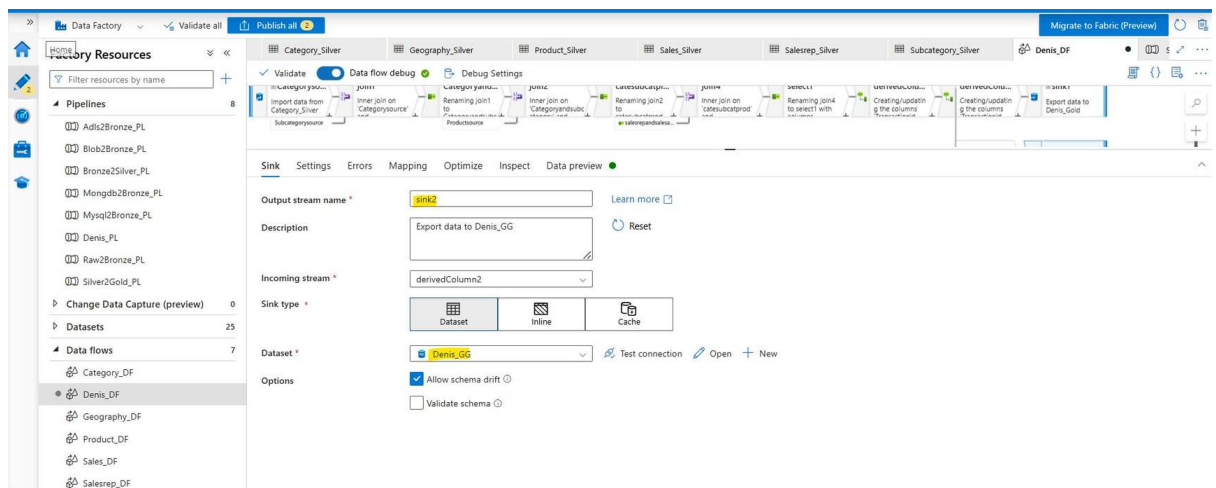
☐ Select from existing table ☒ New table

Schema and table name

Schema name Denis_Gold_TB

> Advanced

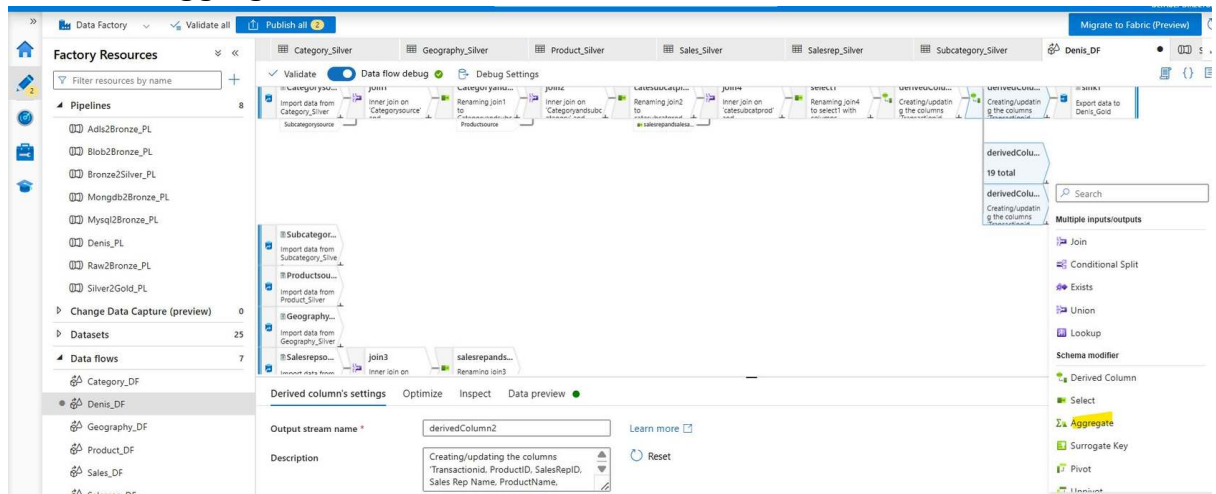
OK Back Cancel



This gives the table format denormalized data of denis

AnalyticalTables:

**Now I wanted to create some analytical tables---I will add a new branch--
-do some aggregations**



Microsoft Azure | Data Factory | nkdenisprojectadf

Factory Resources

- Pipelines (8)
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Denis_PL
 - Raw2Bronze_PL
 - Silver2Gold_PL
- Change Data Capture (preview) (0)
- Datasets (25)
- Data flows (7)
 - Category_DF
 - Denis_DF
 - Geography_DF

Category_Silver | Geography_Silver | Product_Silver | Sa

Validate | Data flow debug | Debug Settings

Aggregate settings | Optimize | Inspect | Data preview

Output stream name: aggregate1

Description: Aggregating data by 'SalesRepID' producing columns 'TotalCost, TotalRevenue, GrossProfit'

Incoming stream: derivedColumn2

Group by: Aggregates

Columns: SalesRepID | Name as: SalesRepID

Aggregate settings | Optimize | Inspect | Data preview

Output stream name: aggregate1

Description: Aggregating data by 'SalesRepID' producing columns 'TotalCost, TotalRevenue, GrossProfit'

Incoming stream: derivedColumn2

Group by: Aggregates

Grouped by: SalesRepID

+ Add | Clone | Delete | Open expression builder

Column	Expression
TotalCost	sum(TotalCost)
TotalRevenue	sum(TotalRevenue)
GrossProfit	sum(GrossProfit)

Add sink

[illegible]

Set properties

Name

AnalyticalSalesrepsummary

Linked service *

AzureSqlDatabase2adf_ls

☐ Select from existing table

☒ New table

Schema and table name

Schema name

AnalyticalSalesrepsummary_TB

> Advanced

The screenshot shows the 'Set properties' dialog for a new table in Azure Data Factory. The 'Name' field is 'AnalyticalSalesrepsummary'. The 'Linked service' is 'AzureSqlDatabase2adf_ls'. The 'Select from existing table' option is unselected, and the 'New table' option is selected. The 'Schema and table name' section shows 'Schema name' and 'AnalyticalSalesrepsummary_TB'. Below the dialog, a preview of the ADF pipeline is visible, showing a data flow from 'Category_Silver' to 'Sink1'.

Analytical table2:

Add new branch from the before sink 1

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines (8)
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Denis_PL
 - Raw2Bronze_PL
 - Silver2Gold_PL
- Change Data Capture (preview) (0)
- Datasets (26)
- Data flows (7)
 - Category_DF
 - Denis_DF

Category_Silver | Geography_Silver | Product_Silver | Sales_Silver | Salesrep_Silver

✓ Validate | Data flow debug | Debug Settings

Categorysource → join1 → Categoryandsu... → join2 → catesubcatprod → join4 → select1 → derived

Aggregate settings | Optimize | Inspect | Data preview

Output stream name: aggregate2

Description: Aggregating data by 'ProductID' producing columns 'GrossProfit, Avg_GrossProfit'

Incoming stream: derivedColumn2

Group by: Aggregates

Columns: ProductID | Name as: ProductID

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines (8)
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Denis_PL
 - Raw2Bronze_PL
 - Silver2Gold_PL
- Change Data Capture (preview) (0)
- Datasets (26)
- Data flows (7)
 - Category_DF
 - Geography_DF
 - Product_DF

Category_Silver | Geography_Silver | Product_Silver | Sales_Silver | Salesrep_Silver

✓ Validate | Data flow debug | Debug Settings

Categorysource → join1 → Categoryandsu... → join2 → catesubcatprod → join4 → select1

Aggregate settings | Optimize | Inspect | Data preview

Output stream name: aggregate2

Description: Aggregating data by 'ProductID' producing columns 'GrossProfit, Avg_GrossProfit'

Incoming stream: derivedColumn2

Group by: Aggregates

Grouped by: ProductID

+ Add | Clone | Delete | Open expression builder

Column	Expression
GrossProfit	sum(GrossProfit)
Avg_GrossProfit	avg(GrossProfit)

Sink this to db

Data Factory | Validate all | Publish all

Factory Resources

- Pipelines (8)
 - Adls2Bronze_PL
 - Blob2Bronze_PL
 - Bronze2Silver_PL
 - Mongodb2Bronze_PL
 - Mysql2Bronze_PL
 - Denis_PL
 - Raw2Bronze_PL
 - Silver2Gold_PL
- Change Data Capture (preview) (0)
- Datasets (26)
- Data flows (7)
 - Category_DF
 - Geography_DF
 - Product_DF
 - Sales_DF
 - Salesrep_DF

Category_Silver | Geography_Silver | Product_Silver | Sales_Silver

✓ Validate | Data flow debug | Debug Settings

Categorysource → join1 → Productsource → join2 → salesrepanddate... → join4 → sink4

Sink | Settings | Errors | Mapping | Optimize | Inspect | Data preview

Output stream name: sink4

Description: Add sink dataset

Incoming stream: aggregate2

Sink type: Dataset

Dataset: Select... | + New

Options: ☒ Allow schema drift | ☐ Validate schema

Set properties

Name: AnalyticalProductSummary

Linked service: AzureSqlDatabase2015

☐ Select from existing table | ☒ New table

Schema and table name: AnalyticalProductSummary_TB

OK | Back | Cancel

[Data Factory](#)
[Validate all](#)
[Publish all](#)

Factory Resources

Filter resources by name

Pipelines 8

- Adls2Bronze_PL
- Blob2Bronze_PL
- Bronze2Silver_PL
- Mongodb2Bronze_PL
- Mysql2Bronze_PL
- Denis_PL
- Raw2Bronze_PL
- Silver2Gold_PL

Change Data Capture (preview) 0

Datasets 27

Data flows 7

- Category_DF
- Denis_DF
- Geography_DF
- Product_DF

Category_Silver
 Geography_Silver
 Product_Silver
 Sales_Silver

Validate
 Data flow debug
 Debug Settings

Categorysource
 Join1
 Categoryandsu...
 Join2
 catesubcatprod
 Join4
 salesrepandsales...

Subcategorysource
 Productsource

Sink
 Settings
 Errors
 Mapping
 Optimize
 Inspect
 Data preview

Output stream name *
 sink4
 Learn more

Description
 Export data to AnalyticalProductssummary
 Reset

Incoming stream *
 aggregate2

Sink type *
 Dataset
 Inline
 Cache

Dataset *
 AnalyticalProductssummary
 Test connection
 Open
 New

Options
 Allow schema drift
 Validate schema

AnalyticalTable3:country level

Add new branch from before the sink 1---add aggregation

Aggregate settings
 Optimize
 Inspect
 Data preview

Output stream name *
 aggregate3
 Learn more

Description
 Aggregating data by 'Country' producing columns 'TotalCost, Avg_Grossprofit'
 Reset

Incoming stream *
 derivedColumn2

Group by
 Aggregates

Columns
 Name as

Columns	Name as
abc Country	Country

Aggregate settings

Optimize

Inspect

Data preview

Output stream name *

aggregate3

Learn more

Description

Aggregating data by 'Country' producing columns 'TotalCost, Avg_Grossprofit'

Reset

Incoming stream *

derivedColumn2

Group by

Aggregates

Grouped by: Country

+ Add

Clone

Delete

Open expression builder

☐	Column	Expression
☐	TotalCost	sum(TotalCost) 1.2 +
☐	Avg_Grossprofit	avg(GrossProfit) 1.2 +

Add sink---to db

Data Factory

Validate all

Publish all

Factory Resources

Filter resources by name

Pipelines

Adls2Bronze_PL

Blob2Bronze_PL

Bronze2Silver_PL

Mongdb2Bronze_PL

Mysql2Bronze_PL

Raw2Bronze_PL

Silver2Gold_PL

Change Data Capture (preview)

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Data flows

Category_DF

Denis_DF

Geography_DF

Product_DF

Sales_DF

Salesrep_DF

Category_Silver

Geography_Silver

Product_Silver

Sales_Silver

Validate

Data flow debug

Debug Settings

Categorysource

Subcategorysource

join1

Categoryandou...

Productsource

join2

catessubcatprod

join4

salesrepdatains...

Sink

Settings

Errors

Mapping

Optimize

Inspect

Data preview

Output stream name *

sink5

Learn more

Description

Add sink dataset

Reset

Incoming stream *

aggregate3

Sink type *

Dataset

Inline

Cache

Dataset *

Select...

+ New

Options

Allow schema drift

Validate schema

Set properties

Name

AnalyticalCountrysummary

Linked service *

AzureSqlDatabase2adf1s

Select from existing table

New table

Schema and table name

Schema name

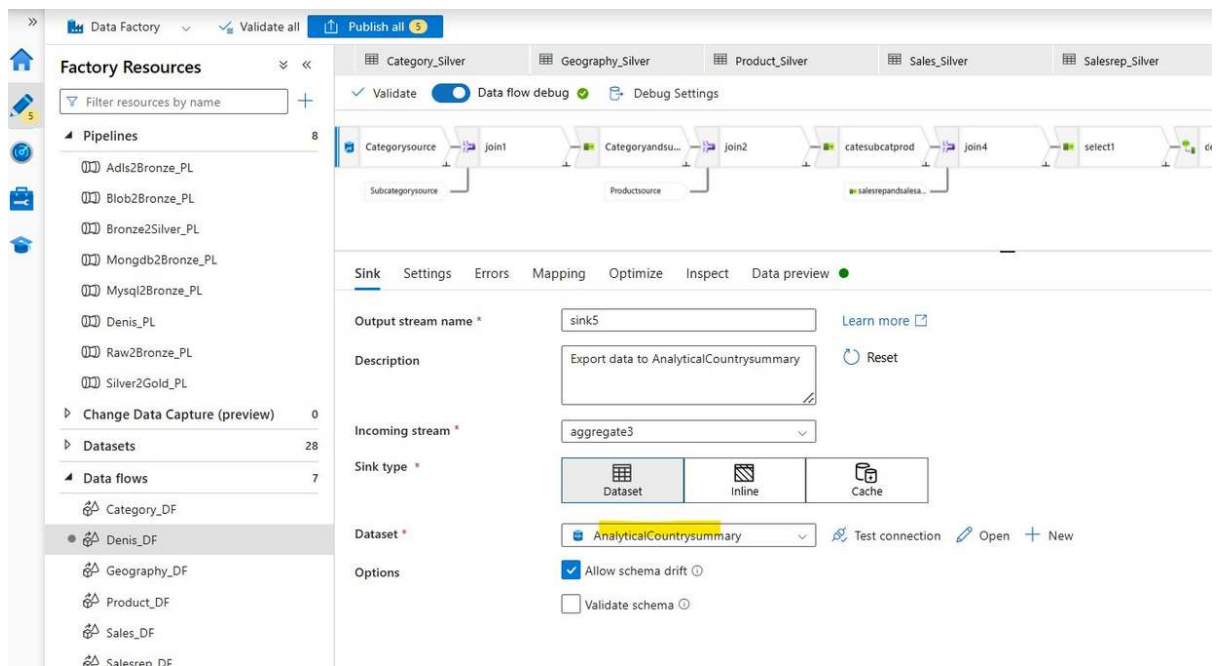
AnalyticalCountrysummary_TB

Advanced

OK

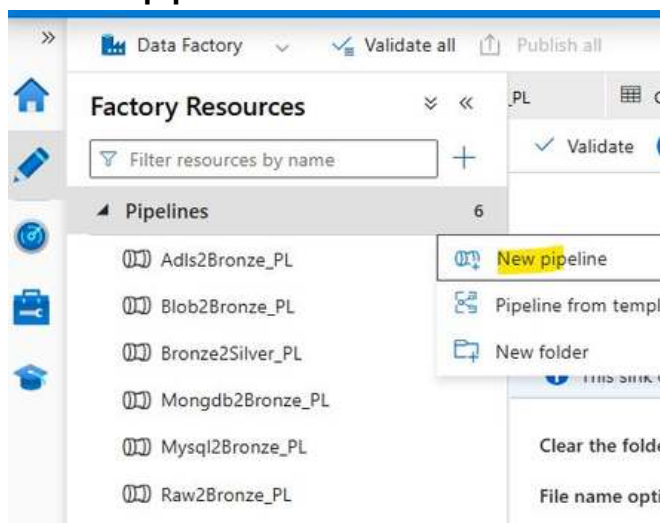
Back

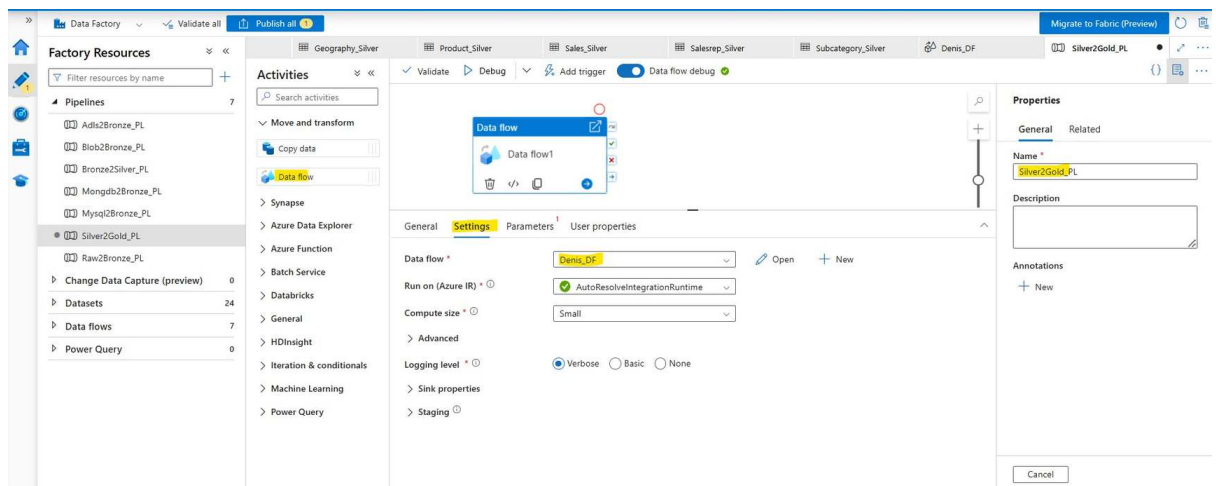
Cancel



Publish all the changes

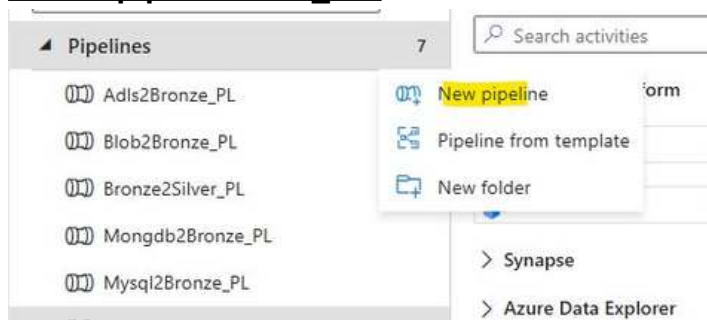
Create a pipeline from silver2 bronze



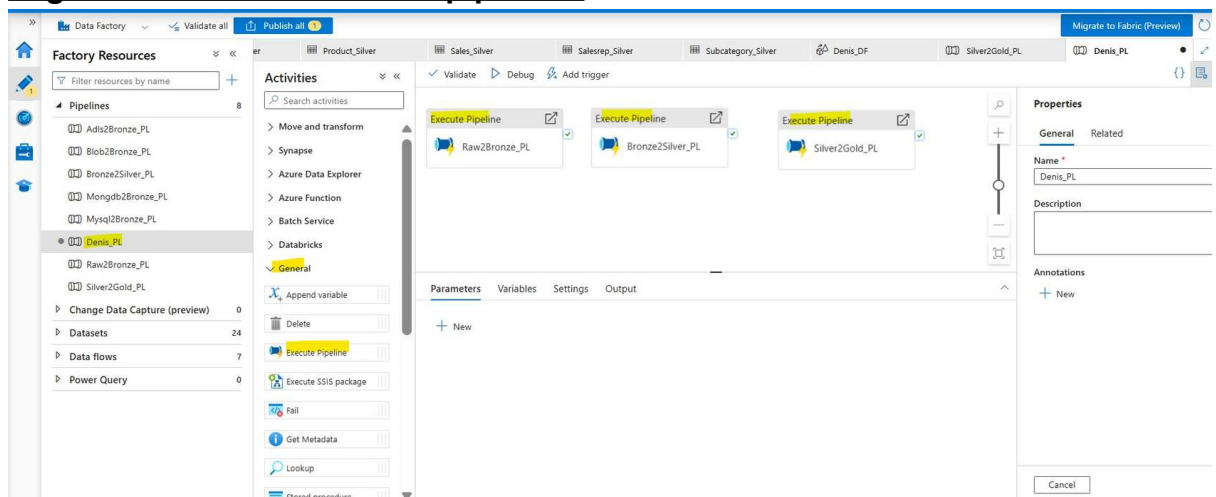


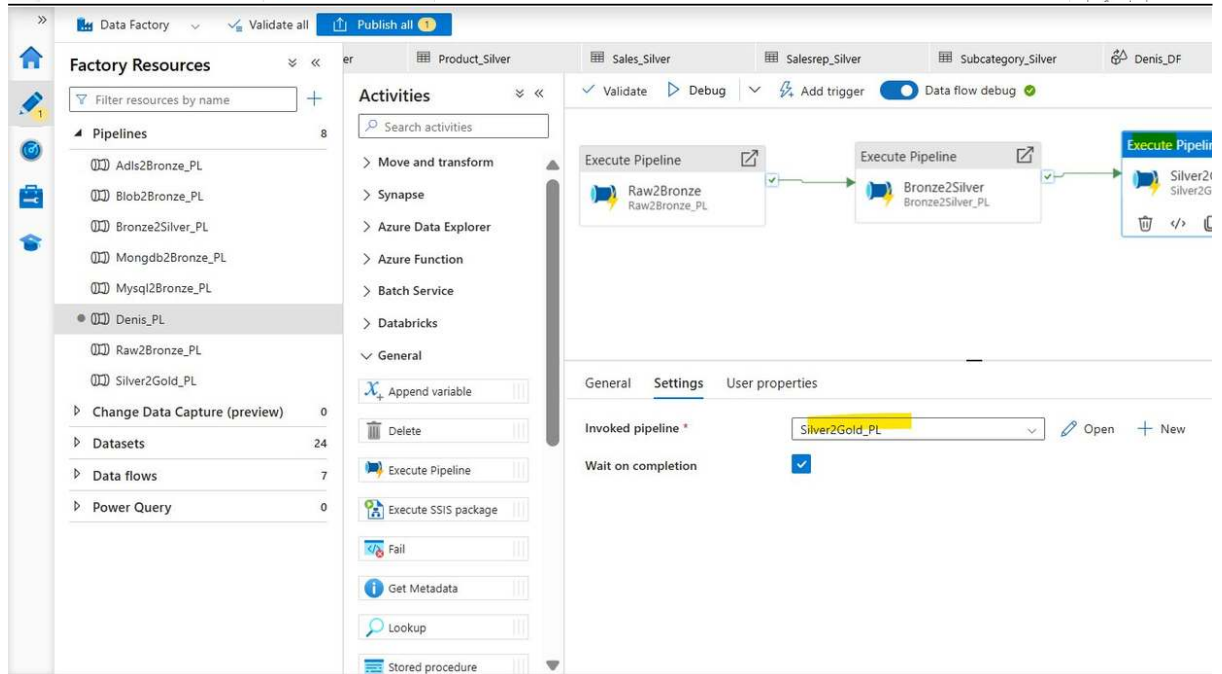
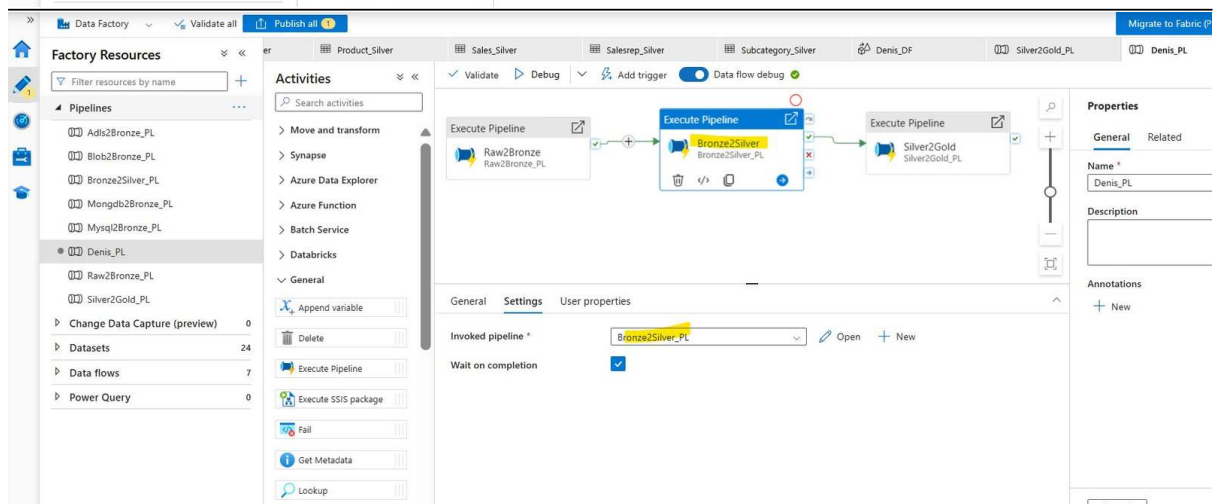
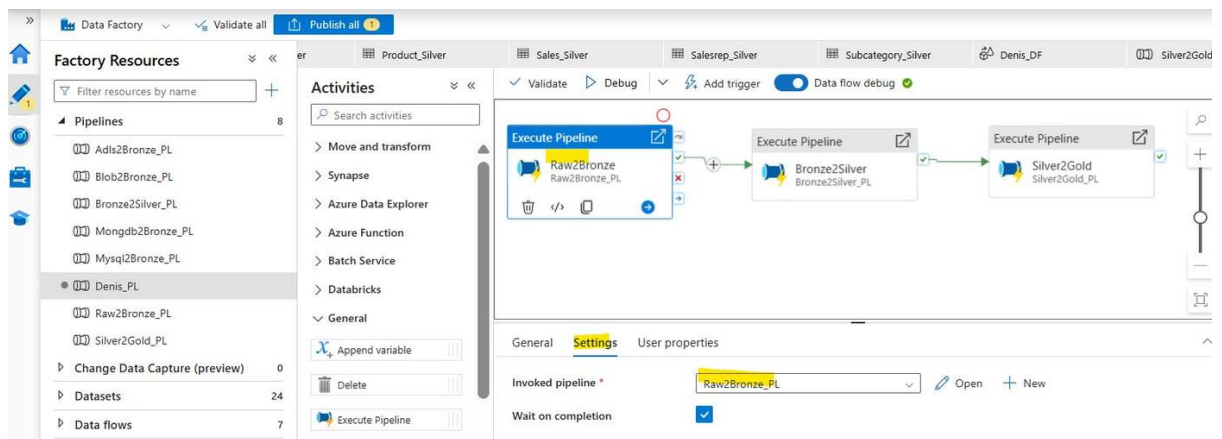
- Publish all changes
- Let's create a new pipeline which has all the bronze --silver---gold

Create pipeline denis_PL:



In general---select 3 execute pipelines

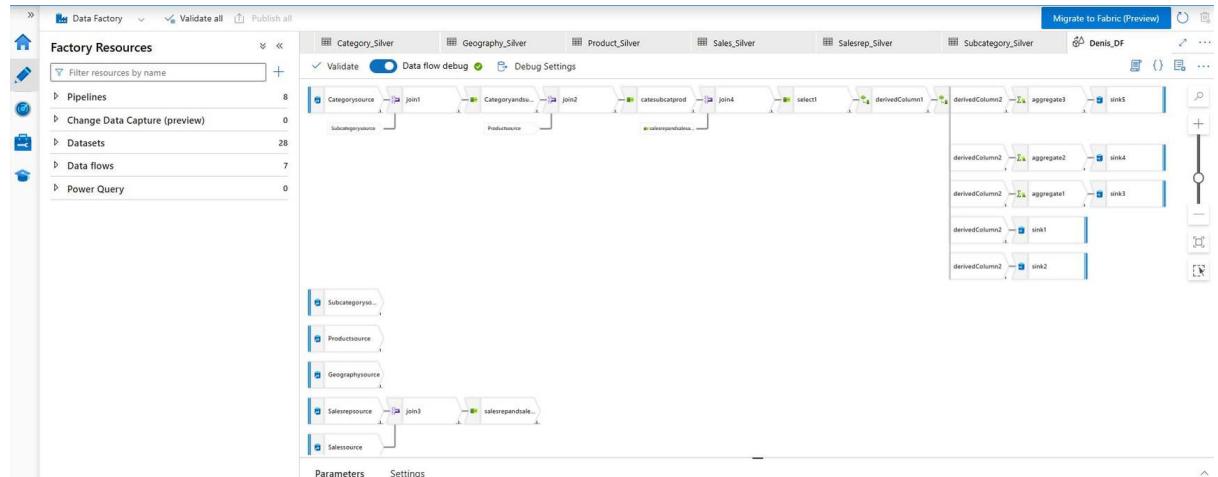




- **Publish all the changes**
- **Before executing the denis_pl---make sure the medallion layers should be empty**

- *******Every execute pipeline is linked with other because they are dependent so we need to connect then on success or else the pipeline fails because of no source files found**
Select denis_PL---add trigger---triggernow---run pipeline

Final dataflow looks like from silver 2 Gold:



Successfully run the pipeline

The screenshot shows the 'All pipeline runs' page for the 'Denis_PL' pipeline. It displays a sequence of three pipeline runs: 'Raw2Bronze', 'Bronze2Silver', and 'Silver2Gold', all of which have completed successfully. Below this, the 'Activity runs' section provides a detailed table of the execution results for the 'Silver2Gold' pipeline run.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Silver2Gold	Succeeded	Execute Pipeline	7/16/2026, 3:44:23 PM	34s			661816fc-bf25-4068-a029-987e388f3149
Bronze2Silver	Succeeded	Execute Pipeline	7/16/2026, 3:38:22 PM	6m 1s			9a04fbcf-7eec-45fe-b91f-af259afdaa28
Raw2Bronze	Succeeded	Execute Pipeline	7/16/2026, 3:37:03 PM	1m 17s			3391837f-aff1-436a-b9c4-0688a37a8b45

Bronze layer(Raw data is dumped)

Home > denisadlssa

denisadlssa | Storage browser

Storage account

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denisadlssa

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+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease

Blob containers > denis > Medallian > Bronze

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Showing all 6 items

Name	Last modified	Access tier	Blob type	Size
[.]				
sales	14/07/2026, 15:44:38			
Categories.xlsx	16/07/2026, 15:37:25	Cold (inferred)	Block blob	17
Geography.Json	16/07/2026, 15:38:06	Cold (inferred)	Block blob	96
SalesRep.csv	16/07/2026, 15:37:25	Cold (inferred)	Block blob	16
SubCategories.csv	16/07/2026, 15:37:25	Cold (inferred)	Block blob	11
'product'.csv	16/07/2026, 15:38:07	Cold (inferred)	Block blob	77

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+ Add Directory Upload Refresh Delete Copy Paste Rename Acquire lease Break lease Edit columns

Blob containers > denis > Medallian > Bronze > sales

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 4 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[.]					..
Sales 2014.csv	16/07/2026, 15:37:27	Cold (inferred)	Block blob	617.38 KiB	Available
sales 2015.csv	16/07/2026, 15:37:27	Cold (inferred)	Block blob	640.35 KiB	Available
sales 2016.csv	16/07/2026, 15:37:27	Cold (inferred)	Block blob	625.03 KiB	Available
sales 2017.csv	16/07/2026, 15:37:27	Cold (inferred)	Block blob	915.34 KiB	Available

Silverlayer: Normalized file format data

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Blob containers > denis > Medallian > Silver

Authentication method: Access key (Switch to Microsoft Entra user account)

Search blobs by prefix (case-sensitive)

Only show active objects

Showing all 6 items

Name	Last modified	Access tier	Blob type	Size	Lease state
[.]					...
Category_Silver.csv	16/07/2026, 15:43:05	Cold (inferred)	Block blob	41 B	Available
Geography_Silver.csv	16/07/2026, 15:41:09	Cold (inferred)	Block blob	167 B	Available
Product_Silver.csv	16/07/2026, 15:43:22	Cold (inferred)	Block blob	394 B	Available
Sales_Silver.csv	16/07/2026, 15:44:12	Cold (inferred)	Block blob	2.74 MiB	Available
Salesrep_Silver.csv	16/07/2026, 15:41:13	Cold (inferred)	Block blob	117 B	Available
Subcategory_Silver.csv	16/07/2026, 15:41:09	Cold (inferred)	Block blob	86 B	Available

Gold layer:denormalized data in file format

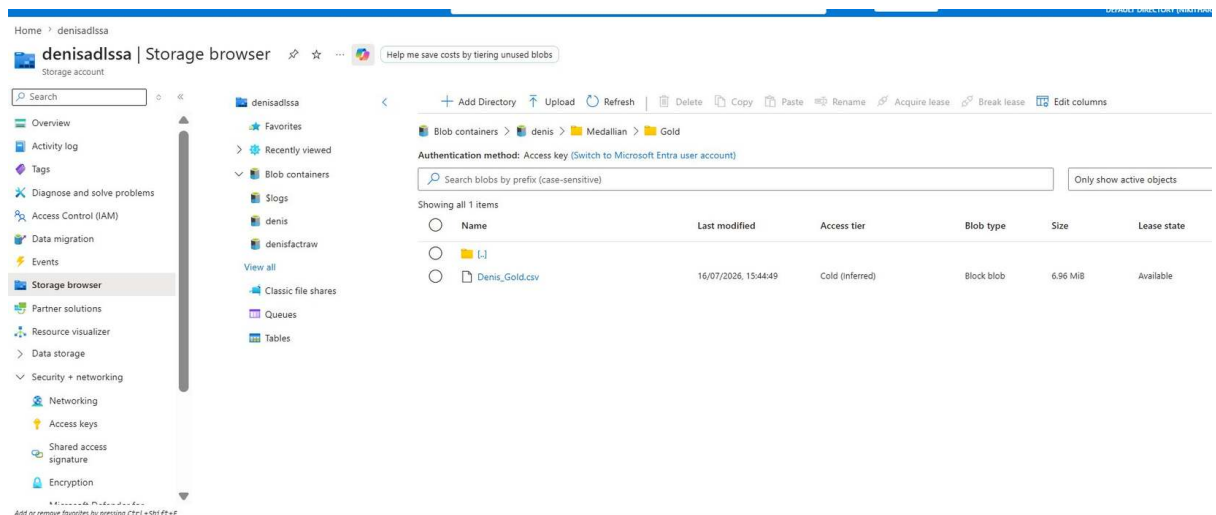
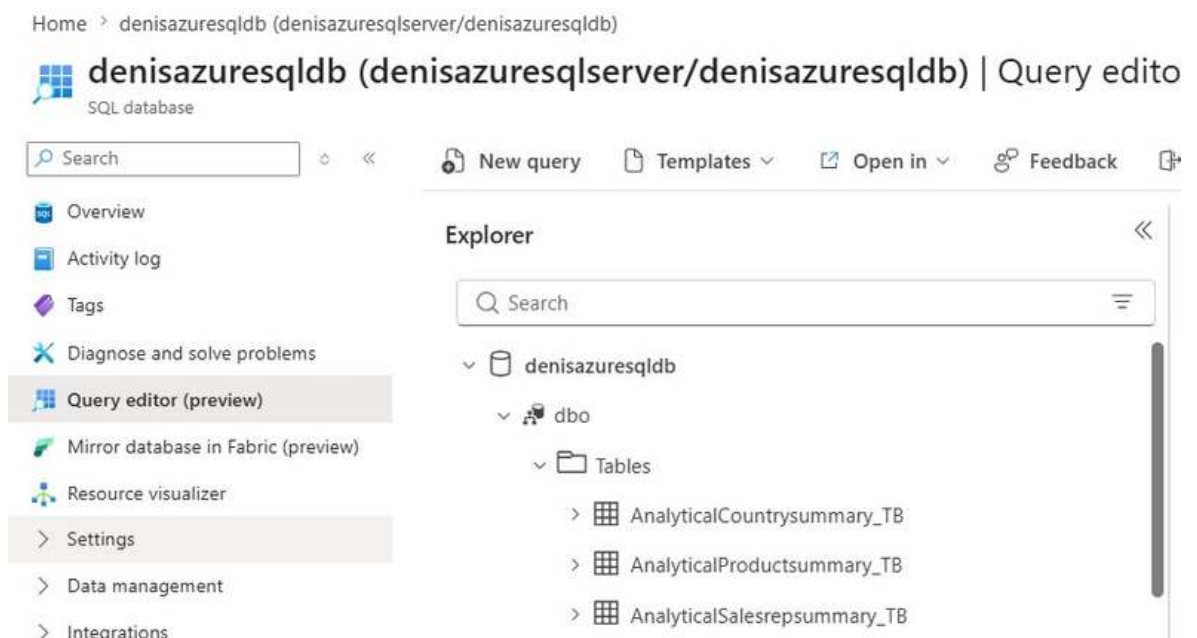


Table format data in azure sql db with analytical tables:



Normalized table data

Microsoft Azure

Home > denisazuresqladb (denisazuresqlserver/denisazuresqladb)

denisazuresqladb (denisazuresqlserver/denisazuresqladb) | Query editor

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 - AnalyticalProductsummary_TB
 - AnalyticalSalesrepsummary_TB
 - Category_TB
 - Denis_Gold_TB**

Normalized table format:

Home > denisazuresqladb (denisazuresqlserver/denisazuresqladb)

denisazuresqladb (denisazuresqlserver/denisazuresqladb) | Query editor

SQL database

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New query Templates Open in Feedback

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- Queries
 - SQL query 1

